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Electrorheological fluids: from historical retrospective to recent trends

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ABSTRACT

The presented review is an attempt to systematize the enormous scientific knowledge accumulated over several decades in the field of electrorheology. The review examines the basic principles of the electrorheological effect and its physical foundations, approaches of rheological description, and widely uses experimental techniques and methods. A roadmap for modern electrorheological fluids has been proposed. Various compositions of electrorheological fluids and the types of used fillers are considered in detail. The exceptional wideness of the operational characteristics of materials with electrorheological activity is shown, as well as their areas of application. The general modern research trends and directions that are still being developed are also noted. Particular attention of researchers is directed to materials with a low concentration of the dispersed phase and a contrasting transition from viscous to elastic behavior under an electric field. In this regard, fillers with high aspect ratio are widely considered for electrorheological fluids. It is assumed that this review will be useful not only for a novice reader who has just become familiar with the concept of electrorheological fluids, but also for an experienced researcher, in order to better understand the nature of the effect. The review can promote to reach the goal of creating materials with tunable and predetermined properties.

1. Introduction

The field of smart or stimuli-responsive materials has great attraction during recent decades in science and technology. They are called "smart" because of capability to reversibly change their properties under external influence, for example, mechanical, temperature, electric or magnetic fields, etc [1]. One of the vivid examples of such materials is electrorheological fluids (ERF), which are disperse systems and can change their physico-mechanical and rheological properties, such as viscosity, yield stress, values of the storage and loss moduli, and others, in a controlled manner by an electric field [2,3]. ERFs can almost instantly pass from liquid to solid state and vice versa under applied or removed electric field. This transition is called the electrorheological (ER) effect. The ER effect was discovered in the 40s of the XX century by American inventor Willis Winslow [4,5]. ERFs usually consist of several components. Basically, they are a dielectric non-conductive liquid media and easily polarizable filler particles. However, the composition may also include various additives that increase the ER effect or sedimentation stability of suspensions. Hao divide ERFs with positive, negative and photo ER effects [6]. Thus, rheological characteristics, such as viscosity, yield stress and storage modulus increase with a positive effect, and a negative effect leads to opposite results [7–10]. A photo ER effect is characteristic of photo-active materials, for example TiO_2 particles, and is the intensification of ER activity due to light irradiation [11,12]. Moreover, photo ER effect can also be positive or negative. It is interesting to note that there is a special class of photorheological fluids exhibiting changes in rheological behavior due to only ultraviolet irradiation. One of the advantages of these fluids is their significantly longer stability compared to ERFs. The rheological behavior is tuned by changing the conformation of polymer micelles. However, the observed viscosity changes are less contrasting in comparison with the ERF, which limits the range of potential applications of photorheological fluids [13,14]. Similar to photo ER effect, it is known fluids with dual-response under influence by both electric and magnetic fields [15,16]. Such effect is achieved by the introduction of magnetically sensitive compounds into the filler. Unfortunately, the number of studies in this field is still limited [17–20]. However, further we will focus on conventional ERFs with a positive effect.

Over more than 75 years of research in the field of electrorheology, various applications of ERFs have been found. However, it

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Abbreviations

AFM	atomic force microscopy
AMPSA	2-acrylamido-2-methyl-1-propanesulfonic acid
BDS	broadband dielectric spectroscopy
BTRU	barium titanyl-oxylate
CCJ	Cho-Choi-Jhon model
CMS	carbo-methyl starch
CNT	carbon nanotube
CTAB	cetyl-trimethylammonium bromide
CTP	calcium and titanium precipitation particles
DD	degree of deacetylation
DLS	dynamic light scattering
DLVO	theory proposed by B. V. Deryagin, L. D. Landau, E. Verway and J. Overbeek
DMSO	dimethyl sulfoxide
DND	detonation nanodiamond
DO	dielectric oil
EDS	energy-dispersive X-ray spectroscopy
ER	electrorheological
ERF	electrorheological fluid
FTIR	Fourier-transform infrared spectroscopy
GO	graphene oxide
HCTO	hybrid calcium titanyl oxalate
HNT	halloysite nanotube
HOPC	hydroxypropyl cellulose
MCC	microcrystalline cellulose
MFR	melamine-formaldehyde resin
MMT	montmorillonite
MCNT	multi-walled carbon nanotubes
PANI	polyaniline
PBMA	poly(butyl methacrylate)
PDES	poly(diethyl siloxane)
PDMS	polydimethylsiloxane
PEG	polyethylene glycol
PGMA	poly(glycidyl methacrylate)
PHEMATMS	poly(2-(trimethylsilyloxy)ethyl methacrylate)
PI-*co*-PTBMA	polyisoprene-*co*-poly(tert-butyl methacrylate)
PMMA	poly(methyl methacrylate)
PMPS	poly(methyl phenylsiloxane)
POA	poly(o-anisidine)
POSS	polyhedral oligomeric silsesquioxane
PPDA	poly(phenylene diamine)
SANS	small-angle neutron scattering
SAXS	small-angle X-ray scattering
SDBS	sodium dodecylbenzenesulfonate
SEC	size exclusion chromatography
SEM	scanning electron microscopy
SO	silicone oil
TEM	transmission electron microscopy
TOC	titanium oxalate
TSA	p-toluenesulfonic acid
TS-1	titanium silicalite-1
TTO	tin titanyl oxalate
W-TTO	wrinkly tin titanyl oxalate
XPS	X-ray photoelectron spectroscopy
XRD	X-ray diffraction

should be noted that all of them are based on the positive ER effect. In another article, Hao examined the classification of ERFs by components, as well as the models describing the observed effect actual at that time [21]. Nowadays, the proposed classification can be significantly expanded (Fig. 1). ERFs containing a liquid dispersed phase are not widely used due to problems with homogeneity, as well as a low ER response. Therefore, the attention of researchers is mainly focused on fluids with a solid dispersed phase. In addition, the dispersed phase is the most variable component of fluids and today a wide number of fillers of various nature are known. In particular, inorganic, which can be divided into oxygen containing, clay, carbon, and others, organic and polymeric, as well as composite. The nature and effectiveness of fillers will be discussed in more detail below.

Experimental studies have shown that the ER effect is associated with the processes of the dispersed phase particles polarization and

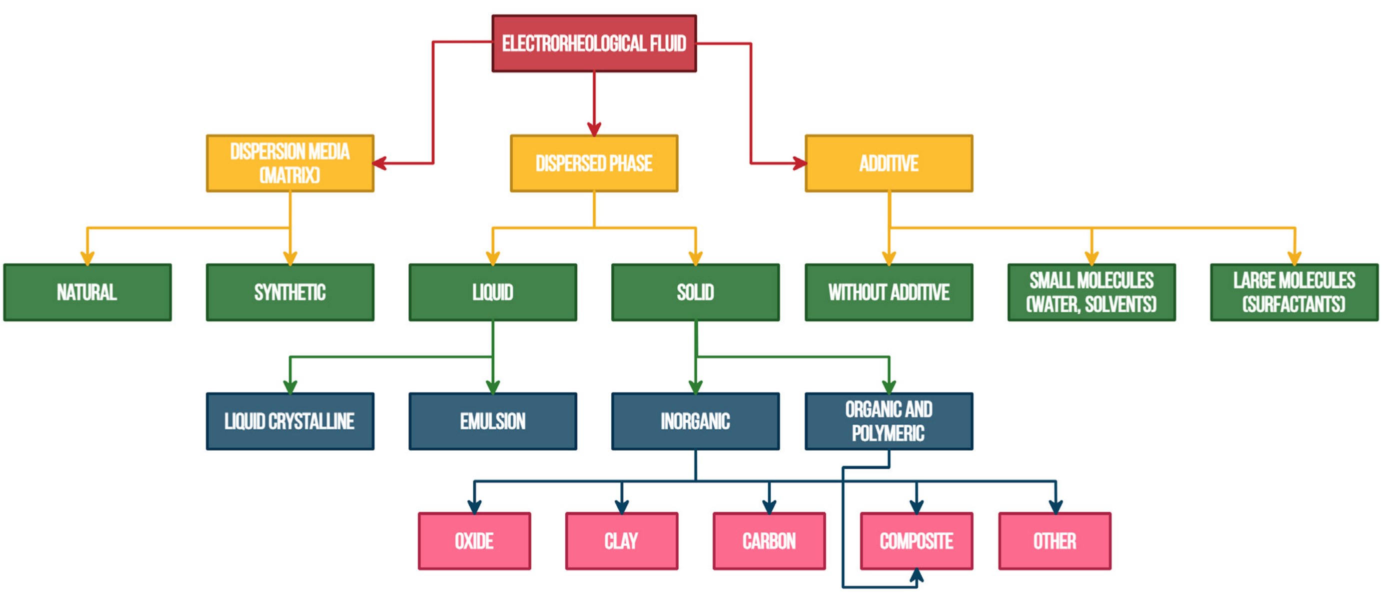

Fig. 1. Road map of ERFs. Adapted from the study by Hao et al. [21] with permission from Elsevier.

their orientation under electric field with the formation of extended chains and columnar structures, which in turn causes a change in the rheological behavior of the material under applied shear stress [22–24]. The idea of external control by material properties is extremely popular. Over the past 20 years, according to the Web of Science database, more than 100 publications are published annually on the request of an “electrorheological fluid” only. ERFs demonstrate their effectiveness in science and technology, for example, vibration control systems, dampers, sensors, microfluidics, and robotics. Potential practical applications of ERFs and some technological processes based on ER effect are described in more detail below in the last part of review.

2. Theory

2.1. Mechanisms of the ER effect

Theoretically, the formation of chains from particles is associated with various interactions that occur in suspension without and under electric field. A deep mathematical analysis of the effect can be found in the book by Ruzicka [25]. As part of our review, we will focus only on the general principles. Main types of mechanisms were discussed by Parthasarathy and Klingenberg [26]. Winslow suggested the mechanism of electrostatic polarization [5]. Its origin is related with the polarization of the dispersed phase particles compared with the dispersion medium. Thus, polarization can occur due to charge transport (electron, atomic, surface polarization) without taking into account the free charge of the particles, which allows considering the charge distribution and leads to ordering of the particles along the applied electric field. Later, Davis showed that along with polarization processes, electrical conductivity also plays a significant role in the appearance of the ER effect, especially in electric fields of low frequency or direct current [27].

Another possible mechanism is based on overlapping double electric layers [28,29]. It is assumed that in the suspension around each particle, a double electric layer is formed and these layers overlap for neighboring particles under the influence of an electric field, which leads to increased electrostatic repulsion between the particles that must be overcome to the particles flow past each other. However, the dispersion medium of ERFs usually has a low dielectric constant, which substantially complicates the formation of a double electric layer. Therefore, this mechanism has undergone significant criticism and is a special case of the electrostatic polarization.

Later, a mechanism for increasing viscosity due to the formation of water "bridges" between particles that are destroyed by shear flow was proposed [30]. The dependence of the ER effect on the electric field strength was explained by the migration of ions through the pores of the particles. The theory was refined, and it was supposed that water migrates into the gap between particles to minimize electrostatic energy [31,32]. Such mechanism was very popular because with a decrease in water content, the observed ER effect also decreased. However, with the discovery of anhydrous ERFs, it became apparent that only the formation of water bridges could not cause an ER response.

When a giant ER effect was discovered, it became clear that it was necessary to attract new ideas that explain the significant ER response of the systems and saturated polarization mechanism was proposed for the description [33,34]. For the first time, a giant ER effect was observed on particles with a core/shell structure. Such structural feature explains the increase in the ER response compared to previously known suspensions. The mechanism consists in the formation of strong dipole interactions between particles in the contact zone under the influence of an electric field due to thin surface layers of molecules with a high dipole moment, which can be supported by the formation of hydrogen bonds also. However, the limitations of this approach were demonstrated later. It was shown in the study by Agafonov et al. [35] that the adsorption of a strongly polar substance on the surface of a filler, for example, dimethyl sulfoxide (DMSO) (dipole moment 3.96 D) and less polar diethyl ether (dipole moment 1.15 D) results to a similar increase in yield stress, while the use of chloroform (dipole moment 1.15 D) and toluene (dipole moment 0.37 D) gave the largest increase in values. In more detail, the effects of additives will be discussed below in the section "Activators".

Thus, the polarization mechanism remains the main one while considering the nature of the ER effect, but may also include additional factors leading to an increase or decrease in the observed experimental parameters.

2.2. Rheological models

As was mentioned above, the formation of particle chains or columns in a suspension under electric field is the driving force for the ER behavior of suspensions resulting in the appearance or increase in the values of the yield stress. Basically, fluids with a yield stress exhibit plastic "Bingham" behavior under shear flow [36]:

$$\begin{cases} \tau(\dot{\gamma}) = \tau_0 + \eta\dot{\gamma}, & at\ \tau > \tau_0 \\ \dot{\gamma} = 0, & at\ \tau < \tau_0 \end{cases}, \tag{1}$$

where, τ is shear stress, τ_0 is yield stress, η is dynamic viscosity, and $\dot{\gamma}$ is shear rate.

The fluid exhibits a Newtonian flow at shear stress exceeding the yield stress and is solid-like when shear stress is lower than the yield point, which in the case of the ERFs is associated with the destruction of the structure formed by filler particles under electric field. It also should be noted that for ERFs the yield stress becomes dependent not only on the shear rate but also on the electric field strength.

However, the rheological behavior of elastic disperse systems can be more accurately described by the generalized flow equation [37]:

$$\tau^{1/2} = \frac{\tau_c^{1/2}}{1 + \chi / \dot{\gamma}^{1/2}} + \eta_c^{1/2}\,\dot{\gamma}^{1/2}, \tag{2}$$

where, τ_c is yield stress by Casson, η_c is viscosity by Casson, χ is parameter related with the type of particles packing in the agglomerate.

A special empirical rheological model for describing the behavior of ERFs was proposed by Cho, Choi, and Jhon, so-called "CCJ − model" [38]:

$$\tau = \frac{\tau_0}{1 + (t_1\dot{\gamma})^{\alpha}} + \eta_\infty\left(1 + \frac{1}{(t_2\dot{\gamma})^{\beta}}\right)\dot{\gamma}, \tag{3}$$

The model contains six variable parameters, where τ_0 is dynamic yield stress, determined by extrapolating the shear stress at low shear rates to zero, α associated with a decrease in shear stress, t_1 and t_2 are the time constants, wherein t_1 is the reciprocal of the shear rate at which the shear stress shows a minimum in the region with low shear rates and t_2 is the reciprocal of the shear rate at which pseudo-Newtonian behavior begins (meaning relaxation times), η_∞ is viscosity at high shear rates close to viscosity without electric field, and β varies within $0 < \beta \leq 1$ at $d\tau/d\dot{\gamma} \geq 0$. The model showed a better description of flow curves, especially in the region of low shear rates compared to other models for ERFs of various composition (Fig. 2a,c) [39–44]. The proposed equation considers the shape of the flow curve regardless of external influences and the effect of electric field indirectly correlates with the parameters of the function.

Later, Seo and Seo proposed a four-parametric equation, and showed good agreement with experimental data. It is noted that in some cases this model fits better than CCJ [45]:

$$\tau = \tau_{sy}\left(1 - \frac{(1-\exp(-a\dot{\gamma}))}{(1+(a\dot{\gamma})^{\alpha})}\right) + \eta_{\infty}\dot{\gamma}, \quad (4)$$

where, τ_{sy} is the static yield stress, a is the reciprocal of the shear rate where the rheological behavior changes, α is fitting parameter, η_{∞} is the viscosity at a high shear rate and $\dot{\gamma}$ is shear rate.

Typically, the yield stress (τ) correlates with electric field (E) as power-law:

$$\tau \sim E^{\alpha} \quad (5)$$

where, α is exponent, typically varies from 1.0 to 2.0.

The yield stress reveals a quadratic dependence on the electric field for the polarization mechanism. However, exponent decreases to 1.5 when conductivity dominates ($\tau \sim E^{1.5}$) [27,46]. A decrease in the proportionality of less than 1.5 down to 1 was observed in a number of experimental studies [47–49]. In this case, the mechanism of the ER effect becomes more complex and includes the possible formation of hydrogen bonds, as well as polarization saturation [50,51].

An important correlation of the yield stress versus electric field strength for representing experimental data was proposed by Choi et al. [52]. Futher equation was modified to generalize model for describing both conventional and giant ERFs [53]:

$$\tau_y\,(E_0) = \alpha E_0^2 \tanh\left(\frac{E_0}{E_c}\right)^{\beta} \Big/ \left(\frac{E_0}{E_c}\right)^{\beta} \quad (6)$$

where, τ_y is yield stress, E_0 is electric field strength, E_c is the critical electric field corresponding to a change in the slope of the yield stress dependence on the field strength during the transition from one mechanism to another (from polarization to conductive or saturated polarization) (Fig. 2b,d), α depends on the dielectric constant of the fluid and β is material parameter.

Later, based on Eq. (6), Seo and Seo proposed the improved model that well describes the behavior of giant ERFs [54,55]. Based on the rheological behavior of the Bingham fluids and the physical basis of the effect, the authors presented a simple equation to describe the behavior of ERF:

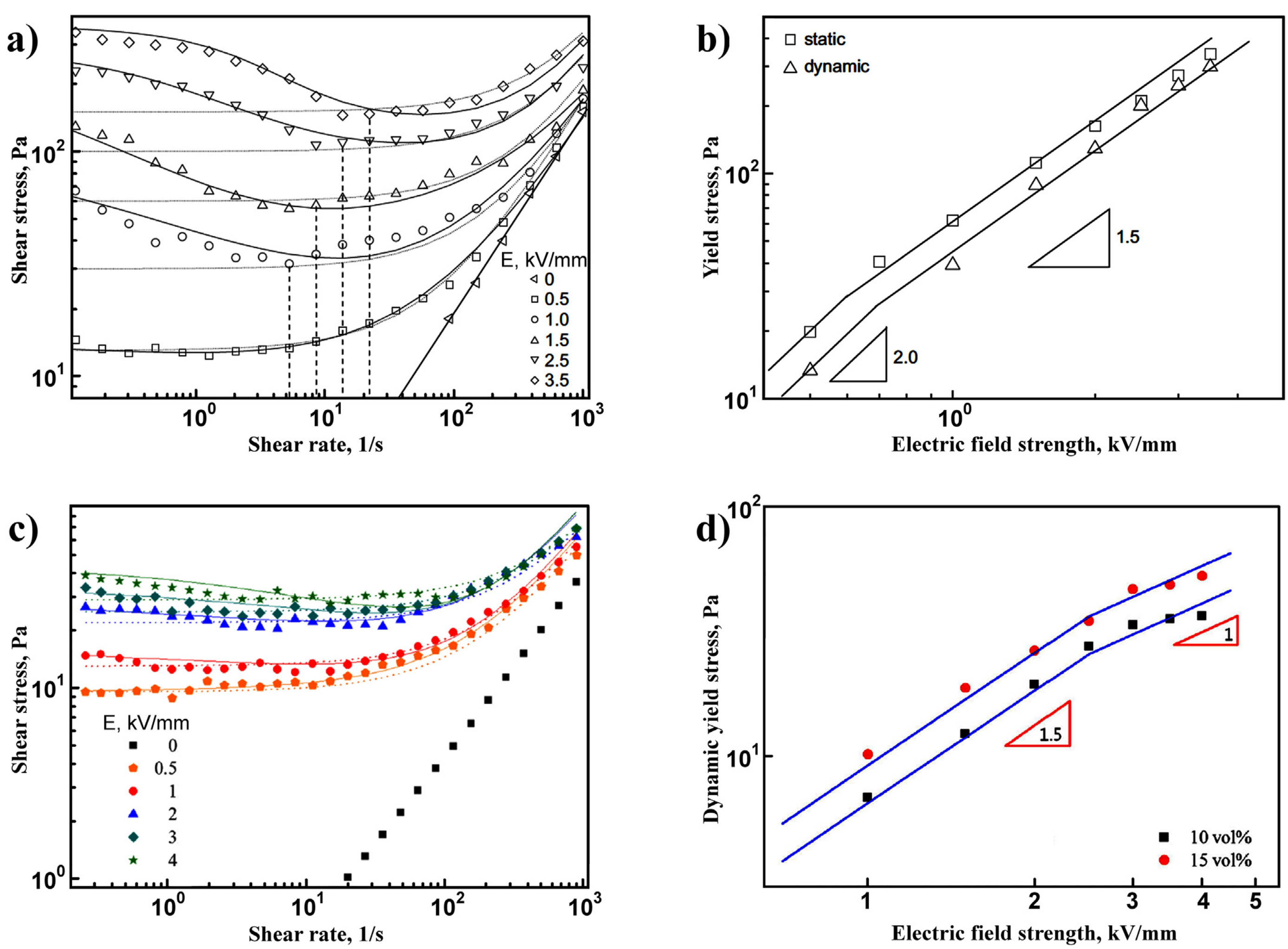

Fig. 2. Flow curves (a,c) and yield stress dependences (b,d) of 15 wt% dodecylbenzene-sulfonic acid-doped polyaniline-based (a,b) [39] and of 10 vol% polypyrrole-wrapped halloysite nanocomposite-based ERFs (c,d) [49] (the dot and solid lines are generated from Bingham and CCJ model, respectively) under various electric fields. The transition from one mechanism to another is revealed (b,d). Adapted from the studies by Kim et al. and Jang et al. [39,49] with permission from Elsevier and Springer Nature, respectively.

$$\widehat{\tau} = \widehat{E}^{3/2}\left(1 - e^{-m\sqrt{\widehat{E}}}\right), \tag{7}$$

where, $\widehat{\tau} = \tau_y/\alpha E_0^{3/2}$, $\widehat{E} = E_0/E_c$, $m = m'\sqrt{E_c}$, τ_y is yield stress, α is a coefficient depending on the dielectric permittivity of the fluid and the volume fraction of the dispersed phase, m' is approximation coefficient. However, the transition from one mechanism of the effect to another is not observed for all compositions of ERFs, and is often absent at a low concentration of the dispersed phase.

Thus, nowadays various models accurately describe the ER behavior of fluids. However, a general model based on strict physical principles still does not exist.

2.3. Forces, acting on a filler particle in an ERF

ERFs are multicomponent systems that may contain activating additives, stabilizing agents, as well as various impurities. So, forces acting on the filler particles in suspension should be considered to interpret the behavior of ERFs.

Electrostatic interactions are the main reason of the ER effect. The electrostatic potential depends on various parameters, such as uneven distribution of charges, the formation of double electric layers, and nonlinear dielectric phenomena. Thus, various simplified models to its description are used [21,26]. The idealized electrostatic polarization model is based on the solution of the Laplace equation, based on the model of solid spheres of a given volume with finite dielectric constant, while the spheres and the media themselves are considered non-conductive. The point dipole approximation considers an uncharged isolated sphere. Then the potential outside the sphere is equivalent to the potential of the dipole with the moment p:

$$p = 4\pi\varepsilon_m\varepsilon_0 r^3\beta E \tag{8}$$

$$\beta = \varepsilon_p - \varepsilon_m/\varepsilon_p + 2\varepsilon_m, \tag{9}$$

where, ε_p, ε_m, and ε_0 are relative dielectric permittivity of the particle and the media, and dielectric constant, respectively. E is electric field strength; r is radius of the particles.

Assuming that the spheres do not change the charge distribution of each other, it is possible to determine the force of action of one particle on another and their organization in chains. Considering only the electrostatic interactions of neighboring particles, the dynamic yield stress can be expressed as [56]:

$$\tau_0 = 18\varphi\varepsilon_0\varepsilon_p E^2 f_m\left[1 - \frac{\sqrt{\pi/6}}{(l/r)\tan\theta_m\sqrt{\varphi}}\right] \tag{10}$$

where, φ is volume fraction of filler particles; f_m and θ_m the maximum in the dimensionless restoring force and the angle at the maximum between the direction of the chain and the electric field, respectively; in the point dipole limit $f_m = 0.057$ and $\theta_m = 21.27°$; l is the gap between electrodes. The idealized structure was used in this model: only linear chains of single particle width corresponding to a limiting case of low volume fractions were considered. The multipole effect takes into account perturbation fields formed by spheres polarizing each other leading to additional electrostatic moments, except for dipole ones. Moreover, multipole effects of the action of distant particles can be observed at high concentrations.

When the particle is anisometric one, the torque starts to act under electric field and for an ellipsoid the moment of torque is determined as follows [57]:

$$M_{el} = \frac{1}{4}\varepsilon_0\varepsilon_p V E_0^2 \sin(2\theta)\frac{\left(\frac{\varepsilon_p}{\varepsilon_m} - 1\right)^2|1 - 3n_z|}{\left(1 + \left(\frac{\varepsilon_p}{\varepsilon_m} - 1\right)n_z\right)\left(1 + \left(\frac{\varepsilon_p}{\varepsilon_m} - 1\right)n_x\right)} \tag{11}$$

where, V is ellipsoid volume; $n_x + n_y + n_z = 1$ are depolarization factors along the corresponding coordinate axes; θ − the angle between the direction of electric field and the ellipsoid symmetry (normal) the axis (Fig. 3) [58]. It follows from the expression that the aspect ratio of the particle plays an important role. In the case of an elongated ellipsoid ($n_z < 1/3$) the arising torque tends to rotate the axis of symmetry along the z direction parallel to the electric field lines and the opposite situation is observed for an oblate ellipsoid ($n_z > 1/3$). The resulting torque is directed to rotate the axis of symmetry along the z-axis perpendicular to the electric field direction. Then the maximum value of the moment will correspond to the maximum value of the sine of the angle, i.e., at $2\theta = \pi/2$. The equilibrium state of the oblate ellipsoid can be observed at the position when electrostatic torque is zero ($M_{el} = 0$). The first position reaches at the orientation along the electric field (z-axis is perpendicular to electric field $2\theta = \pi$), and the second is orientation perpendicular to the direction of the electric field (z-axis is parallel to electric field $2\theta = 0$). Since the particles in the suspension are subject to thermal motion, the second equilibrium state is thermodynamically unstable and as soon as the thermal motion removes particles from this state, the electrostatic torque increases leading to a stable equilibrium state (the first position of the oblate ellipsoid). Thus, two types of anisometric particles packing under electric field are possible depending on the parameters included in Eq. (11), namely, aspect ratio, the dielectric permittivity of the medium and the particle. Obviously, that an ellipsoid is a disk at a high aspect ratio of the semiaxes $a = b \gg c$. Therefore, the formula is also suitable for describing the behavior of plate fillers.

The Maxwell–Wagner model [59,60] is the simplest way to describe the process of bulk or surface polarization at the phase boundary. In an alternating electric field, the accumulation of mobile charges on the particle surface occurs, and at high frequency, the charges do not have enough time for a response leading to polarization, which dominates solely due to permittivity independent of conductivity. In the field of intermediate frequencies, both permittivity and conductivity play a significant role in the polarization process. According to this model, in the point-dipole approximation, the force of influence of one sphere on another is proportional to the square of the amplitude of the electric field:

$$F_{ij}^{el} \sim \beta_{eff}^2 E^2 \tag{12}$$

$$\beta_{eff}^2 = \frac{\beta_c^2 + \beta_d^2(\omega\tau_{MW})^2}{1 + (\omega\tau_{MW})^2} \tag{13}$$

$$\beta_c = \sigma_p - \sigma_m/\sigma_p + 2\sigma_m \tag{14}$$

$$\beta_d = \varepsilon_p - \varepsilon_m/\varepsilon_p + 2\varepsilon_m, \tag{15}$$

where, ω is frequency, τ_{MW} is Maxwell–Wagner relaxation time, σ_p, σ_m − conductivity of the particle and the medium, respectively. Thus, the Maxwell–Wagner polarization takes into account the dielectric permittivity and conductivity of both the particle and the medium. Obviously, the proposed models consider particles in a spherical approximation and do not take into account the influence of shape.

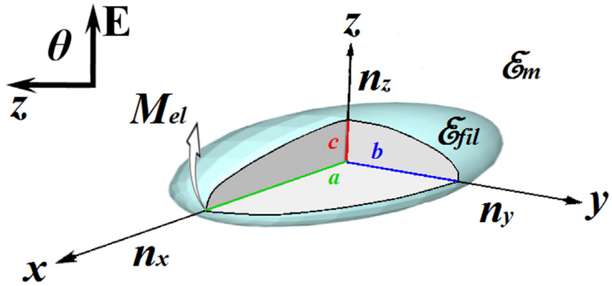

Fig. 3. The model of ellipsoid in electric field. Reprinted from the study by Stolyarova et al. [58] with permission from Wiley Periodicals.

Many researchers analyze the dielectric characteristics of both individual components and suspensions to quantify the effectiveness of a particular filler in polarization processes [61–65]. In particular, the Havriliak–Negami [66] and Cole–Cole [67,68] equations are widely used to describe the electrophysical characteristics of ERFs (Fig. 4).

Havriliak–Negami equation:

$$\varepsilon^* = \varepsilon_\infty + \frac{\Delta\varepsilon}{\left(1+(i\omega\tau)^\alpha\right)^\beta} + i\frac{\sigma_0}{\varepsilon_0\omega^n}, \tag{16}$$

where, ε^* is complex dielectric permittivity; τ is relaxation time, ε_∞ is the real permittivity for high frequencies, $\Delta\varepsilon = \varepsilon_s - \varepsilon_\infty$ is the dielectric intensity or relaxation strength, measuring the difference of the real permittivity at low and high frequencies. Higher $\Delta\varepsilon$ results in stronger polarization. The parameter α is related to the slope of the imaginary part of the dielectric constant and determines the symmetry of relaxation times distribution. The β is a parameter determined by the width of the spectrum. The third term takes into account the contribution of conductivity, where, ε_0 is the dielectric constant and σ_0 is related to direct current conductivity, n describes the width of relaxation times for conductivity.

The Havriliak–Negami equation goes into the Cole–Cole equation at $\beta = 1$. Usually, the first of them is used when the relaxation maximum is observed in the frequency dependence of the dielectric loss modulus. In the case when the maximum position is outside the studied range, it is more reliable to use the Cole–Cole equation.

Hydrodynamic forces start to act on particles under shear flow [36]. The dispersion medium can be considered as extended. Particle sizes of the dispersed phase are in the nanometer or micron range. Therefore, for such systems Reynolds numbers are much less than unity. The flow is laminar and then, omitting the interaction of particles, the fluid dynamics can be described by the Stokes equation:

$$F_h = -6\pi a\eta_m\dot{\gamma}, \tag{17}$$

where, a is effective particle radius, η_m is dispersion medium viscosity (for Newtonian fluid), and $\dot{\gamma}$ is shear rate. The equation does not take into account the rotational motion of particles, which can lead to significant errors if the hydrodynamic forces predominate over electrostatic ones.

In suspension, particles are subjected to chaotic Brownian motion. In the spherical approximation, the diffusion coefficient [69]:

$$D = \frac{k_B T}{6\pi\eta r}, \tag{18}$$

where, k_B is the Boltzmann constant, T is absolute temperature, η is fluid viscosity, and r is particle radius.

As is known, the stability of the dispersed phase from coagulation is determined by the interactions of particles with each other or the macro surface, while the balance between the forces of attraction and repulsion is realized. The most general approach gives the DLVO theory proposed by Soviet scientists Deryagin and Landau [70] and later, independently by Dutch physico-chemists Verway and Overbeek [71]. The energy of attraction between particles is determined as follows:

$$U = -\frac{A^*}{12\pi h^2} \tag{19}$$

$$A^* = A_1 + A_0 - 2A_{01} \tag{20}$$

where, h is distance between particles, A^* is the Hamaker constant for two identical particles interacting across dispersion medium taking into account the nature of interacting bodies, A_1 and A_0 are the Hamaker constants of the dispersed phase and dispersion medium interacting across vacuum, respectively; A_{01} is constant of interaction between the medium and particles.

It can be seen from the equation, that the energy of attraction for colloidal particles is inversely proportional to the square of the distance between them, in contrast to the van der Waals forces for molecules inversely proportional to the sixth power. That is, the attractive forces between dispersed particles decrease much more slowly with increasing distance between them compared with molecules. However, the interaction energy between particles depends on their size and shape and should be taken into account as well.

In addition to the considered above interactions, it is important to take into account short-range forces due to solvation and steric difficulties providing theoretical calculations and modeling. There are also some other approaches describing the observed effects associated with the shape of the particles.

Often the forces acting on particles in the ERF are considered as different groups. Therefore, for example, the ratio of hydrodynamic to polarizing forces is called "Mason's" number and hydrodynamic to thermal motion is "Peclet's" number. The relationship of electrostatic forces to thermal motion (parameter λ) [26], van der Waals forces to polarization and electrostatic to polarization [72] are used as well. Unfortunately, in recent studies, authors neglect such an analysis, apparently due to the complexity of assessments and perceptions. In summary, it can be noted that for a comprehensive understanding of the ERFs operation processes, an experimental study of the structure formation in suspensions is also necessary in addition to a theoretical description of the ER effect. Some extra valuable information can be obtained from recently published theoretical studies [73,74].

2.4. Experimental research methods, measurement modes

The main methods for ERFs studying, as the name implies, are methods of rheology, in particular, the rotational viscometry is widely used. Instruments for measuring the rheological characteristics of substances under the influence of an electric field are currently available in commercial design. For example, the company Anton Paar (German) offers viscometers with an ER device for two geometry types of measuring cell. The first is a double cylinder, which is suitable for low-concentrated suspensions in a liquid state without the electric field. The second type is plane-plane and more suitable for highly concentrated compositions in elastic or solid state (pastes). However, a

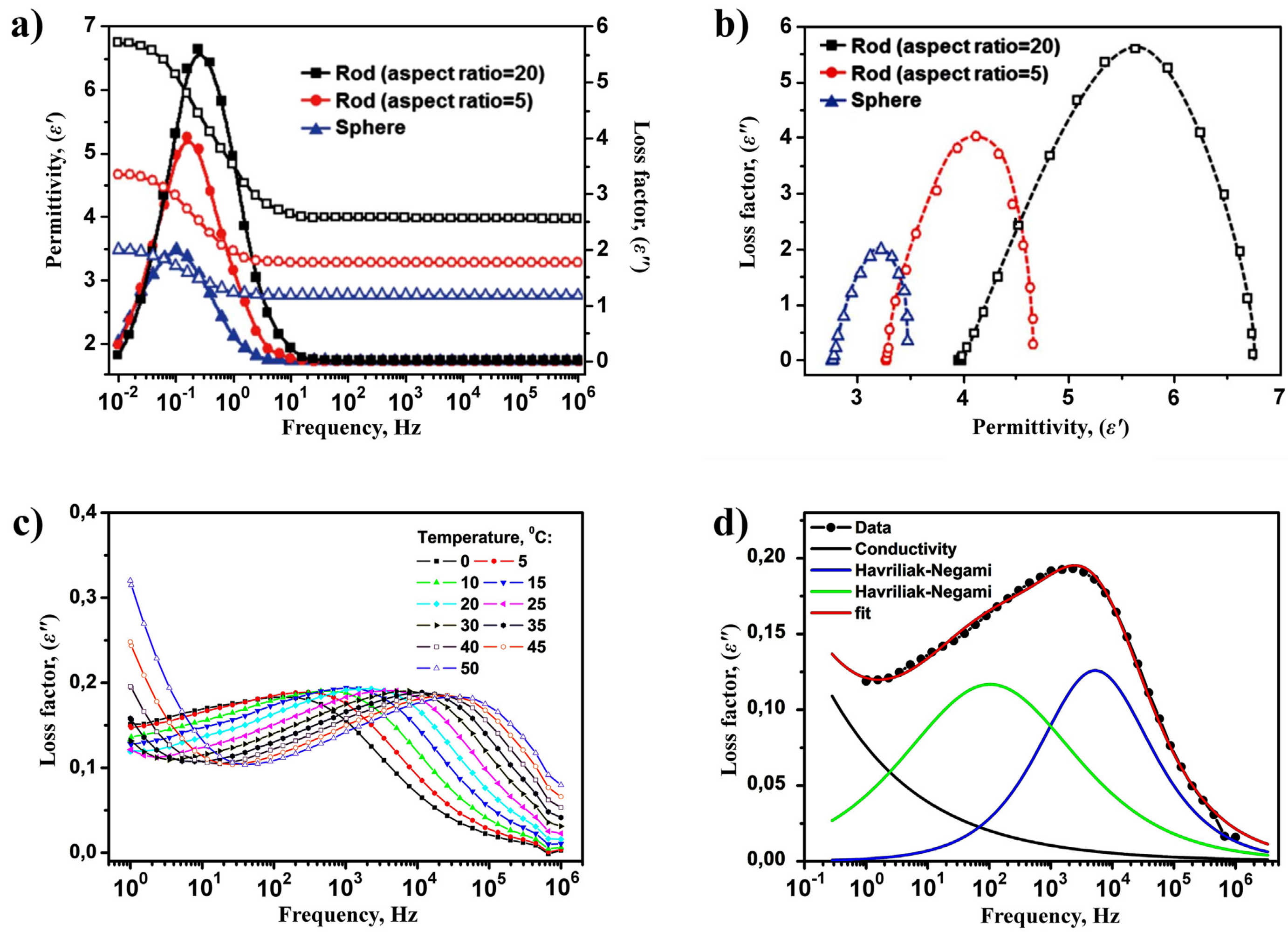

Fig. 4. Dielectric spectra (a) and Cole–Cole diagram (b) of the graphene oxide-wrapped silica-based ERFs (5.0 wt%). Open and closed symbols (a) indicate permittivity and loss factor, respectively [173]; temperature dependences of dielectric spectra (c) and Havriliak–Negami (d) fitting curves of 8 wt% organo-modified montmorillonite-based ERFs [62]. Adapted from the studies by Kuznetsov et al. and Lee et al. [62,173] with permission from the Wiley Periodicals and Royal Society of Chemistry, respectively.

plane–plane geometry can also be successfully used for low-filled ERFs. The problem of the sample leakage can be solved by the gap reducing. Geometries contain insulating gaskets separating the measuring part from the main device, and the electric potential is supplied from an external source. The corresponding physical principles of the rotational viscometers operation with various geometries are well known and are given in the corresponding books and monographs. Therefore, as part of this review, we will dwell in more detail on the measurement modes and types of tests usually performed for ERFs. The main test in shear flow mode is to find the flow curve and its corresponding viscosity curve. The measurement is possible in two modes, namely, controlled shear stress (CSS) and controlled shear rate (CSR). The first mode is better for measuring the yield stress and the second for identifying the behavior of the sample at different shear rates. The presentation of the results in the first case is more evident in linear coordinates, in the second in the double logarithm. Another type of test is measurements in the frequency mode. The frequency dependences of the storage and loss moduli, the complex modulus, as well as the complex viscosity can be obtained. Such tests make it possible to separate the viscous and elastic components in the samples and to observe the transition from a viscous to an elastic state, both depending on the concentration of the filler and the electric field strength. Basically, measurements are carried out in the linear range of viscoelasticity for which an amplitude sweep (the dependence of the moduli on the strain value) at a low frequency (less than 10 Hz) should be firstly obtained. Typically, such dependences demonstrate a decrease in the moduli values with an increase in the strain amplitude after a certain critical value. This critical value is the upper limit of the linear range of viscoelasticity. It is important that this range can change under electric field. Therefore, amplitude tests are preferably carried out at different field strengths to select the appropriate measurement mode of the frequency dependences. Other important rheological test for ERFs is the analysis of the possibility of cyclic operation. The test consists in measuring the viscosity (or shear stress) in time during serial switching on and off of the electric field, while the shear rate is chosen constant and low. The constant shear rate also makes it possible to evaluate the hysteresis of the rheological properties during the formation and destruction of columnar structures and eliminates the mixing effect due to high shear rates, which can be observed in experiments on finding flow curves. Finally, it will not be superfluous to evaluate the leakage current density during the rheological experiments.

It is interesting to note that the geometry of the electrodes can significantly influence on the measured and operational characteristics of the ERFs. Thus, castellated and sawtooth electrode geometries lead to more ordered structure formation under electric field [75]. Geometric singularities (e.g. corners, tips, sharp edges, etc.) locally induce extremely strong electric field gradients and the chain ends are attracted to the sharp tips of the electrodes. Thus, by directionally changing the geometry of the electrode, it is possible to control the structure of the ERF and obtain more ordered and therefore more durable structures. Nevertheless, it is known that the sharpness and tips of the electrodes are voltage concentrators, which can lead to electrical breakdown at lower fields than in the case of flat electrodes. This fact must be taken into account for the practical implementation of devices that are based on detected phenomenon.

In addition, along with the development of technical equipment and its improvement, progress and implementation of new methods for measuring the ER properties of materials take place. In particular, a method of capillary breakup in extensional mode under electric field has recently been proposed [76,77]. The special cell for basic extensional rheometer was created by 3D printing technology. Measurements are carried out in tension mode. The main feature of such experiments is that the electric field is aligned with shear stress. After loading the sample into the cell, the filament thinning process starts and a mid-diameter of filament is analyzed. The strength and length of the filament during extension depend on the concentration of the filler in the ERF and applied voltage. The higher the concentration and the electric field strength, the longer the filament takes to break up. This result is consistent with the shear experiments. The particles form chains oriented in the direction of the field and the chains are under tensile when the fluid under an extensional flow. The advantages of the method are the simplicity of implementation and quickness of the experiment. One measurement procedure, including the duration of the filament thinning process, takes less than 10 s. The disadvantages of the method are the necessity to take into account the sedimentation stability of suspensions and decrease in electric field strength during the experiment due to an increase in the distance between electrodes. This new technic opens the opportunity for validating the models for ERFs under extensional flows. Nevertheless, there are still no practical applications of ERFs for the codirectional action of electric and shear fields.

Other group of methods commonly used in electrorheology relates to characterization of samples and to greater extent to filler. Thus, the AFM, TEM, SEM and optical microscopy, XRD, XPS, EDS, SAXS, SANS, FTIR and Raman spectroscopy, DLS, SEC (for the analysis of the molecular weight distribution and polydispersity index of polymers, which often also includes a dispersion medium), BDS etc. are used, depending on the availability and advisability of the method. Some of them considered in more detail below.

The optical microscopy is widely used to visualize the formation of columnar structures under electric field (Fig. 5a) [78–83]. However, the fine structure of the column, namely, the ordering and orientation of the particles, which is undoubtedly especially important for fillers with a high aspect ratio during the formation of a percolation network, are poorly studied. The main reason is the nanoscale of the filler particles and the insufficient resolution of the microscope. A special case of the optical microscopy method is high-speed shooting in millisecond resolution. The high-speed camera makes it possible to visualize the processes of columnar structures formation and their reorganization during shear flow (a conflict between hydrodynamic and electrodynamic forces), and also allows to identify the features of structuring under altering field [84,85].

Electron microscopy methods are used in the vast majority of publications related to ERFs for morphology analysis of filler particles. These methods allow to quickly and clearly getting information about the shape and polydispersity of submicron and nanometer sized particles. The disadvantages of methods include their locality. Therefore, it is necessary to analyze a large number of images from different areas of the survey to obtain a reliable size distribution of polydisperse filler. An original study was performed by Wu et al. showing SEM images of oriented structures from titanium and calcium oxalate micro rods (Fig. 5b) [86].

SAXS and SANS are used to study the processes of structure formation from dispersed phase particles under electric field. These methods allow analyzing the orientation of individual particles in the structure, especially with pronounced shape anisometry. Detailed studies of the ERFs fine structure were carried out on particles of smectic clays, as well as titanium dioxide [87–93]. It should be noted that the kinetics of column formation and the dynamics of their destruction under shear flow were not considered in these studies. The X-ray scattering method for studying the processes of columnar structure growth in the electric field direction has been applied previously on particles of barium titanate [94] and triglycinesulfate in paraffin films [95]. However, the total number of such studies is not high, and in recent years only several of them were reported [96–98], which, undoubtedly, is a significant omission during researching the structure of ERFs.

Last but not least is the method of BDS. This method allows obtaining the frequency dependences of the conductivity and permittivity, their imaginary parts, as well as the dielectric loss tangent. Such data make it possible to quantitatively evaluate the electrophysical characteristics of suspensions, and get an idea of the ER effect mechanism. The identification of the frequency dependences on temperature for ERFs allows evaluating the activation nature of the polarization and conductivity. The comparison of activation energy values obtained from different electrophysical parameters is a promising tool for analyzing the structural organization of filler particles in suspension [99,100]. Preliminary measurements of the dielectric spectra of the filler can serve to predict of its effectiveness in ERF as well. Modern equipment can measure both liquid and solid samples in a wide temperature range. One of commercially available high-tech instruments for dielectric spectra collecting is Novocontrol Alpha A analyzer (Germany).

Thus, the wide range of modern experimental methods allows to provide a comprehensive analysis of the ERF behavior, as well as to determine the relationship between the nature, size, morphology, and electrophysical characteristics of the filler, its structural organization both without and under the electric field and rheological behavior of its suspensions.

3. Factors affecting the ER effect

The ER behavior of fluids is affected by two types of various factors. The first includes the characteristics of the ERFs components and the second is sensitivity to external parameters. Some obvious factors are the filler concentration, its polarizability, and the electric field strength. An intensification of these parameters typically leads to increase in the ER effect and the elastic response of the material. Other factors are not so obvious, for example, the nature of the dispersion medium and specific interactions with the filler, the effect of activators, the shape of filler particles, etc. Below, we will consider some factors shaping the ER response of suspensions in more detail. Due to the huge variety of fillers, a separate section is devoted to this component of ERFs.

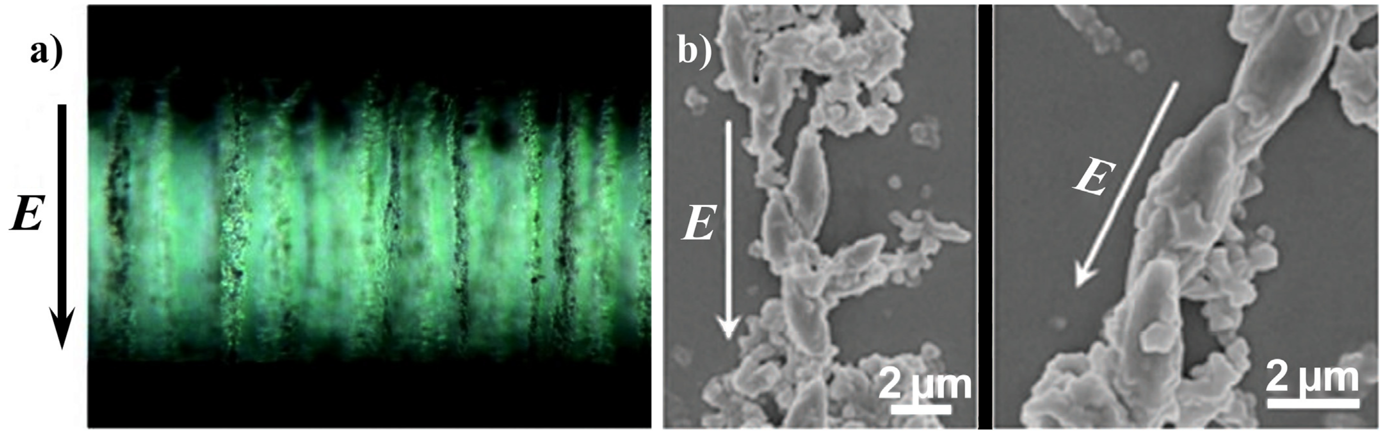

Fig. 5. Optical micrograph of columnar structures formed under electric field by laponite particles (a) and SEM image of oriented structures formed by hybrid calcium titanyl oxalate micro rods. Adapted from the studies by Parmar et al. and Wu et al. [79,86] with permission from American Chemical Society and Springer Nature Limited, respectively.

3.1. Dispersion medium

The ER effect depends on the nature and viscosity of the dispersion medium. The dispersion medium for ERFs is usually non-polar or low polar dielectric liquids. They are oils (mineral, transformer, olive, petrolatum, silicone, etc.), esters (dibutyl sebacinate, diethyl hexyl adipate, di-n-butyl phthalate, diisodecyl phthalate, etc.), hydrocarbons (kerosene, naphthenic, olefinic, paraffins, etc.) [2]. In recent decades, silicone oils have become widespread, in particular polyorganosiloxanes, which are organosilicon high molecular weight compounds with inorganic main chains of macromolecules that consist of alternating silicon and oxygen atoms (side substituents may vary). Such oils have low dielectric permittivity, a wide range of operating temperatures, and are commercially available in a wide viscosity range. Polyorganosiloxanes also have high thermal stability, which is due to the high energy and polarity of the Si—O bond, low saturated vapor pressure, and chemical resistance. In the vast majority of publications during recent years, polydimethylsiloxane is used as a dispersion medium. Polydimethylsiloxanes have a low temperature dependence of viscosity, which is associated with the absence of hydrogen bonds between the chains of molecules, as well as the free rotation of the structure around the siloxane bond [101]. However, they are able to form hydrogen bonds with other substances, including filler particles. Another interesting feature of polydimethylsiloxane is its weak amphiphilicity resulting in the orientation of molecules in thin Langmuir films at the water-air interface [102]. The silica—oxygen skeleton is water-oriented and the methyl groups remain in the air. Such behavior can additionally promote better dispersibility of some fillers.

The viscosity of the siloxane oil depends on the degree of polymerization and can significantly influence on the ER effect. Unfortunately, the authors do not stick to uniform values for viscosity. The values of kinematic or dynamic viscosity are provided in various articles. They are related to each other through the density:

$$\eta_{kin.} = \frac{\eta_{dyn.}}{\rho} \tag{21}$$

Where, $\eta_{kin.}$ is kinematic viscosity, $\eta_{dyn.}$ is dynamic viscosity, ρ is density. However, the density of polydimethylsiloxane is close to 1 g/cm^{-3} (typically about 0.96—0.97 g/cm^{-3}). Thus, the difference between kinematic and dynamic viscosity is rather small.

The effect of the silicone oil viscosity on the ER behavior of suspensions filled by urea coated TiO_2 was considered [103]. Oils with a viscosity of 25, 100, 500 mPa*s were used in this study. It turned out that the lower the viscosity of the oil, the higher the yield stress. Viscosity strongly affects the wettability of particles, as shown by the contact angles. The lower the viscosity, the smaller the angle and the better the filler is wetted. A similar trend was observed in the study by Davydova et al. [104]: suspensions filled by TiO_2 up to 45 wt% demonstrate the highest yield stress in silicone oils of low viscosity (5 cSt) and the values decrease with increasing oil viscosity. The authors attribute this behavior of ERFs in a low-viscosity medium to the lower resistance that particles suffered in dispersion during their translational motion in the process the columnar structures formation along electric field. However, at the same time, the phase separation (sedimentation) is observed in low-viscosity oils, and the homogeneity of the suspension is not restored after electric field removing, which results in instability of the ERFs operation.

It is rather intriguing, that the opposite effect was reported in several studies [105,106]. The yield stress of the samples increases with increasing oil viscosity. The authors explain such behavior by an increase in the degree of the filler particles aggregation and an increase in the average size of aggregates due to a decrease of wettability in oils with large molecular weight polymer chains. The influence of the nature and size of the side groups in the polymer chain on the ER effect was considered as well. It was found that the smaller the group, the stronger the effect. Also, the ER behavior depends on the interaction of the filler particles with side substituents. Thus, the greatest increase in the effect was observed for groups capable to hydrogen bonding, for example, hydroxyl. Such behavior is associated with the features of the ERF structuring mechanism.

However, Sokolov et al. note the weak effect of the dispersion medium viscosity on the ER behavior of halloysite suspensions in silicone oil. The yield stress of 8 wt% fluids under an electric field remains constant during the oil viscosity range from 50 to 400 cSt. However, the sedimentation stability of suspensions increases with an increase in the medium viscosity [107].

The ERFs based on oils of various nature and viscosity filled by urea-coated barium titanyl oxalate particles were studied as well [108]. Dimethylsiloxane, methylsiloxane, white mineral, sunflower seed, corn oils, and liquid paraffin were considered as dispersion media. When the filler was mixed with the medium, a different state of the suspensions from liquid to pasty depending on the nature of the oils was observed. A tendency to deteriorate the dispersion of particles with increasing viscosity of the oil was found for all media, except hydrogenated polysiloxane. It is shown that the ER effect is affected not only by the viscosity of the dispersion medium, but also by its chemical nature.

Hong et al. examined the effect of the oil chemical nature on the ER effect in chitosan-filled suspensions [109]. The oils of the same viscosity (50 cSt), but of different chemical nature, namely corn, soybean, and silicone, were used to maintain the experimental integrity. The highest yield stress was observed in silicone oil for suspension with a particle content of 25 wt%. The result was interpreted by differences in permittivity and conductivity of particles and the medium.

Thus, the molecular structure, viscosity, and interactions of the dispersion medium with the filler can have a significant effect on the behavior of the ERF.

3.2. Activators

The dawn of the ERFs investigation, small amounts of polar substances adsorbed on the surface of the filler particles were added to enhance the ER response. Such additives caused an intensification in the ER effect due to an increase in the dielectric permittivity of the dispersed phase. The most well-known activator is water, but the use of alcohols, amines, acids, alkalis, etc. was also reported [2]. The presence of these activators led to an increase in current density and to limitation of the electric field due to electrical breakdown, as well as to the corrosion of devices based on such ERFs. The next stage of ER development began in 1985 with the discovery of anhydrous ERF arousing new interest in this field, and anhydrous fluids were carefully studied. However, low molecular weight substances are still considered as activating agents. So recently, an intensification of the ER effect of titanium oxide suspensions due to the addition of water [110], an aluminum nitrate and phosphoric acid has been reported [111,112].

In the late 90s and early 2000s, a decline in research activity in the field of electrorheology was observed. The theoretical calculations showed that the maximum yield stress can be as low as about 8 kPa at electric field of 1 kV/mm [113]. This value turned out to be extremely low compared to magnetorheological fluids, which made it possible to reach yield stress of the order of 50–100 kPa [114]. However, one of the serious disadvantages of magnetorheological fluids is the strong limitation of magnetosensitive fillers. Magnetorheological fluids are another broad topic and are beyond the scope of presented review. Therefore, readers interested in the field of magnetorheological fluids should check out some valuable reviews [115,116]. Thus, ERF remained attractive due to the simplicity of control and devices development. Therefore, the researchers again decided to return to activating additives to overcome the upper theoretical value of the yield stress. The idea of enhancing the effect by changing the polarization of the filler particles remains relevant, and the development of such ERFs continues. In 2002, Zhang et al. obtained a fluid with a yield stress of 5–10 times higher than for all systems known at that time [117]. A strontium titanate was used as filler and the yield stress was 27 kPa at electric field of 3 kV/mm. However, a significant breakthrough is associated with the discovery of a giant ER effect in 2003 [50]. Wen and Sheng et al. received a novel nanoscale filler consisting of spherical urea-coated $BaTiO(C_2O_4)_2$ particles. The yield stress of about 130 kPa was achieved at field strength of 5 kV/mm. At present, ERFs filled by silicon dioxide (SiO_2) coated by calcium and titanium compounds; titanium dioxide (TiO_2) with various additives, and mixed calcium titanium oxides (CaTiO) and strontium titanium oxides (SrTiO) with high yield stress reaching tens and hundreds of kPa have been prepared [118–121]. As was noted above, such systems exhibit a linear dependence of the yield stress on the electric field, and not quadratic as for dielectric ERFs.

However, the coated core/shell fillers are not the only trend of development in modern electrorheology, although they reveal a significant improvement in rheological behavior of fluids. For example, Marins et al. examined the effect of a small amount of ionic liquids as additives on the ER effect [122]. Thus, only less than 2% of ionic liquid, namely, triisobutyl(methyl)phosphonium tosylate (CYPHOS®IL106), in silicon dioxide-filled ERF leads to an increase in the yield stress by a factor of 2–3 compared with pristine filler.

Metal ions doping can activate the ER effect as well. Thus, titanium dioxide doped by various cations Na^+, Ba^{2+}, Ca^{2+}, Mg^{2+}, Y^{3+}, Al^{3+}, and Nb^{5+} demonstrates significant improvement in ER characteristics [123]. Moreover, multiply charged cations are the most effective. The complex modulus increases with electric field up to 3 kV/mm for dopants according to the dependence $Nb^{5+} > Al^{3+} > Y^{3+}$ and an inverse relationship $Al^{3+} < Y^{3+} < Nb^{5+}$ was observed at higher electric fields (above 3 kV/mm). Such behavior was associated with irreversible aggregation of the filler; therefore, the aluminum-doping was considered as optimal. The mechanism of the observed effects was explained in terms of crystallography and ion coordination with respect to titanium dioxide particles. Ion-doping changes the dielectric characteristics of the filler and leads to an increase in the ER effect. Additionally, effect of ion-doping was shown by Yoon et al. [124]. The alkaline earth metal doped (Ba, Sr, and Ca) hollow silica/titania spheres reveal both increased dispersion stability and ER activity. The best improvement was revealed by Ca-doped filler. The yield stress reaches 62.3 Pa at 30 vol% filler content and electric field of 3 kV/mm.

Furthermore, doping by ions can qualitatively change the ER effect from negative to positive. Thus, the additive of copper ions in the graphitic carbon nitride varies ER behavior of its suspensions in silicone oil [125]. The graphitic carbon nitride particles in the silicone oil exhibit electrophoresis under electric field, which was demonstrated by optical microscopy. The main reason of such behavior is in trapped electrons in the structure. However, the copper cations oppositely change the fluid behavior, due to possible enhanced population of deep states and possible charge carrier traps in the band gap during copper ion insertion. Moreover, the copper-doping does not significantly affect neither the specific surface nor morphology of the particles.

The unexpected effect was observed by Agafonov et al. [35]. The solvents of various chemical nature with different dipole moments, such as benzene, toluene, chloroform, diethyl ether, acetone, DMSO, and acetonitrile, were considered as additives. There was no direct correlation between the changes in the yield stress of ERFs and the dipole moment of the added substances. Moreover, it has been shown that the non-polar substances, such as benzene and toluene, lead to a more significant effect compared to weakly polar diethyl ether. The authors attribute this behavior to the surface tension forces for solvent additives arising at the phase boundaries due to different chemical nature. Thus, the solvent molecules act as bridges between the filler particles in the dielectric medium, forming an extended network, which leads to an increase in the yield stress. The fundamental limitations of the approach of using only polar molecules as additives have been demonstrated.

One of the recent studies shows that the ER properties of suspensions can also be tuned by the composition of the dispersion medium [126]. The alkanes-additives improve the stability and homogeneity of the ERF reducing the sedimentation rate and polydispersity of aggregates. Although the yield stress under electric field was slightly reduced with the addition of alkane; however, it was shown a lower current density and higher ER efficiency (the ratio of yield stress values under and without electric field). The main reason of such behavior related with non-polarity of alkane molecules resulting in hydrophobic interactions with the polar shell of the filler particles and with much less wettability in contrast with pure silicone oil. The authors believe that alkanes additives have less impact on saturation surface polarization than surfactants and opens the novel way to the liquid phase modifying instead of the filler particles.

3.3. Sedimentation stability

The problem of the colloidal systems stability was considered by Deryagin as a question of their existence. All colloidal systems are thermodynamically unstable, and we can only talk about their kinetic stability [127]. The theoretical foundations of coagulation processes were discussed above and are associated with the DLVO theory. The concentration of the dispersed phase in ERFs is often higher than the percolation threshold; therefore, filler particles interact with each other and are able to form an extended colloidal phase, which leads to an increase in sedimentation stability over time. The classical sedimentation analysis, that is, finding the particle size distribution function, to assess the stability of ERFs is difficult due to the high viscosity of the dispersion medium, as well as high for sedimentation analysis particle concentration, and, therefore, often high turbidity of suspensions. Thus, the concept of sedimentation ratio of the colloidal phase to the total volume is often used to assess the sedimentation stability of ERF (or rather kinetic stability) [128,129]. The dependence of the sedimentation ratio over time allows to assess how sedimentationally stable the ERF is, which is extremely important for practical applications. Obviously, the higher the concentration of particles, the higher the equilibrium sedimentation ratio. Currently, ERFs with outstanding sedimentation stability have been obtained. The sedimentation ratio remains extremely high (over 90%) after several weeks or months, even at a low filler concentration (Fig. 6) [130,363]. However, one cannot overlook the fact of structural reorganization of particles under the influence of an electric field in the case of ERFs. The columnar structures are destroyed under the action of shear stress and thermal motion after the electric field is turned off, which in turn can be expressed in the hysteresis of viscosity during cyclic operation. Therefore, the sedimentation stability of ERFs will depend, among other things, on the quality of redispersion of the filler.

3.4. Temperature

The stability of the ERF response to electric stimuli at different temperatures becomes fundamentally important for the practical application. Several articles were devoted to effect of temperature on ER behavior. However, the number of such studies is still limited. The temperature affects the several factors at once. An increase in temperature leads to an increase in the filler polarizability, an intensification of the Brownian motion, and a viscosity decrease. The balance of these factors largely determines the temperature effect on the ER behavior of fluids [132]. Moreover, for water-activated ERFs, an increase in temperature leads to the evaporation of adsorbed free water resulting in decrease of the ER effect [133].

One of the first reports declares that the yield stress values of polyaniline (PANI) and PANI coated with poly(vinyl alcohol) particles suspensions in chlorinated paraffin oil decreases with temperature (up to 80 °C) [134]. Similar dependence was observed later for block copolymer of poly(methyl methacrylate) (PMMA) and styrene in silicon oil [135]. However, ER properties of 15 wt% of a PANI suspension in silicone oil doped by dodecylbenzene-sulfonic acid revealed the increased yield stress up to 120 Pa at 80 °C and electric field of 1 kV/mm [39]. Later, the temperature effect on suspensions of PANI particles of various sizes and morphologies, namely, nanofibers, micro- and nanoparticles in the range from 5 to 115 °C was studied. All samples reveal the thermal stability; however, some important features were established. The shear stress of the PANI suspensions increases with temperature and then reaches the maximum. The value of the highest shear stress was observed at various temperatures depending on particles morphology, for example, 100 °C for microparticle suspension, broader temperature region for nanofiber suspension (70–100 °C), and the relatively lower temperature of 50–70 °C for nanoparticles suspension. The main reason of such difference is in convection forces induced by heating. These forces disturb ER structures formed by nanoparticles more easily compared to the larger particles. It is also important to note that sedimentation rate of the PANI microparticles significantly increased with temperature compared to nanofibers and nanoparticles. Thus, the stability of microparticles suspension is more sensitive to high temperatures and related with dispersion media viscosity decreasing [136]. An extreme dependence of ER response on temperature passing through a maximum was noted in the pioneering work of Deinega et al. [137,138] and confirmed in some studies, for example, the studies by Gao et al. [139], Lu et al. [140], and Li et al. [141].

However, it was reported that ER behavior of several polysaccarides is slightly sensitive to temperature [142,143], while some other studies declare an increase of ER response [47,144]. Weak temperature effect was also shown for fluids filled by various polymer particles [145], montmorillonite/titania nanocomposite [146] and silica-based ionogels [147]. An increase in temperature typically leads to an increase in the mobility of ions, increase of conductivity and electrical breakdown followed by an abrupt drop of the ER effect [148,149]. Recently, an expansion of the operating temperature range with a slight decrease in yield stress for fluids filled by crosslinked poly(ionic liquid)s has been reported [150]. Such effect caused by an increase of the number of spacer arms in polymer molecule followed by increase of interfacial polarization intensity and the decrease of the electrode polarization.

Thus, the effect of temperature on the ER behavior of suspensions of various compositions is quite diverse. Therefore, the study of the fluid response on electric field at different temperatures remains an urgent task for further practical application of developed materials.

4. The main component of ER fluid is filler

The easily polarizable micro- and nanoparticles of various chemical nature are used as fillers for ERFs. Suspensions with a dispersed phase of inorganic substances, including ceramics, natural and synthetic mineral compounds, clays, polymer particles and composites were reported [2,24]. Furthermore, as noted above, the ER effect can be observed in systems with a liquid dispersed phase, namely in some emulsions, such as a mixture of castor oil and a silicone polymer or a solution of poly-n-hexyl isocyanate in the para-xylene [21]. The problem of particles sedimentation is

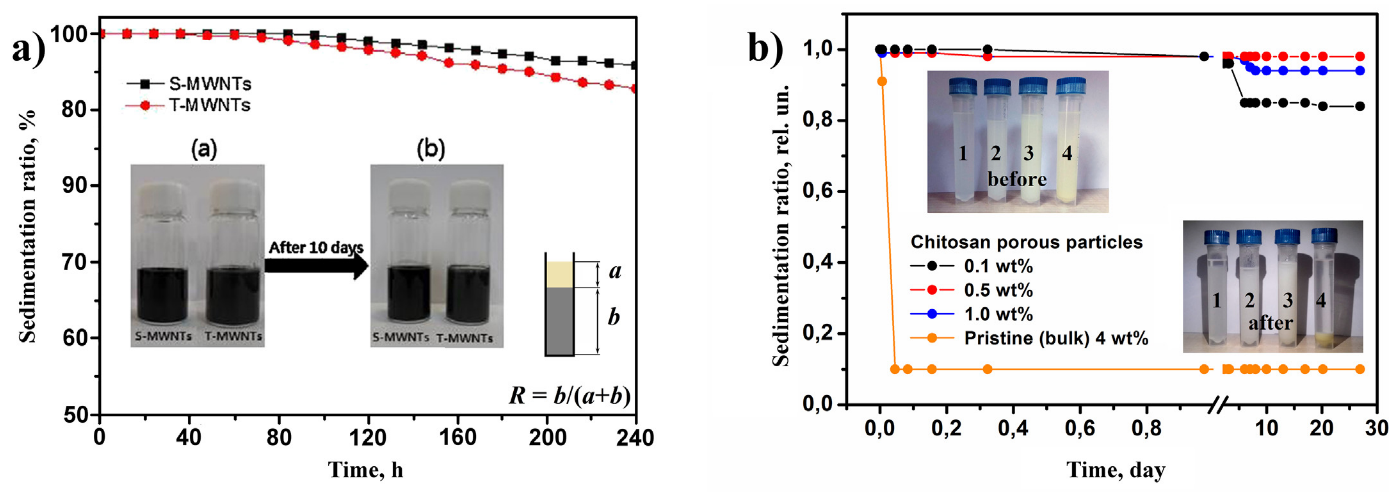

Fig. 6. Sedimentation ratio as a function of time for 5 vol% silica (S-MWNTs) and titania (T-MWNTs) – coated multi-walled carbon nanotubes-based ERFs [130]; and sedimentation properties of porous chitosan particles in polydimethylsiloxane compared with pristine polymer. The inset shows photographs of samples before and after the experiment at concentration 0.1 (1), 0.5 (2), 1.0 (3) and 4.0 (4) wt% [363]. Adapted from the studies by Oh et al. and Kuznetsov et al. [130,363] with permission from Elsevier.

removed for emulsions, but the process of phase separation is observed. Unfortunately, such ERFs are not widely used due to their high viscosity without electric field and low operational characteristics. A giant ER effect was observed on fluids filled by modified particles with a core/shell structure. Therefore, hybrid fillers of this type attract the wide interest of various research groups. Next, we consider some types of commonly used fillers, as well as their modifications, leading to a change in the electrophysical characteristics of the particles and an increase in the ER effect. The authors of various studies do not adhere to uniform values for the concentration of the dispersed phase, and in the literature, you can find notation in both weight and volume percent. The reduction to uniform values is not always obvious and possible; therefore, the concentration values are given in the author's proofs. If suitable, the density for a rough estimation of the volume fraction will be indicated.

4.1. Inorganic fillers, shape effect

One of the first fillers for ERFs was inorganic particles based on silicon oxide (IV), namely, silica, diatomite, aerosil [2]. Nowadays, particles from oxides of aluminum [151], titanium (IV) [104,152], cerium (IV) [153,154], manganese (IV) [155], compounds from iron [156], antimony and bismuth [157], tin [158], molybdenum [159], strontium nickelate oxide with a perovskite-like structure [160], zinc borate [161], carbon nanomaterials [128,162], clays [113,114], and other are considered as fillers.

Pure silicon dioxide as a filler for ERFs is practically not currently used, but fillers based on it remain attractive due to its thermal stability, high porosity, ease of preparation, low cost, and simple control of particle sizes. The most commonly used way for producing micro- and nanoparticles of silicon dioxide is the sol-gel method proposed by Stöber in 1968 [165]. The synthesis includes the hydrolysis of hydrosilicic acid tetraesters in a water-alcohol medium, using liquid ammonia as a catalyst. Tetraethoxysilane is commercially available and widely used as a precursor. The same approach is suitable for titanium dioxide particles. The shape of the resulting particles can be controlled by the addition of various surfactants, for example, cetyl-trimethylammonium bromide [82,166] or sodium bis(2-ethylhexyl) sulfosuccinate (AOT) [167]. Fig. 7 shows the variety of silica particle shapes as an example. Further, surface modification allows to obtain fillers with a high ER response.

Thus, silica coated with poly(o-anisidine) [61] and poly(-diphenylamine) [168] was considered as the dispersed phase. N-[(3-trimethoxylsilyl)-propyl] aniline was used as an activator of the silicon dioxide surface to further creating a shell with o-anisidine or diphenylamine monomers through π-π^* binding followed by polymerization and to obtaining a filler with a core/shell structure. The yield stress in case of poly(o-anisidine) shell was higher, compare ~100 Pa and ~10 Pa at 10 vol% of the filler content and electric field of 2 kV/mm, respectively. Schwarz et al. investigated the ER activity of rod-shaped porous silicon dioxide SBA-15 with intercalated metal supramolecular polyelectrolytes (Fig. 8) [169]. The intercalation of such compounds into the silicon dioxide structure leads to a sharp increase in the ER activity of the filler. Two types of complexes based on iron (II), namely, poly- and monocomplexes were considered in this study. Polyelectrolyte complexes enhance the ER effect by increasing the dielectric permittivity of the filler by 3 times compared to the pristine SBA-15. Intercalation of polycomplexes leads to stronger effects compared to monocomplexes. Thus, the addition of just 1.51 wt% of polycomplex results in an increase in the storage modulus by 53 times and it reaches 158 kPa at electric field of 5 kV/mm and the filler concentration of 10 wt%.

One of the promising areas in electrorheology is the anisometric fillers, which due to their elongated shape can form more durable structures under electric field. The elongated particles also reduce the concentration of filler in the ERF, because of the orientation of the particles and the extended network formation. Significant interest in the shape-effect of the filler particles on the ER behavior appeared only in the last decade. Nevertheless, Kanu and Shaw were among the first to suggest such effect back in 1998. They examined the theoretical difference in the polarization processes of a spherical particle and an elongated ellipsoid and demonstrated that the anisometry of the particle should lead to an increase in the arising dipole moment [170]. The proposed approach was consistent with experimental data on polymer poly-p-phenylene-2,6-benzobistiazole microfibers with various aspect ratios of 2, 4, and 10. An increase in the values of dielectric permittivity and the

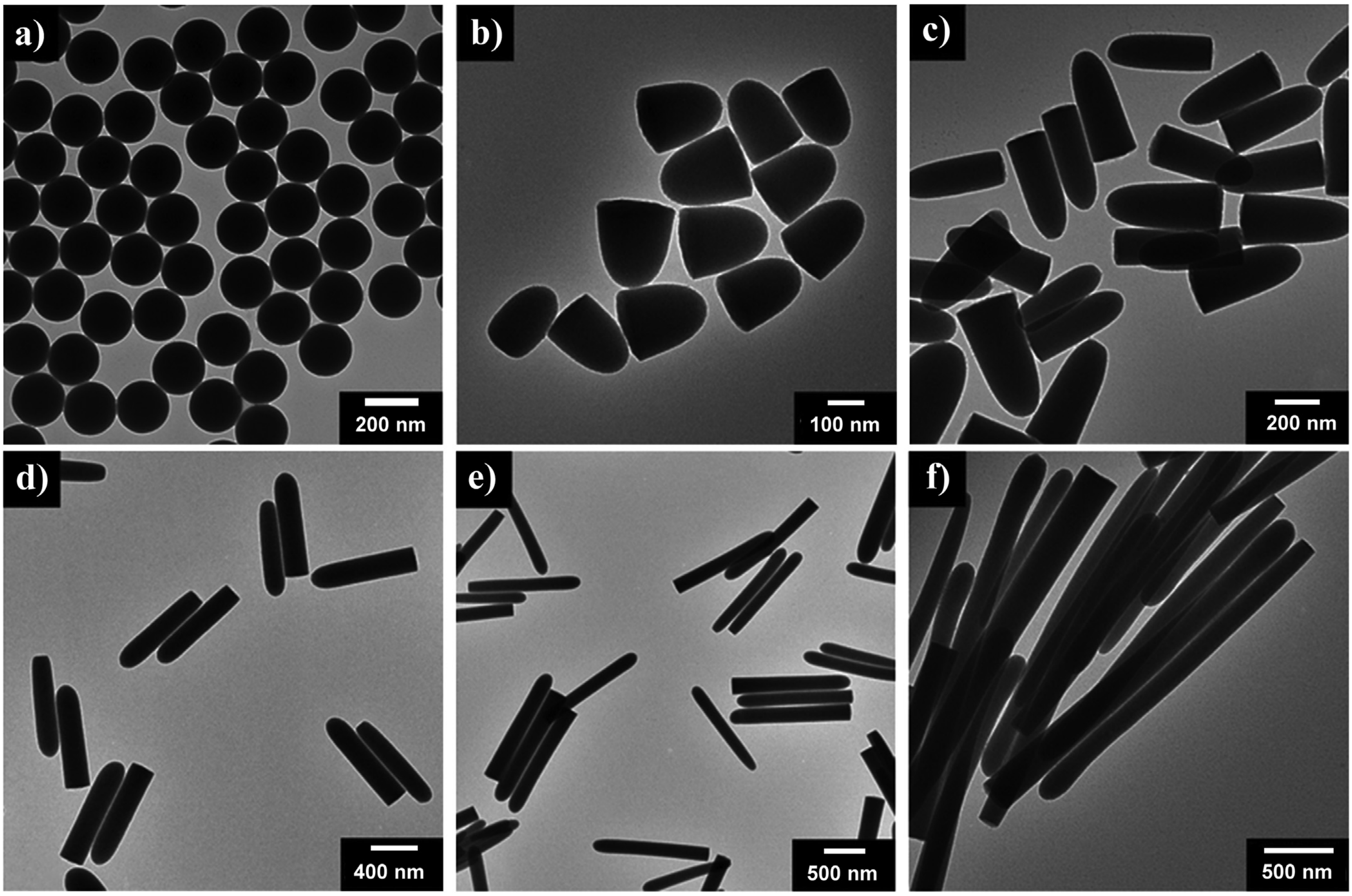

Fig. 7. Variety of silica particles morphology. TEM images of silica spheres with 200 nm diameter (a), silica rods with an aspect ratio of 1.8 (b), 3.1 (c), 5.0 (d), 8.0 (e), and 20 (f). Adapted from the study by Lee et al. [173] with permission from Royal Society of Chemistry.

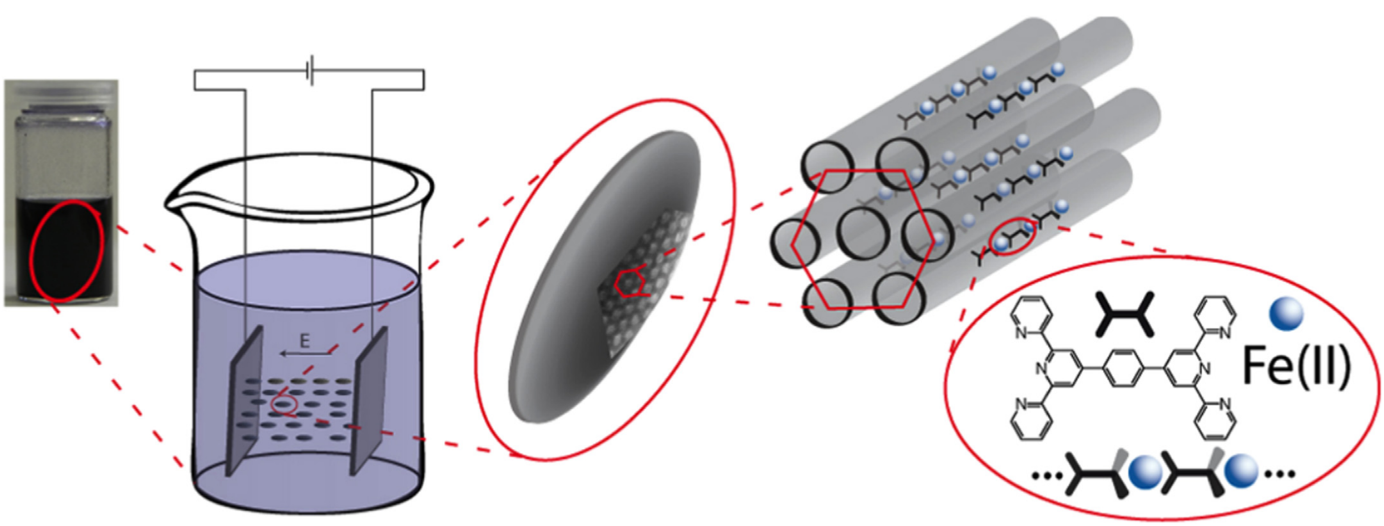

Fig. 8. Model of ERF operation and schematic illustration of a rod-like mesoporous SBA-15 particle with hexagonal orientated mesopores and incorporated metallosupramolecular coordination polyelectrolytes. Reprinted from the study by Schwarz et al. [169] with permission from American Chemical Society.

dynamic modulus of ERF were observed with an increase in the fiber length. However, in 2002, Qi and Wen observed the inverse effect of particle shape on the ER effect [78]. Thus, glass spheres and microfibres of various sizes reveal a decrease of dielectric permittivity and conductivity values, as well as the values of shear stress with an increase in the aspect ratio of the particles. However, it should be noted that further studies have confirmed results by Kanu and Shaw [167,171].

The influence of the shape and size of particles greatly attracted the interest of researchers after manifestation the trend to an increase in the ER effect during the transition from microfillers to nanofillers [24] as evidenced by the rapidly growing number of publications on this topic [81,82,129,152,154,166,172–174]. The nanoscale dispersed phase allows reducing the concentration of particles in suspension and significantly increases the sedimentation stability of ERFs, which is one of the key factors for the stability of such systems.

Interesting results were obtained by Yoon et al. [82]. The authors used a mixture of spheres and rods based on silicon dioxide, so-called bimodal fillers. Three types of rods with aspect ratio of 2, 3, and 5 (with lengths of ~100, 200, 800 nm and diameters of ~50, 70, 160 nm, respectively) were obtained to create filler compositions. The sizes of the spheres were 50, 100, 150 and 350 nm. The ER behavior of fluids with 12 variants of spheres and rods combinations with different content of particles were studied (60 samples). The total filler concentration in the ERFs was 3 wt%. An increase in the ER effect with a decrease in particle size was observed for fluids with spherical filler. At the same time, the ER effect increases with an increase in the aspect ratio. Anisometric particles reveal a more significant effect than spherical ones, which confirms the previously obtained results. A bimodal filler consisting of rods with a high aspect ratio and small content of 50 nm spherical particles (3–6 wt%) showed the highest ER effect, which turned out to be even higher than on fluid filled exclusively by rods. A similar effect was observed for mix with larger spheres of 100 nm. Moreover, the bimodality effect does not affect the rheological behavior of the suspensions without electric field. The observed phenomena, according to the authors, was not related to polarization mechanisms and dielectric characteristics of particles. The dielectric permittivity of rods is higher than the spheres, but for bimodal systems, its value is lower than for the most anisometric rods. Therefore, the enhancement of the effect was associated with the geometry of the filler. In the proposed model of bimodal fluids (Fig. 9), the nanospheres are located in the contact zone of the rods (at the ends) aligned along the lines of electric field, which leads to a strengthening of the structure and an increase in the ER effect. The results demonstrate the synergistic effect of bimodal systems, and indicate that the polydispersity of the filler leading to the distribution of the aspect ratios of the particles can have a significant effect on the ERFs behavior. Additionally, the shape-effect of the filler will be discussed in other subsections by the example of particles of various nature.

Another widely used filler these days is titanium oxide (IV)-based nanoparticles. Suspensions filled by titanium dioxide under electric field exhibit yield stress of tens of kPa, but the degree of filling remains very high about 40–60 wt%. As was noted above, a giant ER effect was discovered on particles of barium oxalate titanate, that is, a derivative of titanium oxide [50]. The combination of titanium and silica oxides to create fillers with a core/shell structure leads to ERFs with extremely high operational characteristics.

Cheng et al. co-precipitated silica particles with a mixture of calcium and titanium precipitate consist of oxalates and titanium hydroxyl oxide [119]. The yield stress for ERF with such filler was 109 kPa at electric field of 5 kV/mm and concentration of 67 wt%. Wu et al. later confirmed the effectiveness of titanium compounds as a coating [86,175]. Silicon dioxide particles coated by titanium dioxide exhibit increased ER activity and the yield stress increases by 4.5 times. Replacing the shell by hydroxyl titanium oxalate leads to an even greater increase in the effect by a factor of nine compared with silicon dioxide. The yield stress reaches 45 kPa at 5 kV/mm. An intriguing fact is the absence of a yield stress of fluids without electric field at such a high degree of filling (44 wt%). Moreover, the viscosity of the suspensions remains low about units of Pa*s [175]. The hybrid micro- and nanoparticles of titanium and calcium compounds as a filler also leads to high ER activity. Thus, the yield stress was 35 kPa at 5 kV/mm for ERF with a dispersed phase concentration of 29 vol% and reaches 255 kPa at a concentration of 49 vol% [86].

Titanium silicalite zeolites were studied as fillers for ERFs as well. ER effect strongly depends on the particle size [176]. Particles with size of 50–100 nm show in more than 23 times higher yield stress (compare 199.48 and 8.56 at 3 kV/mm) at 10 wt% concentration than of 100–200 nm. Additional doping with titanium dioxide resulting in core/shell structure enhances the ER effect [177]. Nevertheless, all considered zeolite-based fluids has perfect sedimentation stability.

The efficiency of particles with a core/shell structure at low fillings was demonstrated by Plachy et al. [178]. Suspensions with particles of lithium titanate at 5 wt% concentration demonstrate in three times higher values of yield stress than particles of pristine titanium dioxide. The urea shell on the surface of lithium titanate leads to an even more significant increase in values. The yield stress reaches about 100 Pa at filling up to 15 wt% and electric field of 3 kV/mm.

Extremely high ER activity has been found for double shell SiO_2/TiO_2 hollow nanoparticles [179]. The yield stress rises up to 302.4 kPa under an electric field of 3 kV/mm at filler concentration of 10 wt%. It was assumed, that additional electrostatic interactions are formed between the two shells in particle leading to stronger shear stress. The effectiveness of the filler increases with decreasing particle size.

The morphology of titanium dioxide has a significant influence on ER behavior of its suspensions. Thus, $H_2Ti_2O_5$ nanotubes with shell of titanium dioxide reveal enhanced ER effect due to stronger interfacial polarization and lower leaking current density [180]. The flower-like TiO_2–MoS_2 nanocomposite with a hierarchical and core/shell structure shows high ER effect [181]. A facile synthesis of the filler was proposed during this study. The possibility of cyclic operation and the conducting mechanism of the effect were shown as well. The yield stress reaches about 90 Pa at a particle concentration of 10 wt% and electric field of 3 kV/mm, which is more than three times higher than for pristine titanium dioxide. The synthesis glucose solution acted as a binder and a protective agent on the particles surface. Moreover, it was shown that particles with glucose reveal enhanced ER performance. Thus, activation properties of glucose were observed. Titanium dioxide nano-whiskers with high aspect ratio also reveal enhanced ER effect [182]. The two different regimes of yield stress as a function of electric field were observed, due to transition from polarization mechanism to conducting. The yield stress about 1 kPa was observed for 10 vol% suspension under electric field of 3 kV/mm. Recently, there were several reports about the ER activity of TiO_2-based particles of various morphology, such as bowl-like [152], cactus-like [183], eggshell-like [184] and nanobox-like [185]. The variety of titanium dioxide particle shapes presented in Fig. 10. Changing particle structure opens up possibilities for producing highly efficient ERFs with tunable properties.

The ER performance of TiO_2 particles with unusual structure was studied [186]. The nanoparticles of spherical shape with carbon clusters incorporated inside forms the jujube cake structure. The size of particles was about 500 nm and of carbon clusters was about 1–10 nm. Such carbon-doped titanium dioxide particles ensure the excellent performance of the ERFs resulting in a high yield stress, long lifetime, small current leakage, and high thermal stability. The yield stress was 120 kPa at 38 vol% and 5 kV/mm field strength, which corresponds to the giant ER effect. Such efficiency achieved by the nanoconductor particles acting similar to the polar molecules and inducing dipole moments under electric field.

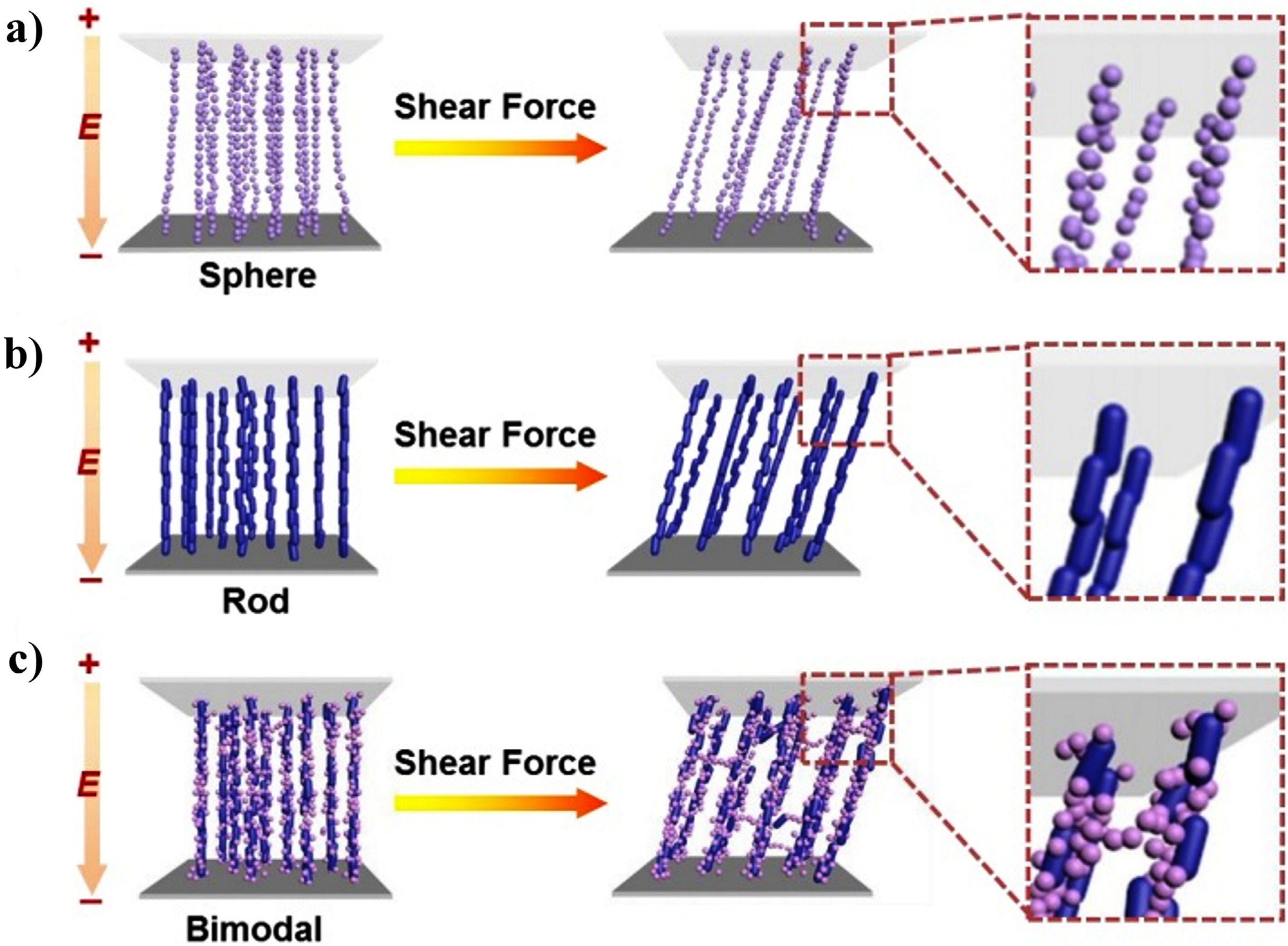

Fig. 9. Tentative model of chain-like structures formed from sphere-based (a), rod-based (b) and bimodal (c) ERFs under electric field. Adapted from the study by Yoon et al. [82] with permission from American Chemical Society.

One of the novel types of inorganic fillers for ERFs is two-dimensional MXenes. They are transition-metal carbides/nitrides. 2D-Layered C/TiO_2 hybrids in silicon oil shows high ER effect [187]. The yield stress was about several hundred Pa even at 4 vol% of filler content. The specific lamellar structure of conductive carbon layers provides electron transport and reveals fast response under electric field and the titanium dioxide nanoparticles reveal high surface polarization. Nevertheless, despite the facile synthesis, the conditions and reagents have a critical effect on the ER behavior of the resulting filler. In the absence of optimal conditions, the structure of hybrid MXenes is violated leading to drop of ER properties and electrical breakdown at extremely low field strengths less than 1 kV/mm.

Thus, inorganic particles containing oxygen are widely used as fillers for ERFs. Moreover, one can vary the characteristics of fluids by controlling the chemical composition and shape of the filler particles, and adjust in this way the parameters of the material for a specific task. Finally, the main ERFs parameters of reviewed studies are summarized in Table 1.

4.1.1. Aluminosilicate fillers

Other well-known fillers illustrating the shape effect are various aluminosilicate clays. In the literature, there are studies of suspensions filled by particles based on laponite [87,89], fluorohectorite [88,90,92], montmorillonite (MMT) [146,188,189], kaolin and halloysite [93,190,191]. The rheological characteristics under electric field turned out to be not very high; however, interest in these fillers has not dried up, as evidenced by recent publications [164,192–195]. Several reviews are devoted to the ER behavior of clays and their composites [163,164]; therefore, within the framework of this publication, we briefly dwell on the most significant points and some studies of recent years. A significant advantage of nanoclays is usually a high aspect ratio, which allows the formation of a percolation network in suspension at an extremely low concentration in the order of several weight percent. Such liquids are called "empty" [196]. MMT and halloysite are one of the most promising aluminosilicate clay fillers for modern ERFs. They are natural resources, and therefore cheap enough.

MMT is a clay mineral, an aluminosilicate belonging to the class of smectics. The chemical formula of MMT is $M_x(Al_{4-x}Mg_x)Si_6O_{20}(OH)_4$, where x changes between 0.5 and 1.3 and M is monovalent metal cation [197]. MMT has a layered structure and forms agglomerates of particles with sizes of 1–10 μm. A particle contains up to several hundred aluminosilicate platelets collected in stacks and formed tactoid phase. The distance between the platelets is called the interlayer space or gallery. Particles are randomly located in the agglomerate. An individual MMT-plate has

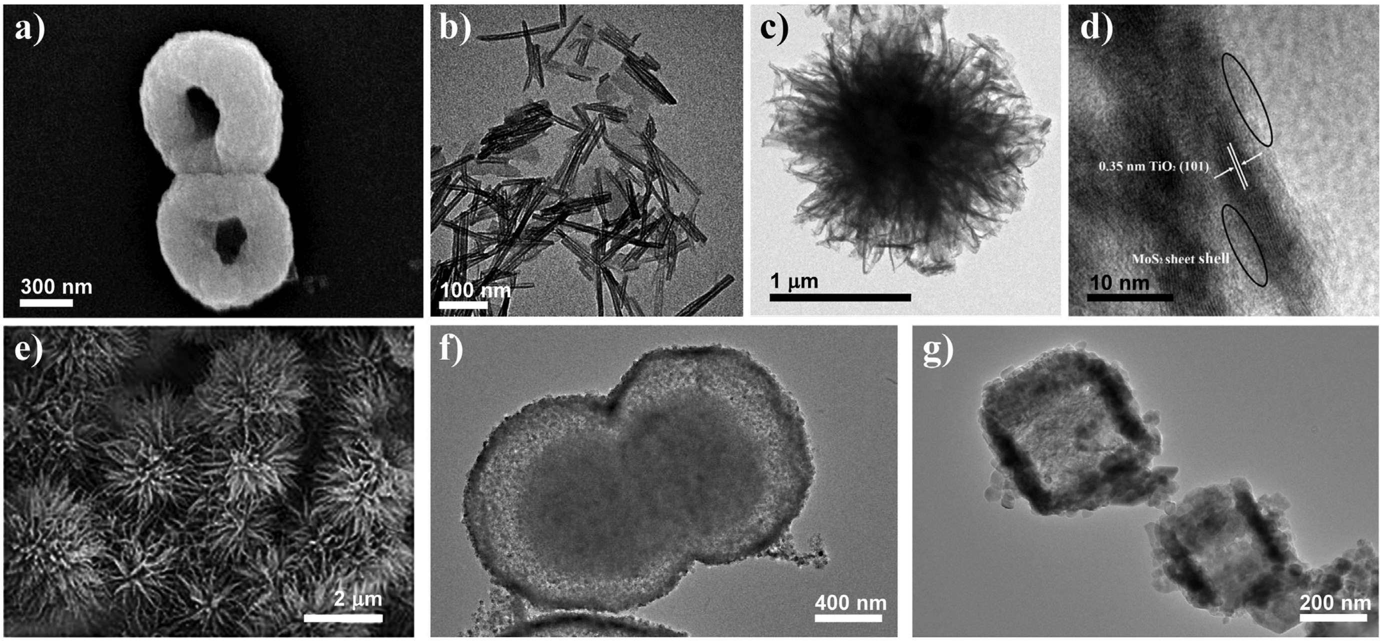

Fig. 10. Variety of TiO_2 particles morphology. SEM images of bowl-like particles after 6 h of solvothermal time (a) [152] and double-shell structured SiO_2@TiO_2 cactus-like particles (e) [183], TEM images of titanium oxide@$H_2Ti_2O_5$ core/shell nanoparticles (b) [180], of flowerlike TiO_2−MoS_2 nanocomposite (c) with core/shell structure of an individual needle (d) [181], of eggshell-like TiO_2 hollow spheres (f) [184] and of TiO_2 nanobox particles (g) [185]. Adapted from the study by He et al. [152] with permission from Royal Society of Chemistry and from the studies by Wang et al., He et al., Zhang et al., Sun et al., and Li et al. [180,181,183−185] with permission from Elsevier, respectively.

a thickness close to 1 nm and average dimensions of about 200 nm and 500 nm, respectively [198]. The aspect ratio of the individual plate reaches an incredible value of 200. No other ERF filler has such anisometry. MMT this compares favorably with synthetic fluorohectorite and laponite. The MMT crystal lattice consists of three layers and has 2:1 type of structure. The two outer layers are formed by silicon-oxygen tetrahedra with a central silicon atom, and the inner layer consists of oxygen octahedra with a central aluminum atom (Fig. 11a) [199]. The crystal lattice of clay may contain defects. Thus, heteroisomorphic substitution of atoms, namely trivalent Al^{3+} by divalent Mg^{2+} or tetravalent Si^{4+} by Al^{3+}, Mg^{2+} or Fe^{2+}, leads to the excess negative charge on the layer surface, which is typically compensated by cations of alkali or alkaline-earth metals (Li^+, Na^+, K^+, Ca^{2+}) and is characterized by cation exchange capacity. This capacity can slightly vary and depends on the mineral deposition. Hydration of compensating cations leads to swelling of the filler and an increase in the interlayer spacing in the clay stack (intercalation). The tactoid can be destroyed into individual plates (exfoliation) with a strong increase in the gallery space, that is, at a sufficiently high swelling degree. Such structural transition is essential for polymer composites to achieve a more uniform distribution of the filler in the matrix [200]. Substitution of exchange cations by cations of the same kind, for example, Na^+ or Ca^{2+}, results in a modified form of MMT and then it's called sodium or calcium MMT, respectively. Moreover, ammonium compounds with hydrocarbon substituents, namely quaternary ammonium cations (quats) can be often used as modifiers instead of alkali and alkaline earth metal cations. Such forms of MMT are called organoclays [201]. The introduction of quats increases the interlayer spacing and ensures the penetration of the dispersion medium into the clay structure, which in turn leads to an increase in the degree of exfoliation of the filler and the formation of a percolation network at a clay concentration of less than 5 wt%. The various chemical structures of quats substitutions affect the hydrophobicity of the filler, as well as the type of their packaging between MMT layers and determine the inter-gallery space. Quats can form mono- or bilayers located in the interlayer space of MMT at an angle to the surface of the silicate, forming a "paraffin" type of packaging [202−204]. The type of packing is determined experimentally by X-ray diffraction by changing the position of the small-angle reflection d_{001}, which is responsible for the interplanar distance in the tactoid [205] (Fig. 11b and c).

It is interesting, that the fundamental questions of the influence of concentration, changes in dielectric characteristics in the context of intercalation and exfoliation processes and their effect on the ER behavior were considered only recently, although fillers for ERFs based on MMT have long been known [58]. In particular, the introduction of modifiers (quats) in the clay structure on the one hand changes the quality of the filler dispersion (exfoliation degree), and on the other hand influences the dielectric characteristics and expands the potential range of electric fields for stable fluid operation (up to 7 kV/mm). The yield stress of the organomodified suspension at a low concentration of 8 wt% reaches about 90 Pa at maximum electric field. However, MMT is usually used as part of polymer composites acting as the dispersed phase of the ERFs.

Various groups of authors studied ERFs filled by composites of MMT in the concentration range of 5−30 wt%. It was showed that the composite filler increases the ER effect compared to the pristine one [146,188,189,206−208]. The introduction of metal and metal-polyelectrolyte complexes into the MMT structure as a modifier allowed obtaining a significant increase in the ER effect due to changes in the dielectric characteristics of the filler increasing the difference between dielectric permittivity of the medium and particles [193,194]. The storage modulus reaches 2.7 kPa at electric field of 1 kV/mm and MMT concentration of 10 wt%. Thus, MMT can be easily modified and act as the template for the new type of fillers with increased ER activity.

Another natural material related to nanoclay is halloysite [209]. The chemical formula is Al_2O_3*$2SiO_2$*nH_2O, where n = 2, 4. The morphology of halloysite, depending on the crystallization conditions and geological environment, can be different. It crystallizes in tubular, spherical, fiber-like, cylindrical, lamellar, bulbous, and

Table 1
Comparative analysis of ERFs filled by inorganic-based materials. Yield stress values are from experimental data unless an appropriate rheological model is specified.

Materials	Medium type and viscosity	Morphology	ρ_{filler}, g/cm^3	Conc.	$\Delta\varepsilon_{fluid}$, rel.un.	E_{max}, kV/mm	J_{max}, μA/cm^2	Yield stress at 3 kV/mm or E_{max}, Pa	Slope $\log(\tau_0) = f(\log(E))$	Sedimentation ratio, %	*Eff*, rel.un.[b]
Al_2O_3 [151]	SO	Irregularly shaped particles with size of 21 ± 4 nm forming loose open nanoparticle clusters	3.6	5 wt%	–	2	–	13.15 ± 0.628[a] (Herschel–Bulkley)	1.00	–	1300
CeO_2 [154]	SO, 20 cSt	Needle-shaped particles of 75 nm in length and 20 nm in width	–	40–60 wt% (15.8–26 vol%)	–	5.25	–	60–140	–	50	6
		Isotropic particles of 9 nm						55–100		43	5
α-MnO_2 [155]	SO, 300 cSt	Needle-shaped particles of 150–200 nm in length and 15–20 nm in width	4.2	10–20 wt% (1.5–4.5 vol%)	262–2096	5.6	–	60–160	–	54–60	4
δ-MnO_2 [155]		Plate-like particles (~4 nm width) were fused into spheroid aggregates ~ 100 nm	–		62–76	5.6		140–170		24–30	4
FeC_2O_4 [156]	SO, 194 mPa s	Rod-like morphology with thickness of 1–2 μm (aspect ratio 7–30)	2.35	1–10 vol%	–	1.5	–	1.5–80[a]	1.5–2	~85	530
$Bi_{1.8}Fe_{1.2}SbO_7$ [157]	SO, 300 cSt	Uniform spherical aggregates (140–240 nm) of monodisperse 20 nm particles	6.77	40–80 wt%	1.52–19.3	5.6	0.05–0.5	46–50		65	1
MoS_2 [159]	SO, 100 cSt	Nanosheets, the size of the few-layer is 200–400 nm	–	10 vol%	–	3.5	–	61.0[a]	–	~97	16
SiO_2 [166]	SO, 100 cSt (PMPS)	Nanorods with diameter of 60 nm (aspect ratio 3.3)	2.77	3 wt%	1.27	3	–	~30	2 (< 1 kV), 1.5 (> 1 kV)	81	320
SiO_2/TiO_2 [166]		Core/shell nanorods with diameter of 66 ± 4 nm (aspect ratio ~3.2)	3.01		1.98			~70		77	380
		Hollow nanorods with diameter of 72 ± 4 nm (aspect ratio ~3.2)	2.71		3.05			~100		90	540
$SrTiO_3$ [117]	SO, 50 cSt	Spherical particles with diameter of 0.5–10 μm	2.28	16–36 vol%	–	4	< 20	2-27·10^3	–	~100	7
TiO_x/1.9% SDBS/acrylamide [118]	SO, 800 mPa s	Spherical particles with diameter of 60–300 nm	2.48	66.7 wt%	21	5	27	~40·10^3	–		90
SiO_2/TiO_2 [167]	SO, 100 cSt (PMPS)	Nanospheres with diameter of 20–25 nm (aspect ratio 1)	–	15 vol%	3.65	5	–	12·10^3	–	–	42
		Nanorods with diameter of 20–25 nm (aspect ratio 8)			7.90			21·10^3			80
		Nanotubes with diameter of 150–200 nm (aspect ratio 33)			9.72			40·10^3			74
TiO_2 [171]	SO, 50 mPa s	Nanowhiskers with diameter of 10 nm and length of 10–1000 nm	1.85	10 vol%	–	3	~1	1.1·10^3	–	~96	3330
TiO_2 [172]	SO, 50 mPa s	Spheres of about 270 nm	–	30 vol%	6.68	5	6	~8·10^3	1.59		32
TiO_2 [174]	SO, 50 mPa s	Spheres with diameter of ~500 nm	–	55 wt%	–	5	~6.5	~1.1·10^3	–	11.5	8
TiO_2/MCNTs [174]							< 1	~1.7·10^3		43.2	22
SiO_2/TiO_2 [175]	SO, 50 mPa s	Core/shell spheres of 419 nm and the shell thickness of 12 nm	–	44 wt%	–	5	–	~5·10^3	–	84 (for 20 wt%)	67
SiO_2/TOC [175]		Core/shell spheres of 417 nm, and the shell thickness of 11 nm						~8·10^3		92.5 (for 20 wt%)	52
SiO_2/TiO_2 [179]	SO, 100 cSt (PMPS)	Single shell spheres of 240 nm, and the shell thickness of tens nm	–	10 wt%		3	0.10	11·10^3	–	–	1660
		Double shell spheres of 120–240 nm, and the shell thickness varied from 10 to 30 nm	2.68–2.74		3.38–8.24		0.12–0.36	~45–300·10^3		88–98	4540
TTO [158]	SO, 10 mPa s	Spherical particles of 500 nm with smooth surface	4.3	19–41 vol%	8.73	4	~15	12–45·10^3	–	82.7	52
W-TTO [158]		The tremella-like particles with diameter of 0.5–3 μm	4.2		18.84		~6	12–62·10^3		91.1	71
Urea-coated $BaTiO(C_2O_4)_2$ [50]	SO	Spherical particles with diameter of 50–80 nm	–	15–30 vol%	–	5	~20–100	12–70·10^3	–	~100	71
	SO		–	~77 wt%	–	5	–	90·10^3	–	–	22

(*continued on next page*)

Table 1 (*continued*)

Materials	Medium type and viscosity	Morphology	ρ_{filler}, g/cm^3	Conc.	$\Delta\varepsilon_{fluid}$, rel.un.	E_{max}, kV/mm	J_{max}, μA/cm^2	Yield stress at 3 kV/mm or E_{max}, Pa	Slope $\log(\tau_0) = f(\log(E))$	Sedimentation ratio, %	*Eff*, rel.un.[b]
Urea-coated $Ba_{0.8}Rb_{0.4}TiO(C_2O_4)_2$ [121]		Spherical particles with diameter of 10–20 nm									
Urea/polar groups-coated TiO_2 [120]	SO	Spherical particles with diameter of 50–100 nm	–	35 vol%	–	5	< 20	$125\cdot10^3$	1	–	104
Urea/polar groups-coated $CaTiO_3$ [120]								$110\cdot10^3$		–	44
Urea/polar groups-coated $SrTiO_3$ [120]		Spherical particles with diameter of 2–6 μm						$50\cdot10^3$		–	23
Li_2TiO_3 [178]	SO, 194 mPa s	Spherical particles with diameter up to 150 nm	–	5 wt%	0.10	3	–	~1	–	–	60
Urea-coated Li_2TiO_3 [178]					0.20			~2.5	1.15		160
				15 wt%	–			~100	1.87		550
TiO_2/MoS_2 [181]	SO, 500 cSt	Core/shell flower-like with diameter of ~1–1.5 μm	–	10 wt%	0.41	3	–	~70	1.4	~85	310
SiO_2/POA [61]	SO, 50 cSt	Core/shell spheres with diameter of ~1 μm	–	10 vol%	3.90	2	–	100 (Bingham)[a]	1.5	50	1730
SiO_2 [81]	SO, 100 cSt (PMPS)	Sphere with diameter of 70 (aspect ratio 1)	–	3 wt%	0.66	10	< 4	~15	1.5	–	1650
		Rod with diameter of 60 and aspect ratio 2			1.22			~26			2880
		Rod with diameter of 60 and aspect ratio 4			1.85			~47			5210
HCTO [86]	SO, 50 mPa s	Spindly particles with diameter of 4 μm and aspect ratio ~3.3	2.45	29–49 vol%	–	5	4	~10–150$\cdot10^3$	–	~84–95	54
SiO_2 (2.3 wt%) –CTP [119]	SO, 10 mPa s	Spherical morphology with diameter of ~40 nm	–	67 wt%	24	5	30	$50\cdot10^3$	–	–	27
Ionic-liquid-doped SiO_2 [122]	SO, 500 cSt	Spherical morphology with diameter 170–600 nm	–	30 wt%	–	3	Up to 117	Up to $1.2\cdot10^3$	–	–	89
SiO_2-based ionogels [147] –	SO, 50 cSt	Spherical morphology with diameter 260–370 nm	1.92	10 vol%	0.96	4	–	~70	1.50	–	6660
0.5 TESM-IL		Rough and irregular surfaces with nanometer and sub- micron bumps	1.62		1.01			~200	1.95		16700
0.8 TESM-IL			1.63		1.80			~400	2.00		20000
1.0 TESM-IL			2.00		0.86			~100	1.92		11600
1.88 TESM-IL			2.13		1.03			~50	1.65		2000
Alkaline earth metal-doped SiO_2/TiO_2 [124]	SO, 100 cSt (PMPS)	Core/shell spheres with diameter of ~130 nm	2.65–2.74	30 vol%	1.77–4.27	3	–	8–55	1.5	85–90	55
$H_2Ti_2O_5$ [180]	SO, 500 cSt (PMPS)	Nanotubes with diameters in the range of ~10 nm and length of several hundreds of nm	–	10 wt%	~0.9	2	148.7	~100[a]	2.45	–	410
$TiO_2/H_2Ti_2O_5$ [180]		Core/shell structured nanotubes with diameters in the range of ~10 nm and length of several hundreds of nm			~1.3	3	50.92	~200	1.57		1390
Al-doped TiO_2 [112]	DO	Spherical particles with diameter of tens nm	–	20 wt%	1.3	2	~15	~150	–	–	35
Al and P double doped-TiO_2 [112]					2.4	3.5	~2	~270[a]			55
$Sr_{2.8}Ba_{0.2}TiAl_{0.8}Ni_{0.2}O_7$ [160]	SO	Perovskite-like structure, spherical particles with diameter of 5–10 mm	–	30 wt%	0.6	3.5	3	~21	–	–	10
Zinc borate + 1 vol% Triton X-100 [161]	SO, 1 Pa s	–	–	15 vol%	3.64	3.5	–	21	–	–	9
TiO_2 [152]	SO, 500 cSt	Bowl-like particles of 470–1400 nm	2.3	10 wt% (4.3 vol%)	0.3	3	~23	266	1.72	~85	8860
SiO_2/TiO_2 [183]	–	Cactus-like double-shell structured microspheres of 300 nm, shell thickness of 20 nm	–	12 wt%	–	4	–	~20	1.5	–	550
TiO_2 [184]	SO (PDMS)	Eggshell-like particles of ~1 μm and wall thickness of ~30–50 nm	–	10 wt%	0.71	3	–	~150	1.6	–	730
$TiOF_2$ [185]	SO, 500 cSt (PDMS)	Nanocubes particles of 300 nm	–	10 wt%	0.64	3	–	~35	1.2	–	134
TiO_2 [185]		Nanobox-like particles of 50 nm			1.03			~100	1.5		163

$C{-}TiO_2$ [186]	SO, 300 cSt	Spherical particles of 500 nm with carbon of 1–10 nm	3.8	33 vol%	~1	5	1.2	~$20 \cdot 10^3$	2	–	1210
$Ti_3C_2T_x$ MXenes [187]	SO, 100 cSt	TiO_2 nanoparticles anchored homogeneously on and between the ultrathin C nanosheets	3.57	4–8 vol%	3.02	4	–	~200–350	2 (< 1 kV), 1.5 (> 1 kV)	–	14500
TS-1 zeolites [176]	SO, 500 cSt	Spherical particles (50–100 nm) Spherical particles (100–200 nm)	–	10 wt%	~2 ~0.5	3	–	~200 ~10	1.31 0.75	80 100	550 24
TS-1 zeolites/TiO_2 [177]	SO, 500 cSt	Core/shell spherical particles of ~100 nm	2.04	10 wt%	0.69	3	4.074	~200	1.61	~99	460

[a] Yield stress values are given at maximum electric field.
[b] Hereinafter *Eff* is relative efficiency calculated as $\boldsymbol{Eff} = (\boldsymbol{\tau_E} - \tau_0)/_{\tau_0 \boldsymbol{E_{rel}}\,\boldsymbol{\omega}}$, where, $\boldsymbol{\tau_E}$ (Pa) is the yield stress under electric field, $\boldsymbol{\tau_0}$ (Pa) is the yield stress at zero-field, $\boldsymbol{E_{rel.}}$ (rel.un.) is normalized electric field strength, $\boldsymbol{\omega}$ (rel.un.) is filler content. The values calculated for electric field of 3 kV/mm except of ERFs with a lower limiting field strength. The differences between weight and volume filler content were not taken into account. If $\boldsymbol{\tau_0}$ is not specified in the study, a value of 0.1 Pa was used for calculation.

other forms. However, typically halloysite is multilayer nanotubes with a cavity inside [210]. The tube is formed by a rolled aluminosilicate sheet. The elementary layer has a 1:1 structure and is formed by tetrahedral silicon-oxygen and octahedral alumina-hydroxyl networks. The bonding between the layers performed by hydrogen forces and interactions of isomorphic substitution defects with compensation cations [197]. Two types of halloysite nanotubes are known with various interlayer spacings of 10 and 7 Å. The first with increased spacing is a hydrated form. The second can be obtained by heating of the hydrated form at the temperature above 50 °C (Fig. 11d–f). Thus, nanotubes easily lose a significant part of water and pass from one form to another. Nanoscale, a specific tubular shape of particles with a cavity inside allowing the penetration of various compounds, and a high aspect ratio as well opens the opportunity for specific halloysite applications, such as drug carriers [211–214], the catalysts template [215,216], and, along with other layered aluminosilicates, as a filler for polymer composite materials [217–219].

Halloysite nanotubes were also considered as fillers for ERFs. Kuznetsov et al. studied the effect of a specific tubular particle shape on the ER effect in low-filled suspensions (up to 8 wt%) [220,221]. The activating effect of water adsorbed on a high surface area of particles was shown. It was found, that halloysite forms a percolation network even at concentration of 4 wt% in polydimethylsiloxane media. The yield stress increases with electric field and filler concentration and reaches about 50 Pa at 8 wt% filler content and electric field of 7 kV/mm. In other study, the yield stress of about 7.5 Pa was observed at an electric field of 2 kV/mm, when the suspension filled by halloysite up to 5 wt% [93]. In this case, unlike the layered aluminosilicates, namely kaolin, MMT, laponite and fluorohectorite, the halloysite nanotubes do not show strict orientation along the electric field lines according to X-ray scattering data [93,96]. One of the possible reasons for the lack of orientation is the high polydispersity of halloysite, which explains the reduced operational characteristics of halloysite compared to MMT and is a disadvantage of this filler. However, more often halloysite is considered as a template for composite fillers [49,222]. The ER effect of halloysite suspensions and its composites with PANI and polypyrrole in silicone oil was considered. ERF filled by composite with PANI (halloysite content of about 91 wt%) reveals the yield stress of 60 Pa with a filler concentration of 15 wt% and electric field of 2 kV/mm. The authors note the classical quadratic dependence of the yield stress on the electric field [222]. A composite based on polypyrrole showed a yield stress of 22.6 Pa at 10 vol% and the same electric field (2 kV/mm). The dependence of the yield stress on the electric field was not quadratic and varied from 1.5 to 1 at critical electric field value of 2.5 kV/mm [49].

Obviously, the same aluminosilicate nature of clays with different morphology makes them an ideal object for analyzing the particles shape-effect on the ER behavior of fluids. Thus, Ramos-Tejada et al. analyzed the ER behavior of suspension filled by MMT, laponite, and sepiolite, and their organomodified forms [195]. The sizes and morphology of particles were evaluated by TEM and were of 500 ± 200 nm for MMT and of 21 ± 4 nm for laponite platelets at thickness of 1 nm. The length of sepiolite fibers was of 1100 ± 600 nm and diameter of 27 ± 7 nm. It was found that morphology and modification significantly affect the homogeneity and type of columnar structures formed under electric field. The main result was as follows: platelet-shape particles show higher values of yield stress and storage modulus. Moreover, large MMT plates reveal the best properties. However, the effect of the modifier was ambiguous. The authors investigated mixed systems with equal volume fractions of each of the clays searching for combined effects also. It was shown that any kind of clay is able to reinforce MMT suspensions, and that platelet particles weakly improve the

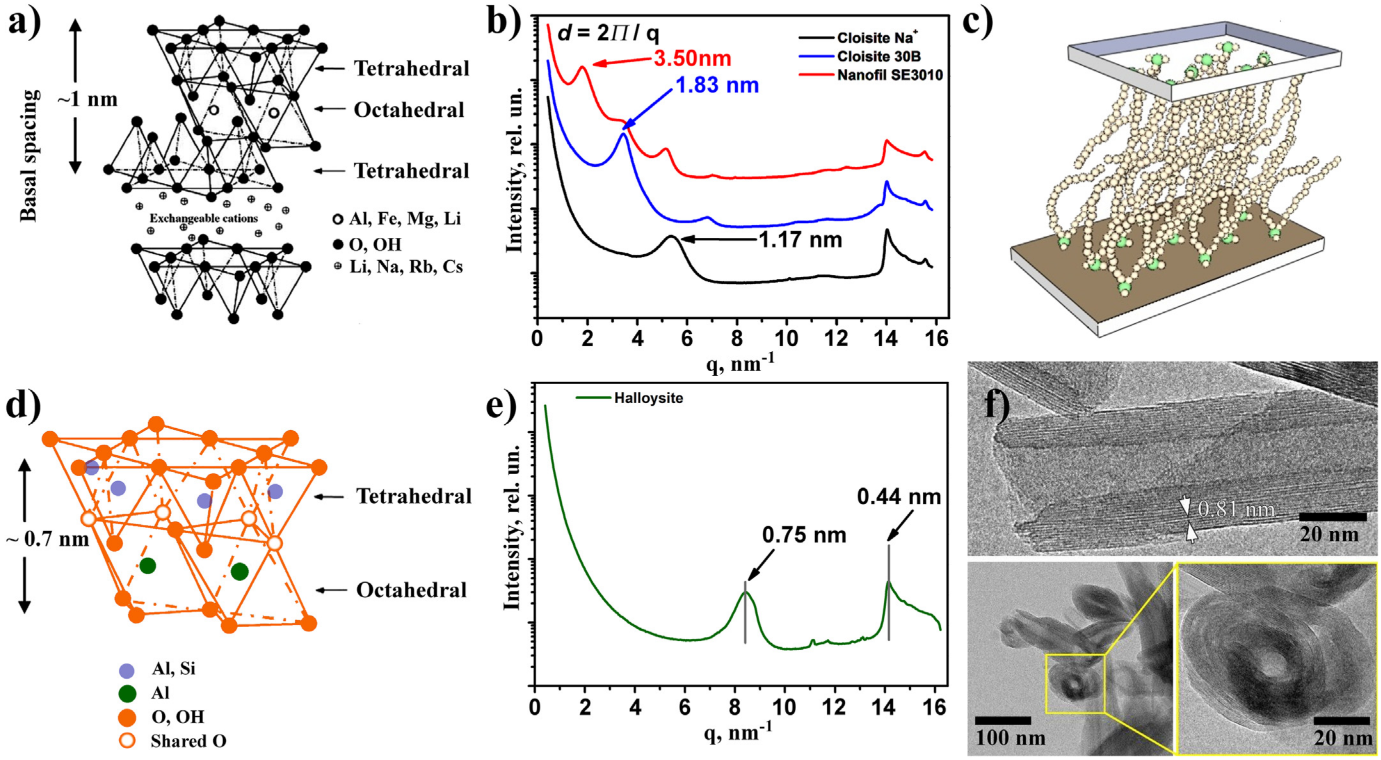

Fig. 11. Atomic structure of montmorillonite (2:1 phyllosilicates) (a) [199], and halloysite (d). X-ray patterns of variously modified montmorillonite (b) [62] and halloysite (e) [221] powders. A model of a “paraffin” type of packaging of quats in the interlayer space of montmorillonite (c) [62]. Typical HR TEM image of halloysite nanoparticle (f) [221]. Adapted from the studies by Kuznetsov et al. and Sinha Ray et al. [62,199] with permission from Elsevier and Wiley Periodicals, respectively, as well as reprinted from the study by Kuznetsov et al. [221] with permission from eXPRESS Polym. Lett. (Open access).

behavior of sepiolite fluids. One of the recent studies by Kutalkova et al. shows high sedimentation stability of low filled sepiolite fluids. The sedimentation ratio remains 100% after 200 h for 15 wt% suspension [223].

Thus, the availability of raw materials, high shape anisometry, and the ability to be modified make MMT and halloysite promising templates for creating new hybrid fillers for high-performance ERFs. The main characteristics of ERF filled by clays and their composites are given in Table 2.

4.1.2. Carbon fillers

One of the popular nanomaterials is graphene. Graphene has a two-dimensional structure formed by carbon atoms arranged in the form of regular hexagons [224]. In 2010, Novoselov and Geim were awarded the Nobel Prize in Physics "for groundbreaking experiments regarding the two-dimensional material graphene". Despite the fact that the pioneer work of Novoselov and Geim were published in 2004 [225], in less than 20 years of research on graphene, it has become an important material for many potential and practical applications due to its unique electronic characteristics, good mechanical strength, high thermal conductivity, and specific surface area [226]. The area of graphene applications includes touch displays, electronic devices, transparent conductive films, catalysts, and capacitors. However, preparation of high quality graphene sheets is still challenging. One of the promising methods for inexpensive and industrial production of graphene is exfoliation of graphite in an oxidizing medium to produce graphene oxide and its subsequent annealing [227,228]. The most famous are the methods suggested by Staudenmaier [229], Hofmann [230,231], and Hummers [232] et al. These methods are mainly distinguished by oxidizing agents.

Graphene oxide has a lot of defects compared to pure graphene. The presence of surface oxygen-containing groups, such as epoxy, carboxyl, hydroxyl [224], provides facilities for using the graphene oxide as a filler for ERFs, due to suitable conductivity and polarizability [162], along with pure graphene with a conjugated π-system. One of the first articles on the ER response of suspensions based on a graphene oxide composite with PANI in silicone oil was published by Zhang et al., in 2010 [233]. The authors showed the organization of particles into columnar structures under the influence of an electric field by optical microscopy. Later, studies of the ER and dielectric characteristics of the ERFs based on graphene oxide and its composites with PANI and polystyrene were performed [41,162]. Graphene-based ERFs in recent years were the subject of several reviews [234,235]; therefore, in the framework of this article, we will not dwell on many different compositions and will restrict by the most interesting publications in our opinion, and will present the main parameters of the known fluids.

Shin et al. studied the influence of the mechanical grinding conditions of graphite (with its subsequent oxidation) on the particle sizes of the obtained graphene oxide and the ER response of its suspensions [131]. The best ER effect was observed for particles obtained after 2 h of grinding. The particle sizes were of 400–700 nm in length with thickness of 4–8 nm. The yield stress at 3 vol% of graphene oxide in silicone oil was 78.5 Pa at electric field of 1 kV/mm. Mechanical grinding generates heat leading to changes in the morphology of graphene oxide. There was a transition from a flat to a spherical shape. Moreover, an increase in the thickness and volume of particles was observed. Such morphological transformation through the mechanochemical process plays a key role in reducing the density of graphene oxide.

Dhar et al. used an unusual dispersion medium. The significant ER response was observed on low molecular weight polyethylene

Table 2
Comparative analysis of ERFs filled by clay-based materials. Yield stress values are from experimental data unless an appropriate rheological model is specified.

Materials	Medium type and viscosity	Size	ρ_{filler}, g/cm^3	Conc.	$\Delta\varepsilon_{fluid}$, rel.un.	E_{max}, kV/mm	J_{max}, µA/cm^2	Yield stress at 3 kV/mm or E_{max}, Pa	Slope $\log(\tau_0) = f(\log(E))$	Sedimentation ratio, %	*Eff*, rel.un.
PANI-MMT [188]	SO	–	–	20 wt%	–	1.2	~0.4	~100[a]	1.86	–	410
PANI-MMT [206]	SO	~100 nm	–	30 wt%	–	3	< 30	$7.19 \cdot 10^3$	–	98	80000
DMSO-doped kaolinite [190]	SO, 500 mPa s	2 µm	–	30 wt%	–	3	< 20	~600	–	< 90	6670
Glycerol-doped kaolinite [190]							< 30	~560	–		6220
MMT/22.7 wt% TiO_2 [146]	SO, 500 mPa s	2 µm	–	25 vol%	~8	4	< 7	$8 \cdot 10^3$	–	< 85 (at 5 vol%)	200000
MMT/TiO_2 [207]	SO, 500 mPa s	Tens of µm	–	20–30 wt%	–	4		~$2.8–4.3 \cdot 10^3$[a]	–	–	48000
CTAB-MMT [208]	SO, 200 mPa s	< 45 µm	1.035	25 wt%	–	0.9	< 28	< 40[a]	–	52	33
Polyindole/CTAB-MMT [208]			0.931							82	22
Laponite [79]	SO, 100 mPa s	Thickness of 1 nm and diameter of 30 nm	2.65	48.8 vol%		3.5		~800	1.72–1.94		6150
CTAB-laponite [89]	SO, 100 mPa s	200–400 nm	2.65	10 wt%	~1.7	3.5	18	~230	–	~55	7660
Kaolinite/CMS [191]	SO, 20 cSt	< 2 µm	–	25 wt%	–	3.7	–	~700	–	–	260
Fluorohectorite [90]	SO, 100 mPa s	< 50 µm	–	10 wt%	–	1	80	~25[a]	1.66–2.12	–	270
Polyindole/CTAB-MMT 92.8%/7.2% [189]	SO, 1 Pa s	4.6 µm	1.08	25 vol%	–	3	–	10.5	–	95	0.3
PANI/HNT [222]	SO, 30 cSt	< 1 µm	–	15 wt%	–	4	–	~100	2	–	440
Kaolinite [93]	SO	0.2–4.0 µm	–	5 wt%	–	2	15	18.5 ± 1.3[a]	–	–	298
HNT [93]		Diameter of 50–100 nm					305	6.3 ± 0.1[a]			95
Polypyrrole/HNT [49]	SO, 30 cSt	Diameter of 30–70 nm and length of 1–3 µm	–	10 vol%	–	4	–	25 (Bingham) 33.6 (CCJ)	1.5 (< 2.5 kV) 1 (> 2.5 kV)	–	1290
Laponite [195]	SO, 500 mPa s	21 ± 4 nm	–	10 vol%	–	0.8	–	< 10[a]	1.17 ± 0.13	–	1240
CTAB-laponite [195]						2.6		~200[a]	1.27 ± 0.04		7690
MMT [195]		500 ± 200 nm				2.2		~130[a]	1.54 ± 0.15		5900
CTAB-MMT [195]						2.5		~80[a]	1.49 ± 0.03		3200
Sepiolite [195]		Length of 1100 ± 600 nm and diameter of 27 ± 7 nm				1.6		~100[a]	1.37 ± 0.04		6240
CTAB-sepiolite [195]						2.5		~510[a]	1.97 ± 0.12		20400
Sepiolite [223]	SO, 194 mPa s	Length of 1700 nm and diameter of 40 nm	–	5 wt%	1.84	3	–	~11.2	–	75	68
				10 wt%	4.52			~21.5		93	56
				15 wt%	–			~70.0		100	2
HNT [64,221]	SO, 100 mPa s	Diameter of 20–20 nm and length of 50 nm–2 µm	–	8 wt%	~0.5	7	–	~5	–	–	429
MMT (Cloisite Na^+) [58,62]	SO, 150 mPa s	Thickness of 2–100 nm and length of 200 nm–10 µm	–	8 wt%	5.6	2.5	–	< 20[a]	–	–	1000
MMT (Cloisite 30B) [58,62]					1.2	7		11.7			3750
MMT (Nanofil SE3010) [58,62]					0.8			3.8			2500
Chitosan/bentonite 1%/99% [192]	SO, 1 Pa s	14 µm	1.158	25 wt%	–	1	–	150[a]	2	~40	36

[a] Yield stress values are given at maximum electric field.

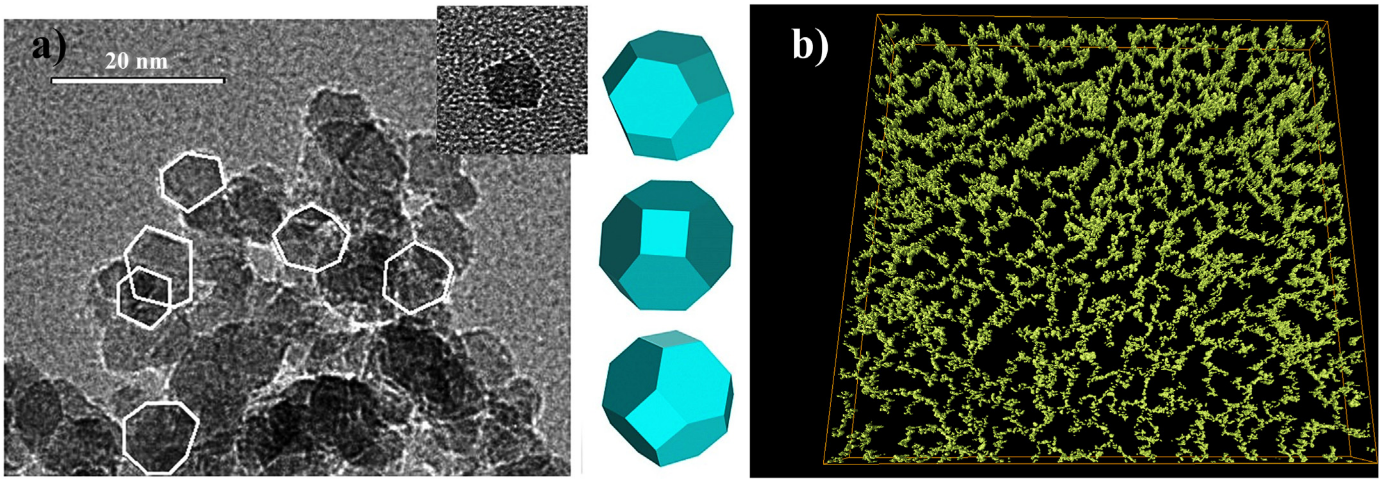

Fig. 12. TEM image of DND particles on the graphene membrane (a) [255]. The individual particles denoted by white lines and the views of regular truncated octahedrons are shown for comparison. The insert shows individual nanodiamond particle. The cryo-electron tomography 3D reconstruction of 1 wt% hydrogenated DND hydrosol (b) [258]. The native structure of hydrosol reveals the percolation network formation. Adapted from the studies by Dideikin et al. and Kuznetsov et al. [255,258] with permission from Elsevier.

glycol gels (400 Da) filled by graphene particles [236]. Gels were prepared by mixing the graphene with the polymer through an aqueous suspension, followed by distillation of the water. The final concentration of water in the gel was not controlled. Despite the low filler concentration (2 wt%), it should be noted, that the gel has a high viscosity of 5 kPa*s and a yield stress of about 1.5 kPa without an electric field. The yield stress reaches about 13 kPa under electric field of 4.8 kV/mm.

Several recent studies report about ERFs filled by graphene oxide composites with poly(methyl)- and poly(butyl)methacrylate [63,237]. The both fillers show improved ER response and reveal conduction mechanism of ER effect. The yield stress reaches several tens of Pa at relatively low field strengths of up to 3 kV/mm.

Another modern carbon material that has found wide application is carbon nanotubes (CNTs) [238], discovered by the Japanese scientist Iijima in 1991 [239]. Nanoscale, high anisometry of particles, facile synthesis, and unique electrophysical properties make CNTs promising filler for ERFs. However, suspension of CNTs exhibits a negative ER effect under direct electric field. Thus, particles migrate to one of the electrodes and electrophoresis occurs [240,241]. Therefore, more often multi-walled CNTs in electrorheology are used as composite materials, and nanotubes can act both as a base (core) [130,242,243] and as a surface agent (shell) [40,244]. The CNTs shell of filler particles significantly increases the sedimentation stability of fluids, due to prevention of direct contact between the initial nanoparticles and, as a result, minimization of aggregation and sedimentation. The presence of CNTs on the particles surface introduces effective repulsive interactions. These additional interactions can also lead to an increase in the energy barrier that must be overcome in order to achieve the ER effect and to form columnar structures under electric field. Thus, it is possible to obtain more stable ERFs finding a balance between the number of nanotubes on the particle surface and the sedimentation stability of suspensions [128]. The adsorption of CNTs to the surface of titanium dioxide particles enhances the filler's ER activity and allows reaching yield stress of several kPa at various electric fields (up to 5 kV/mm). However, the concentration of particles in such fluids remains high of 55 wt%. Furthermore, the modified filler exhibits lower leakage current density resulting in more attractiveness for practical applications [174].

In recent years, the growing interest in the attractive applications of nanodiamonds produced by detonation synthesis (DND) from carbon explosives is observed [245–247]. The method of detonation synthesis is quite cheap, and the world stock of explosives is huge, allowing obtains DNDs on a commercial scale. Among the applications of nanodiamonds, including detonation ones, it should be especially highlighted biomedicine and polymer composites [248]. A certain obstacle for application such DNDs is their agglomeration. Typically, sizes of solid aggregated particles produced by industrial synthesis vary from several hundred nanometers to units of micrometers, although according to XRD data the individual crystalline region of DND is about 4–5 nm [249,250]. The first method of DND disaggregation was the mechanical grinding by zirconium oxide balls [251,252]. This deagglomeration method is widely used by many researchers until now; however, it leads to graphitization of the DND surface and to the appearance of an onion-like sp^2 phase. Later, another method based on the thermal treatment of the DND powder and subsequent centrifugation of their aqueous dispersions was proposed for producing individual DND particles [253]. As a result, a stable hydrosol of individual particles with sizes of 4–5 nm can be obtained. Further, the optimization of the deagglomeration process allowed obtaining DNDs without the sp^2 phase on the surface. If annealing is carried out in the air, then in water the particles have a negative zeta potential, and if in a stream of hydrogen, then potential is positive. Dideikin et al. showed that, according to theoretical calculations [254], faceted particles are formed in this way (Fig. 12a) [255]. It is interesting to note that the electrostatic potential of such particles will be spherically asymmetric, compared with ball-milled particles having a surface sp^2 phase. Moreover, the hydrosols of such annealed DNDs are sedimentally stable for an unbelievably long time (tens of months) and exhibit unusual rheological behavior, namely, the yield stress even at 1 wt% concentration and a pronounced sol-gel transition at 5–6 wt% was revealed [256,257]. Such behavior of DND hydrosols was explained from the standpoint of the DLVO theory due to the interaction between individual particles with the chain formation. Recently, the percolation network formation from DNDs has been directly proved by cryo-electron microscopy and tomography (Fig. 12b) [258]. Apparently, the percolation network is dynamic, which explains the high sedimentation stability of hydrosols. Moreover, it is noted that the structuring processes are different for carboxylated and hydrogenated DNDs. Such behavior was only recently explained in the framework of the continual model and is associated with the

Table 3
Comparative analysis of ERFs filled by carbon-based materials. Yield stress values are from experimental data unless an appropriate rheological model is specified. Adapted from the study by Dong et al. [235] with permission from MDPI (Open access).

Materials		Medium type and viscosity	Size	ρ_{filler}, g/cm^3	Conc.	$\Delta\varepsilon_{fluid}$, rel.un.	E_{max}, kV/mm	J_{max}, μA/cm^2	Yield stress at 3 kV/mm or E_{max}, Pa	Slope $\log(\tau_0) = f(\log(E))$	Sedimentation ratio, %	*Eff*, rel.un.
0.49% MCNT-BTRU [128]		SO, 20 cSt	Roughly spherical particles of 500 nm	–	25 wt%	–	5	–	$>194\cdot10^3$	–	Up to 92	332
Silica-coated MCNTs [130]		SO, 50 cSt	Core/shell tubes with diameter of 60–70 nm and lengths of 2–3 μm	–	5 vol%	0.56	2	–	6.5[a]	1.02	95.8	510
Titania-coated MCNTs [130]						2.37			24.2[a]	1.42	92.6	232
Silica-coated MCNT [242]		SO, 50 cSt	Core/shell tubes with diameter of 20–40 nm and shell thickness of 40 nm	–	20 vol%	0.35	4	1	~410	1.49	~88	755
Silica/titania-coated MCNTs [242]						0.64		16	~820	1.65	~80	1610
Titania-coated MCNTs [242]						0.62		24	~1100	1.71	~74	1870
Titania/silica-coated MCNTs [242]						0.38		10	~480	1.50	~83	872
Nanodiamonds [262]		SO, 1000 cSt	~9.2 nm	3.5	8 wt% (2.19 vol%)	–	4	–	~130	0.8–1.3	–	18
Hydrogenated DNDs [263]		SO, 105 mPa s	4–5 nm		4 wt% (1.10 vol%)		7	–	~34	1.3	96	1410
Carboxylated DNDs [263]									~5	–	90	408
Hydrogenated DNDs [100]		Mineral oil, 135.1 mPa s	3 nm	3.5	3 wt%	~5	6	–	~20	1.0 ± 0.1	74	211
			5 nm			~14			~27		95	100
GO [131]	Ball-milling time = 1 h	SO, 100 cSt (PMPS)	Sheets with diameter of 0.7–1.5 μm and thickness of 1.5–3 nm	1.78	3 vol%	–	1	< 10	~60[a]	2	~80	170
	2 h		Sheets with diameter of 0.4–0.7 μm and thickness of 3–8 nm	1.56					~80[a]		~97	210
	3 h		Sheets diameter of 0.2–0.3 μm and with thickness of 4–10 nm	1.53					~45[a]		~98	150
GO [162]		SO, 100 cSt	Sheets with size of < 100 μm	1.78	5 wt% (2.77 vol%)	0.45	2.5	–	~30	–	–	2390
GO/TiO_2 [425]		SO, 30 cSt	Sheets with size of tens of μm	–	15 wt%	–	3	–	160 (CCJ)	–	–	3550
GO-wrapped titania [426]		SO, 50 mPa s	Spheres with diameter of 800–1000 nm	–	10 vol%	2.7	3	1	600	1.47	45	1000
TiO_2/GO [427]		SO	Spheres with diameter of 400–500 nm	1.978	37 vol%	–	5	< 10	$8\cdot10^3$	–	50	14
Fe_3O_4/GO [19]		SO, 100 cSt	Spheres with diameter of 10 nm	–	15 wt%	–	3	–	100	1.5	–	2660
Fe_3O_4/SiO_2/GO [428]		SO	Sheets with size of 20–40 nm	–	25 vol%	–	3		15	1.5	82	200
GO/Al_2O_3 [429]		SO, 30 cSt	Spheres with diameter of ~600 nm	–	20 wt%	–	5	–	52 (CCJ)	1	–	290
GO-wrapped silica [173]		SO, 100 cSt	Spheres with diameter of 200 nm	–	5 wt%	0.71	5	–	~100	2 (<1 kV) 1.5 (>1 kV)	–	20000
			Rods with diameter of 200 nm and lengths of 1 μm			1.40			~250			40000
			Rods with diameter of 200 nm and lengths of 4 μm			2.78			~800			133300
GO-coated silica spheres [430]		SO, 100 cSt	Spheres with diameter of 270 nm	2.03–2.67	3 wt%	0.94–2.65	4	–	9.88–30.18	–	45–65	3100

(*continued on next page*)

Table 3 (*continued*)

Materials	Medium type and viscosity	Size	ρ_{filler}, g/cm^3	Conc.	$\Delta\varepsilon_{fluid}$, rel.un.	E_{max}, kV/mm	J_{max}, μA/cm^2	Yield stress at 3 kV/mm or E_{max}, Pa	Slope $\log(\tau_0) = f(\log(E))$	Sedimentation ratio, %	*Eff*, rel.un.
GO/SiO_2 [431]	SO, 50 cSt	Sheets with lengths of 1–5 μm and thickness of 30 nm	–	3 vol%	0.9	2	–	~680[a]	–	–	5650
Si-GO [432]	SO, 30 cSt	Sheets with size of several μm	–	9 wt%	2.3	3	–	264 (CCJ)	2	–	974
GO-Si [433]	SO	Core/shell micro spheres	2.00	15 wt%	0.06	2.5	–	61.6[a] (CCJ)	1.5	–	654
GO/PANI [434]	SO, 30 cSt	Sheets with size of < 20 μm	–	10 vol%	–	2	–	630[a] (CCJ)	2 (< 0.7 kV) –1.5 (> 0.7 kV)	–	31500
GO/co-PANI [435]	SO, 30 cSt	Sheets with size of < 100 μm	–	10 wt%	1.98	5	–	~200	2 (< 1 kV) 1.5 (> 1 kV)	–	16660
GO/poly(2-methylaniline) [436]	SO, 30 cSt	Sheets with size of < 20 μm	1.43	10 vol%	–	2	–	35[a] (Bingham)	1.5	85	1750
Poly(p-phenylenediamine)/GO [437]	SO	Sheets with lengths of ~150 nm and thickness of ~4 nm	–	8 wt%	1.71	2.5	–	~100[a]	–	–	245
POSS-GO [438]	SO, 50 cSt (PMPS)	Sheets with lengths of ~1 μm and thickness of ~3.5 nm	–	~3 wt%	–	3	< 0.5	~600	1.53	–	2660
Polystyrene/GO [439]	SO, 30 cSt	Core/shell spheres with diameter of < 300 nm	–	10 vol%	2.54	2	–	114[a] (CCJ)	1.5	–	5700
PMMA/GO [237]	SO, 50 cSt	Spheres with diameter of ~200 nm	1.27	10 vol%	0.41	2.5	–	~30[a]	1.5	–	1200
PGMA/GO [440]	SO, 50 cSt	Roughly spherical particles of 200–300 nm	1.41	10 vol%	1.6	2	–	10.5[a] (Bingham)	1.5 (< 0.8 kV) 1 (> 0.8 kV)	–	520
GO-PGMA [441]	SO, 200 mPa s	Sheets with size of < 55 μm	–	10 wt%	1.81	2.5	–	~100[a]	1.46	90	4000
GO-PBMA [63]	SO, 200 mPa s	Sheets with lengths of hundreds nm and thickness of ~5 nm	2.21	5 wt%	1.37	2.5	–	110[a]	1.48	60	8800
GO-PHEMATMS [442]	SO, 194 mPa s	Sheets with size of < 45 μm	–	5 wt%	2.11	2.5	–	200[a]	1.44	90	16000
GO/polypyrrole [280]	SO, 50 cSt	Micro sheets	–	8 wt%	–	1	–	~40[a]	1.35	–	5000
GO/polypyrrole/poly(ionic liquid) [280]					0.38	2		~250[a]			6240
Nano graphene [236]	Water-PEG (400)	Sheets with size of 100–400 nm up to 2 μm	–	2 wt%	–	4.8	–	~10·10^3	–	–	77

[a] Yield stress values are given at maximum electric field.

ble separation of charge in hydrogenated particles and localization of charge in carboxylated ones [259].

The rheological properties of DND suspensions in silicone oil also attract the attention of researchers to their further application as new lubricants due to a possible improvement in tribological characteristics [260,261]. The secondary agglomeration processes substantially depend on the dielectric characteristics of the medium and the filler, and it is obvious that it significantly differs for water and oil; therefore, the sedimentation stability of DND-oil suspensions become of great importance. Recently, it was shown that the processes of secondary DND aggregation in a polydimethylsiloxane medium are qualitatively similar to the behavior of their hydrosols. Moreover, the electrophysical characteristics of suspensions substantially depend on the structural organization of DNDs and related with concentration as well [99]. It is surprising that only several studies are devoted to ERFs filled by nanodiamonds. Suspensions of nanodiamonds in polydimethylsiloxane exhibit ER activity even at low concentrations of less than 8 wt% (less than ~ 1 vol%). The yield stress reaches about 233 Pa at 8 wt% and electric field of 4 kV/mm. Moreover, the proportionality of the yield stress on electric field is close to unity. The authors associate such result with the giant mechanism of the ER effect [262]. Recently, the effects of the DND surface functionalization and size on the ER behavior of suspensions were considered. Hydrogenated DNDs in silicon oil reveal ER effect, while DND particles with predominantly carboxyl groups exhibit electrophoresis under electric field [263]. The reasons for such behavior remain ambiguous. However, a small amount of water may be critical for the behavior of particles functionalized by groups capable of dissociation under electric field. Thus, the type of functional groups on DND surface plays a crucial role in the behavior of ERFs. Additionally, the role of DND particles size (3 and 5 nm) on the ER properties of their suspensions in mineral oil has been revealed [100]. The yield stress and sedimentation stability for 5 nm DNDs suspensions turned out to be higher than for 3 nm particles due to formation of less branched agglomerates from 3 nm DNDs in suspension. Nanoscale, high dielectric permittivity of particles, simple way to surface modification, makes DNDs promising filler for ERF along with other nanocarbons, namely graphene and nanotubes. The parameters of ERFs filled by various types of nanocarbons are given in Table 3.

4.2. Polymer fillers

Anhydrous semiconducting polymers are widely used to create high performance ERFs [264,265]. The common properties of these polymers are electronic conductivity due to the presence of a conjugated π-system, thermal stability, as well as facile synthesis. Today, ERFs filled by polythiophene [266], polypyrrole [267,268], poly-p-phenylene [269], polyphenylenediamine [270], polyindol [43,271,272] etc. [44,273,274], as well as their derivatives were obtained. PANI [275] deserves special attention and will be discussed additionally below. Here, we consider several recent studies of particular interest.

Poly(ionic liquid)s is a type of functional polyelectrolytes and can exhibit strong ER effect in dry state. Crosslinking is a general strategy to improve the mechanical and thermal properties of polymer. Thus, team of scientists led by Yin studied the role of counter ion type, alkyl spacer length on ionic transport and consequently ER effect of linear and self-crosslinked poly(ionic liquid)s particles [276,277]. It was shown that ER activity of linear poly(ionic liquid)s can be varied by type and nature of counter ions and does not depend exclusively on the ion size. In case of self-crosslinked poly(ionic liquid)s the ER effect increases with the length of alkyl spacer because of increased ion mobility and induced interfacial polarization. However, the operating temperature range narrowing due to enhanced leaking current density. It also should be noted, that composites of poly(ionic liquid)s with PANI [278,279] and polypyrrole-coated graphene oxide [280] were considered as fillers as well.

Recently, novel metal-containing polyimides particles as disperse phase of ERFs were considered [149]. Thus, Na-containing polyimide reveals ER activity in silicon oil. ERFs up to 15 wt% particles content were studied. Typically, ER properties increase with concentration. The behavior of 10 wt% suspension was considered in more detail. It is important, that at this concentration the fluid has a yield stress of the order of 1 Pa. Therefore, the efficiency of the filler was evaluated by the ratio of the relative increase of the yield stress value under electric field to the value without field. Nevertheless, the yield stress reaches 80–90 Pa at electric field of 4 kV/mm. However, metal-containing polyimides are both difficult to synthesize and to characterize.

One of the interesting fillers is the functionalized organosilicon particles of the polyhedral oligomeric silsesquioxane (POSS) with ladder structure [281]. The main feature of such polymer is the same chemical nature with dispersion media (polydimethylsiloxane). Thus, an empirical rule for predicting solubility, that is, "like dissolves like" (lat."Similia similibus solventur"), should result in good dispersibility of the filler. Group of McIntyre et al. provides a series of studies of ERFs filled by POSS. Variously functionalized POSS structures were considered, namely, modified by tri-sulfonic acid ethyl groups (sulfonated), octa-isobutyl groups [282], and cyanopropyl groups [283,284], as well as mixtures of sulfonated POSS with polystyrene [285]. POSS-based ERFs reveal pronounced ER effect even at low fillings (up to 10 wt%). The filler with octa-isobutyl fragments does not show an ER effect. The authors attribute this result to its dielectric inactivity compared to sulfonated POSS, which, in turn, demonstrates a significant ER response. Moreover, the dependence of the ER activity on the concentration passes through a maximum at 7 wt%. The reduced ER effect at higher fillings was not associated with the dielectric characteristics of the suspensions, but was explained by steric effects during the dipole organization under electric field in the time range of the experiment. However, the yield stress of suspensions at such low fillings reaches about 100 Pa at electric field of 2 kV/mm [282]. The mixture of polystyrene with small sulfonated POSS content (~1 wt%) exhibited more intense ER effect. Thus, the increase in an order of magnitude was observed compared with pristine 10 wt% sulfonated POSS-based ERF. Obviously, polystyrene particles make a major contribution to the behavior of the fluid, and POSS acts as an activator [285]. The cyanopropyl functional POSS reveal ER activity as well. However, the operational characteristics, such as yield stress, storage and loss moduli under electric field were lower than for sulfonated POSS. It is interesting to note that the effectiveness of cyanopropyl functional POSS depends on the number of substituents in the ladder structure. The more cyanopropyl substituents, the higher the ER activity (up to eight groups). The range of applied electric field strengths up to 4 kV/mm was expanded as well [283,284]. The main parameters of ERFs filled by various polymers are presented in Table 4.

4.2.1. Polyaniline

PANI is one of the most promising polymer fillers for ERF, attracting much attention due to its facile synthesis, good stability, low material cost, and conductivity control [275]. PANI particles can be obtained by various synthesis strategies, such as bulk synthesis [286], electrospinning [287], interfacial polymerization [288], self-assembly processes [289], and polymerization in solution [290]. Moreover, depending on the synthesis conditions and additives, it is possible to obtain particles of different morphology, namely

Table 4
Comparative analysis of ERFs filled by polymer-based materials (except PANI). Yield stress values are from experimental data unless an appropriate rheological model is specified.

Materials		Medium type and viscosity	Size	ρ_{filler}, g/cm^3	Conc.	$\Delta\varepsilon_{fluid}$, rel.un.	E_{max}, kV/mm	Yield stress at 3 kV/mm or E_{max}, Pa	Slope $\log(\tau_0) = f(\log(E))$	Sedimentation ratio, %	*Eff*, rel.un.
$FeCl_3$/polypyrrole 1/2 [267]		SO, 100 mPa s	< 20 μm	–	15 wt%	–	1.5	< 300[a]	–	–	13330
Poly(p-phenylene) [269]		SO, 30 cSt	20 μm	–	10 wt%	–	3	~5	–	–	197
$FeCl_3$-doped poly(p-phenylene) [269]								~60			2000
Protonated PPDA (hydrochlorides) [270]		SO, 200 mPa s	5–10 μm	1.43	10 wt%	0.35	2.5	~70[a]	–	–	89
PPDA (bases) [270]				1.26		0.45		> 100[a]			129
PI-*co*-PTBMA-Li [273]		SO	~45 μm	–	25 wt%	–	1	~8[a]	–	33–5	60
Poly(o-toluidine) [274]		SO, 500 mPa s	10 μm	–	20 wt%	–	5	79.94 (Bingham)	2	–	166
TSA-poly(o-toluidine) [274]			15 μm					366.20 (CCJ) 454.64 (Bingham)			617
Polyimide-Na [149]		SO, 400 cSt	Microparticles	–	10 wt%	–	4	~70	–	–	2330
Tris sulfonic acid ethyl-POSS [282]		SO	–	–	10 wt%	~1.1	2	~8[a]	1	–	395
8 CN-POSS [283,284]		SO	1–200 μm	–	8 wt%	–	4	~27 (Herschel-Bulkley)	0.7	–	4
1 CN-POSS [284]			5–100 μm					~16 (Herschel-Bulkley)	0.15		1
Tris sulfonic acid isobutyl-POSS/polystyrene [285]		SO	~41 μm	–	1 wt% POSS 10 wt% PS	0.92	2	~100	2	–	223
Polyindole [43]		SO, 50 cSt	300–400 nm	1.47	5 vol%	1.01	2.5	166[a] (CCJ)	1.5	79	13300
Polyindole [271]		Light crude oil	90 nm	–	5 vol%	–	1.2	44.43[a] (Herschel–Bulkley) 41.41[a] (Bingham)	–	–	2450
		Heavy crude oil					0.3	2.06[a] (Herschel–Bulkley) 2.17[a] (Bingham)			1310
Polyindole at various oxidant-to-indole mole ratio [272]	1	SO, 194 mPa s	Globular-like shape, 1–2 μm	1.29	10 wt%	1.34	3	100	2.00	–	1330
	1.75			1.32		1.65		20	1.55		330
	2			1.31		1.20		10	1.46		182
	2.25			1.29		0.88		6	1.52		122
	2.5			1.33		0.77		5	1.48		108
SiO_2- poly(diphenylamine) [168]		SO, 100 mPa s	Core/shell spherical particles of 1 μm and shell of ~30 nm	2.65	10 vol%	–	2.5	10.5[a] (Herschel–Bulkley)	1.5	–	416
Triazine-based polymer [44]		SO, 100 cSt	1.0–2.5 μm with pores of 1–2 nm	1.46	5 vol%	2.44	2	156[a] (CCJ)	2	–	15600

[a] Yield stress values are given at maximum electric field.

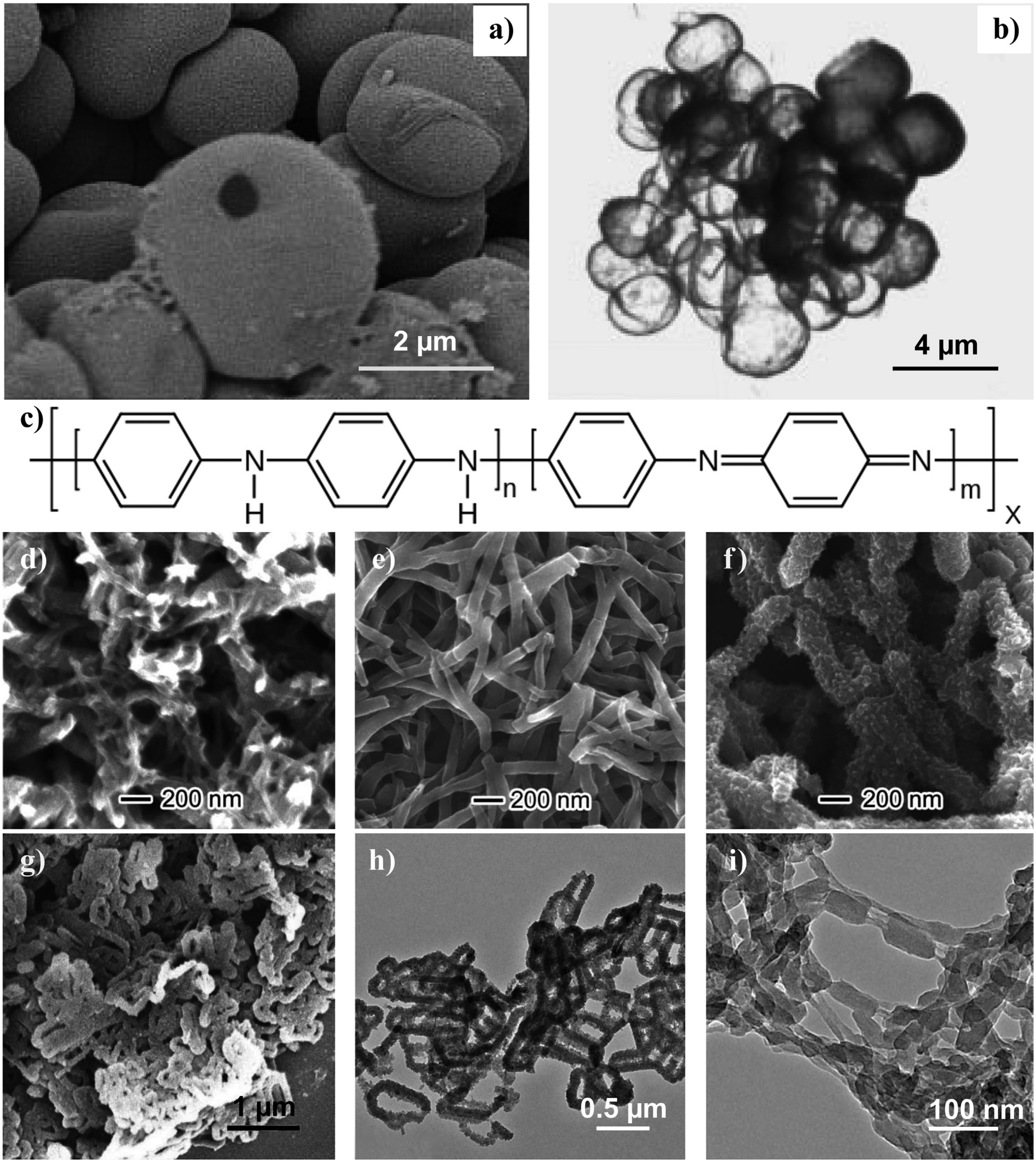

Fig. 13. SEM and TEM images of PANI particles with various morphology (a,b,d-i) and chemical structure (c). Particles were obtained at various conditions: (a,b) hollow microspheres synthesized by salicylic acid doping at acid/PANI ratio of 0.1; (d,e,f) PANI nanofiber seeds of ~120 nm in diameter obtained by bulk synthesis in presence of perchloric acid. Images show the effect of stirring, where (d) no stirring and (f) under rapid stirring; (g,h) nanocliplike PANI particles synthesized with CTAB and polyvinyl pyrrolidone at content ratio of 1:3; (i) PANI nanofibers produced via rapidly mixed polymerization. Fibers are soft and easy aggregated together. Adapted from the studies by Zhang et al., Wang et al., Dan et al., and Wen et al. [289] (a,b), [309] (i), [286] (d,e,f), and [296] (g,h) with permission from Wiley Periodicals, American Chemical Society and Elsevier, respectively.

spherical micro- and nanoparticles, nanofibres, and nanotubes, etc. (Fig. 13).

The high PANI conductivity in the ERFs leads to high leakage current density; therefore, its value is controlled by a simple protonation process [291,292] or doping [293], as well as monitoring the conditions of the synthesis [294]. The PANI degree of protonation has a significant effect on the ERFs behavior. For example, the yield stress of suspensions with PANI particles processed at different pH values goes through a maximum at pH = 8 (electric field strength was 2.8 kV/mm) [292]. The effect of doping agents was widely studied as well. Thus, PANI nanofibers doped by indole-2-carboxylic acid in silicone oil revealed a yield stress of 200 Pa at 3.5 kV/mm and a concentration of 10 vol%. That exceeds the ER activity of PANI nanofibers doped by hydrochloric acid. The authors attribute such result to the higher polarizability of indole-2-carboxylic acid [293]. The level of PANI conductivity can be controlled by coatings of various non-conductive polymers. PANI coated by melamine-formaldehyde resin exhibits an ER effect allows to adjust the properties without pH control [295]. It is important, that the polymerization temperature also significantly affects the ER response of PANI suspensions. Thus, in the study by Choi et al. [294] effect of temperature in the range from 5 to minus 10 °C was studied. The highest yield stress was demonstrated by suspensions of particles obtained at minus 10 °C, due to higher molar mass and higher conductivity of the longer main chains of polymer molecules. At the same time, the dielectric permittivity and dielectric loss coefficient had maximum value. Such result was consistent with the ER properties. Mrlik et al. found that ERFs filled by oligomeric aniline particles exhibit greater ER activity than PANI-based suspensions [290]. ER characteristics depend on the molar concentration of the methanesulfonic acid during oxidation. Particles with low conductivity were obtained at a concentration of 0.1 M. However, such particles were not capable of forming sufficiently long chains in suspension under electric field. The optimum acid concentration was determined as 0.5 M providing the ER material with suitable conductivity, dielectric characteristics, and the ability to create compact structures in suspension under electric field.

The size and shape of PANI particles are also significant factors affecting ER activity. Yin et al. showed that the suspension of nanofibers exhibits the greatest ER effect [136]. The yield stress was in 2.5–3.0 and 1.3–1.5 times higher than in ERFs filled by spherical nano- and microparticles, respectively. The most likely reason is in the ability of nanofibers to form a stronger oriented structure under electric field. The flow curves analysis showed that the shear stress of the ERF filled by PANI nanofibres is stable over a wider range of shear rates (0.1–1000 s^{-1}) compared with suspensions of nano- and microparticles. Thus, suspensions of PANI nanofibres revealed high ER properties. PANI ellipsoid nanofibres can be obtained by oxidative polymerization in the presence of surfactant mixtures (Fig. 13g and h) [296]. The most stable nanofibers were obtained by adding a mixture of cetyl-trimethyl ammonium bromide (CTAB) (quat) and polyvinylpyrrolidone in ratios of 3:1 and 1:3, respectively. ERFs of these particles showed electrical conductivity of the order of 10^{-11} S/cm and yield stress of more than 200 Pa at electric field of 3 kV/mm and a concentration of 10 wt%. The effect of the oxidizing agent concentration on the morphology and conductivity of PANI nanofibers was studied as well [297]. This study confirmed effect of particles shape anisometry on ER response. Nanofibers provide high ER effect and their suspensions were more sedimentation stable compared to spherical PANI particles.

Hollow spherical PANI particles were obtained by synthesizing spherical polystyrene particles coated by a PANI layer with further dissolving of the polystyrene core in tetrahydrofuran [298,299]. Suspensions of PANI hollow capsules with a narrow particle size distribution (diameter of 200 nm and a layer thickness of 50 nm) showed high sedimentation and thermal stability. Their suspension in silicone oil was ER active and stable at different electric field strengths (current density at 3 kV/mm was 0.07 μA/cm^2). Moreover, several studies show that hollow PANI nanotubes exhibit higher ER response compared to spherical particles [300,301].

The properties of ERFs filled by various PANI copolymers have also been studied. Suspensions of PANI-*co*-o-ethoxyaniline and poly-2-dodecyloxyaniline particles in silicone oil revealed a significant ER response in concentration range of 10–30 vol%, namely, the yield stress reaches several hundred Pa at electric field of 3–4 kV/mm [302,303]. In another study, the yield stress calculated from the Seo–Seo model was 106 Pa at 2.5 kV/mm for suspension of monodispersed PANI-diphenylamine copolymer spherical nanoparticles dispersed in silicone oil at a concentration of 10 vol% [304].

A large number of studies are devoted to studying of PANI particles with a core/shell structure, PANI composites, and copolymers as well. Thus, p-phenylenediamine functionalized graphene oxide-*g*-PANI fibers show enhanced sedimentation stability. Nonetheless the ER performance was smaller than many other PANI-based ERFs [305]. Suspensions of PANI microparticles coated by polymers, poly(vinyl alcohol), poly(methyl methacrylate), and polystyrene, in chloroparaffin oil has a yield stress in the order of several kPa at a relativity low electric field strength up to 2.8 kV/mm and a filling degree of 25 vol% [134]. ERFs of such polarized nanosized particles can form a high-strength chain structure leading to high yield stress. ER properties of suspensions depend on coating thickness also. Several studies were devoted to the ER activity of PANI coated poly(methyl methacrylate) particles [42,306,307]. Modified monodisperse particles with an ideal spherical shape were obtained. The study of poly(methyl methacrylate) microspheres of various sizes coated by PANI layer with the same shell thickness showed that the ER effect of suspensions increases with increasing core size [306]. Poly(methyl methactylate) particles coated with poly-2-ethylaniline were later synthesized [42]. The yield stress of 10 vol% suspension reaches 458 Pa at electric field of 3.5 kV/mm. Recently, application of core–shell structured metal–organic framework with PANI [308] and poly(diphenylamine)/PANI [65] nanocomposites were reported. Both fillers reveal enhanced ER activity compared with pristine core materials.

PANI composites with various clays also showed increased ER activity [140,309,310]. The PANI composite with MMT leads to the formation of an extremely strong structure with a high yield stress of the order of several kPa at electric field of 3 kV/mm and the filler concentration of 20–30 wt% [140]. The ERFs filled PANI composites with laponite reveal superior properties compared to fluids filled by PANI nanofibers with kaolinite composite particles [309]. However, at the same time, the laponite-based particles show decreased efficiency than pure PANI nanoparticles or nanofibres, and PANI/MMT composite as well [310]. Such difference is probably due to the various particle morphology and polarizability of the composites.

PANI coated silica particles were also used as fillers for ERFs. It was shown that the PANI layer on the silica surface significantly increased the ER effect [311]. *In situ* polymerized PANI in mesoporous silica (MCM-41) as filler for ERF led to enhanced ER activity compare to pure PANI [312,313]. Moreover, the ER effect was higher than for SBA-15 silica matrix [314]. The shape effect of PANI coated mesoporous silica in silicone oil was considered as well [315]. The aspect ratio was varied as 1, 5, and 10. The low concentration of 3 wt

Table 5
Comparative analysis of ERFs filled by PANI-based materials. Yield stress values are from experimental data unless an appropriate rheological model is specified.

Materials	Medium type and viscosity	Size	ρ_{filler}, g/cm^3	Conc.	$\Delta\varepsilon_{fluid}$, rel.un.	E_{max}, kV/mm	J_{max}, μA/cm^2	Yield stress at 3 kV/mm or E_{max}, Pa	Slope $\log(\tau_0) = f(\log(E))$	Sedimentation ratio, %	*Eff*, rel.un.
Aniline oligomers [290]	SO, 108 mPa s	Flake-like particles in μm range	1.35	10 vol%	1.65	2	–	48.2[a]	1.5	–	236
PANI [292]	SO, 45–60 cSt (PDES) Chlorinated paraffin oil	Grains of < 38 μm	1.30–1.45	25 vol%	–	2.8	~18 ~180	$\sim 9 \cdot 10^3$[a] $\sim 11 \cdot 10^3$[a]	–	–	129000 157000
HCl doped PANI [293]	SO, 50 cSt	Fibers with diameter of 50 nm and length of	1.51	10 vol%	2.1	3.5	–	~120	2 (< 2.04 kV) 1.5 (> 2.04 kV)	–	800
Indole-2-carboxylic acid doped PANI [293]		1 μm	1.50		3.5			~200	2 (< 1.25 kV) 1.5 (> 1.25 kV)		663
PANI [294]	SO, 30 cSt	Grains of < 38 μm	1.3	20 vol%	~2	3	–	~1200	2	–	332
PANI [136]	SO, 50 mPa s	Fibers, diameter of 150–250 nm and length of 1–5 μm	–	15 wt%	–	4	–	> 750	1.26 ± 0.03	~95	183
		Particles, diameter of 150–250 nm						~200	1.43 ± 0.09	~92	53
		Particles, diameter of 3.2 μm						~600	1.46 ± 0.06	~70	220
PANI [298]	SO, 50 cSt	Hollow spheres with diameter of 200 nm and shell thickness of 50 nm	–	10 vol%	–	3	0.085	~60	–	~63 (0.35 vol%)	200
PANI [299]	SO, 50 cSt	Hollow spheres with diameter of 560 nm and shell thickness of 100 nm	1.32	10 vol%	1.65	2.5	–	27.3 (CCJ)[a]	1.5	–	1090
PANI [300]	SO, 50 mPa s	Hollow nanotubes with diameter of 150–200 nm and length of 1–5 μm	–	10 vol%	4.50	5	< 200	~400	~1.3	–	12000
PANI [301]	SO, 50 cSt	Hollow nanotubes with diameter of 230–270 nm and shell thickness of 130–150 nm	–	10 vol%	1.1	3	–	44.2 (CCJ)	1.5	–	1470
PANI [296]	SO, 485 mPa s	Clip-like nanofibers with diameter of 50–100 nm and length of 300–500 nm	–	10 wt%	~2.7	3	2.03	277.07 (CCJ)	1.6	–	112
PANI/MFR [295]	SO, 50 cSt	Core/shell particles in μm range	–	15 wt%	–	3	–	~110	–	–	4440
PANI/AMPSA [297]	SO, 50 cSt	Fibers with diameter of 34–49 nm	–	15 vol%	–	1.5	–	210 (CCJ)[a]	2	–	129
PMMA/PANI [306]	SO	Core/shell particles with core of 9 μm and shell of 261.4 nm	–	10 vol%	–	3	–	~60	–	–	2000
PMMA/PANI [307]	SO	Core/shell microspheres of ~11 μm	–	10 vol%	–	3	–	~60	–	–	2000
PMMA/poly(2-ethylaniline) [42]	SO, 100 cSt	Core/shell spherical particles of 1.6 μm	–	10 vol%	0.89	3.5	–	~313 (CCJ)	2	–	15300
PANI-*co*-o-ethoxyaniline [302]	SO, 30 cSt	Particles of 10–15 μm	1.23	30 vol%	–	4	–	~650	1.4–1.8	–	221
Poly(2-dodecyloxyaniline) [303]	SO, 10 cSt	Particles of 20 μm	–	20 vol%	–	4.2	–	~100	2.2	–	109
PANI-*co*-diphenylamine [304]	SO	Core/shell spherical particles of 60–100 nm	–	10 vol%	–	2.5	–	106 (Seo–Seo)[a]	1.5	–	4240

(*continued on next page*)

Table 5 (*continued*)

Materials	Medium type and viscosity	Size	ρ_{filler}, g/cm^3	Conc.	$\Delta\varepsilon_{fluid}$, rel.un.	E_{max}, kV/mm	J_{max}, μA/cm^2	Yield stress at 3 kV/mm or E_{max}, Pa	Slope $\log(\tau_0) = f(\log(E))$	Sedimentation ratio, %	*Eff*, rel.un.
PANI/poly(diphenylamine) [65]	SO, 100 cSt	Core/shell spherical particles of 60–120 nm and shell of 13 nm	1.3	10 vol%	1.64	2.5	45	92.1 (CCJ)[a]	1.5	–	3680
Poly-N-methaniline/MMT [140]	SO, 500 mPa s	Spherical particles of < 100 nm	–	20 wt%	~45	3	30	$2.52 \cdot 10^3$	–	97	15
PANI/kaolinite [309]	SO, 500 cSt	Sheet-like particles of 1–10 μm coated by fibers with diameter of 20 nm and length of 200–400 nm	–	10 wt%	–	3	< 10	~40	1–2	~89	1330
PANI/laponite [310]	SO, 50 cSt	Nanoparticles with rough surface in μm range	1.56	10 vol%	1.57	3	–	82.6 (CCJ)	2	–	2750
SiO_2/PANI [315]	SO, 100 cSt (PMPS)	Spheres with diameter of 100 nm	–	3 wt%	1.44	3	1.14	84.1	2 (< 1 kV) 1.5 (> 1 kV)	–	9330
		Rods with diameter of 100 nm and length of 500 nm			1.71		1.09	229.8			25500
		Rods with diameter of 100 nm and length of 1 μm			1.86		0.97	365.2			40600
PANI/SiO_2 [313]	SO	Twisted rectangular bar particles of < 10 μm	–	5 vol%	–	3	–	~65 (CCJ)	–	–	4330
TiO_2/PANI [316]	SO, 500 mPa s	Peanut-like shape with core/shell structure with sizes of < 1 μm	–	20 wt%	0.62	3	–	~250	2.37	–	103
$BaTiO_3$/PANI [317]	SO, 110 mPa s	Nanotubes with core/shell structure with diameter of 90–140 nm and length of 1–3 μm	–	10%	–	3	–	~100	1.8	~95	330
Fe_2O_3/PANI [318]	SO	Sheets with size of 2–5 μm	–	30 vol%	–	4.3	–	~2 10^3	–	–	13
Metal-organic framework/PANI [308]	SO, 500 cSt	Core/shell spheres of 200–300 nm	–	10 wt%	~0.8	3	8.12	~75 (CCJ)	1.4	–	97
PANI/Zn-Fe [20]	SO, 100 cSt	Raspberry-like core/shell particles in μm range	5.4	5 vol%-	–	1	–	~5[a]	1	–	980
Poly(ionic liquid)/PANI [279] HCl	SO, 50 cSt	Irregular particles with an average size of about 4–10 μm	–	20 vol%	3.02	3	–	~600	1.8	–	332
NH_3					3.58			~730	1.7		404
N_2H_4					3.12			~780	1.7		323
Poly(ionic liquid)/PANI [278]	SO, 50 cSt	Core/shell spheres of 1.17 μm with a thin shell	–	20 vol%	3.97	4	~4	~600	1.74	–	265
		Core/shell spheres of 1.45 μm with a thicker shell			5.52		~2	~700	2.07		2500
Poly(p-phenylenediamine)/GO-g-PANI [305]	SO, 100 cSt	Jellyfish-shaped particles of < 20 μm	1.40	8 vol%	2.22	3	–	26.2 (CCJ)	–	85	1090

[a] Yield stress values are given at maximum electric field.

% was chosen for studies. The yield stress at electric field of 3 kV/mm decreases as 365.2, 229.8, and 84.1 Pa, with decreasing aspect ratio of particles, respectively. Such result related with the influence of particle geometry on flow resistance and mechanical stability. Moreover, ERF filled by particles with higher aspect ratio showed improved dielectric properties with higher polarizability and shorter relaxation times. Therefore, the synergistic contribution of steric effects and dielectric properties results in an increased ER activity.

It should also be noted the PANI nanocomposites with TiO_2 [316], $BaTiO_3$ [317], and Fe_2O_3 [318] particles. Suspensions of the anisotropic nanocomposites based on titanium dioxide coated with PANI revealed a yield stress of the order of several hundred Pa at 3 kV/mm and 20 wt% filling degree [316]. ERFs of $BaTiO_3$/PANI composite nanotubes with a core/shell structure demonstrated high sedimentation and thermal stability, as well as a significant increase in yield stress, storage and loss moduli with electric field [317]. The PANI/Fe_2O_3 composite nanoparticles were considered as a novel dispersed phase of ERFs. The results showed that the yield stress reaches 10 kPa at electric field of 4.3 kV/mm and particles concentration of 30 vol%. Such value is much higher compared with suspensions containing pure PANI, Fe_2O_3 or their mixture as filler. The result confirms the synergetic effect of Fe_2O_3 and PANI on ER response [318]. Interesting study was performed by Kim et al. [20]. The authors synthesized raspberry-like core–shell particles with PANI core and magnetic zinc ferrite shell. The suspensions of such particles in silicon oil revealed ER and magneto-rheological effect, due to both electric and magnetic activity of the filler. Thus, the transition from a viscous to an elastic state can be achieved due to the action of one of the fields (magnetic or electric) and demonstrates the fundamental possibility of creating materials with improved characteristics due to the simultaneous application of two different external fields.

Thus, PANI is a semiconductor polymer with special physical properties finding its application as filler for ERFs. In addition to facile ways of the particles' conductivity controlling, the size and shape of the particles play an important role affecting the ER activity. Numerous studies of PANI-based ERFs confirm the promising of fillers with a high aspect ratio. Conductive PANI nanoparticles coated with a non-conductive shell can provide suspensions with high ER activity and low current density in electric field. Suspensions of such particles can form a very strong structure under electric field resulting in a high yield stress. Recent studies show the promise of PANI intercalated with nanoscale inorganic material. High-performance PANI composites open up the possibility of creating a new class of polymer and composite fillers for ERFs. The main properties of PANI-based ERFs are summarized in Table 5.

4.2.2. Biocompatible materials

Other special types of polymer fillers are biocompatible, biodegradable, and renewable natural polymers belonging to the class of polysaccharides – chitin, chitosan, cellulose, and starch [319]. Such compounds do not have a conjugated π-system, and, most likely, the presence of side polar groups is a key factor in particle polarization under electric field leading to the ER effect. The ER behavior of cellulose-based suspensions was reviewed in the

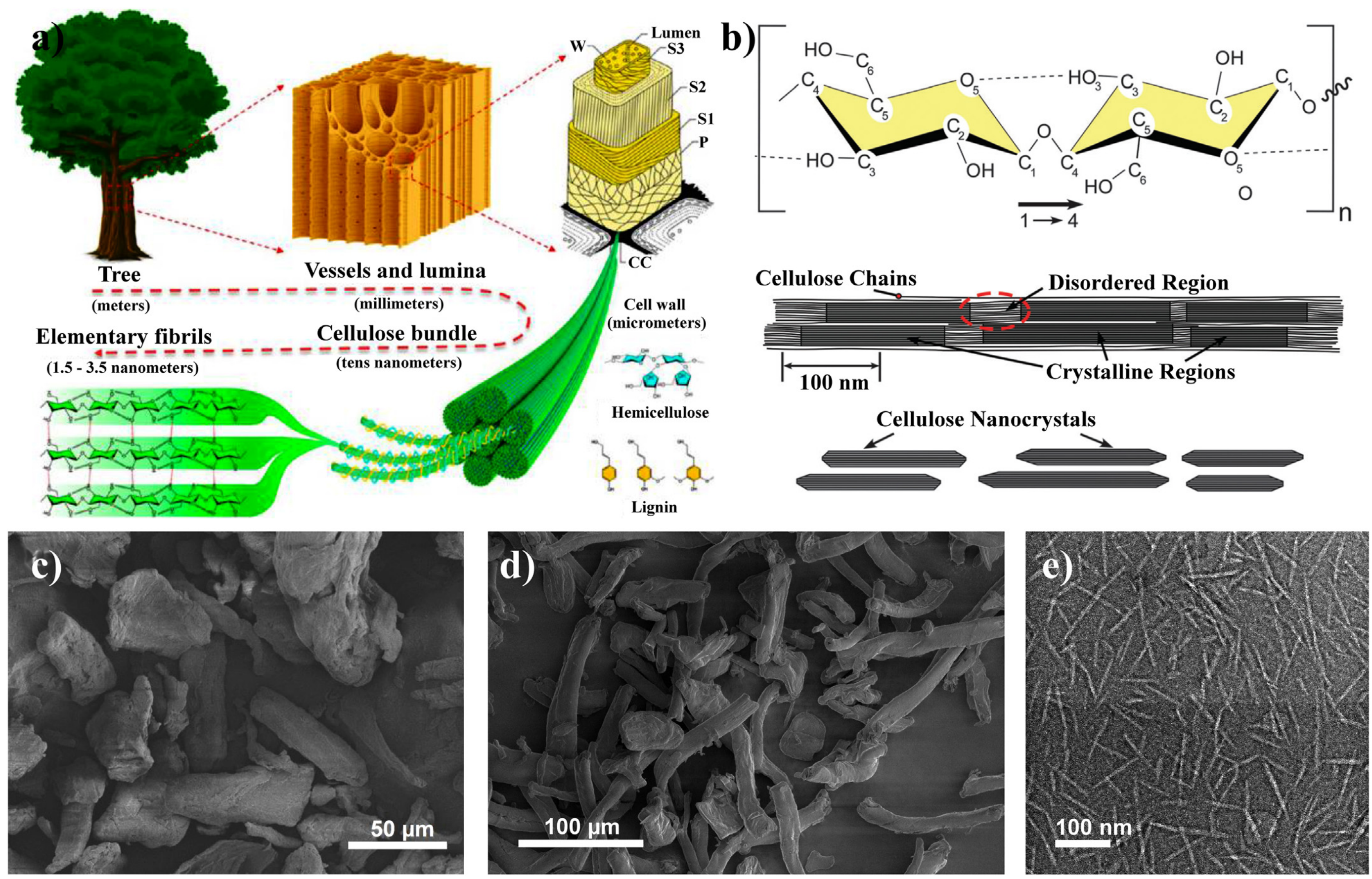

Fig. 14. Illustration of the hierarchical structure of wood, from macroscopic to molecular scale (a) [322]. Scheme of single cellulose chain repeat unit and intrachain hydrogen bonding (dotted line), model of cellulose microfibrils containing the crystalline and amorphous regions and cellulose nanocrystals (b) [321]. SEM images of cellulose microfibers obtained from wood of various sources (e,d) and TEM image of nanocrystalline cellulose with high aspect ratio (e) [340,366]. Adapted from the studies by Choi et al., Chen et al., Liu et al., Kuznetsov et al. [321,322,340,366] with permission from American Chemical Society, Royal Society of Chemistry, MDPI (Open Access) and Springer Nature, respectively.

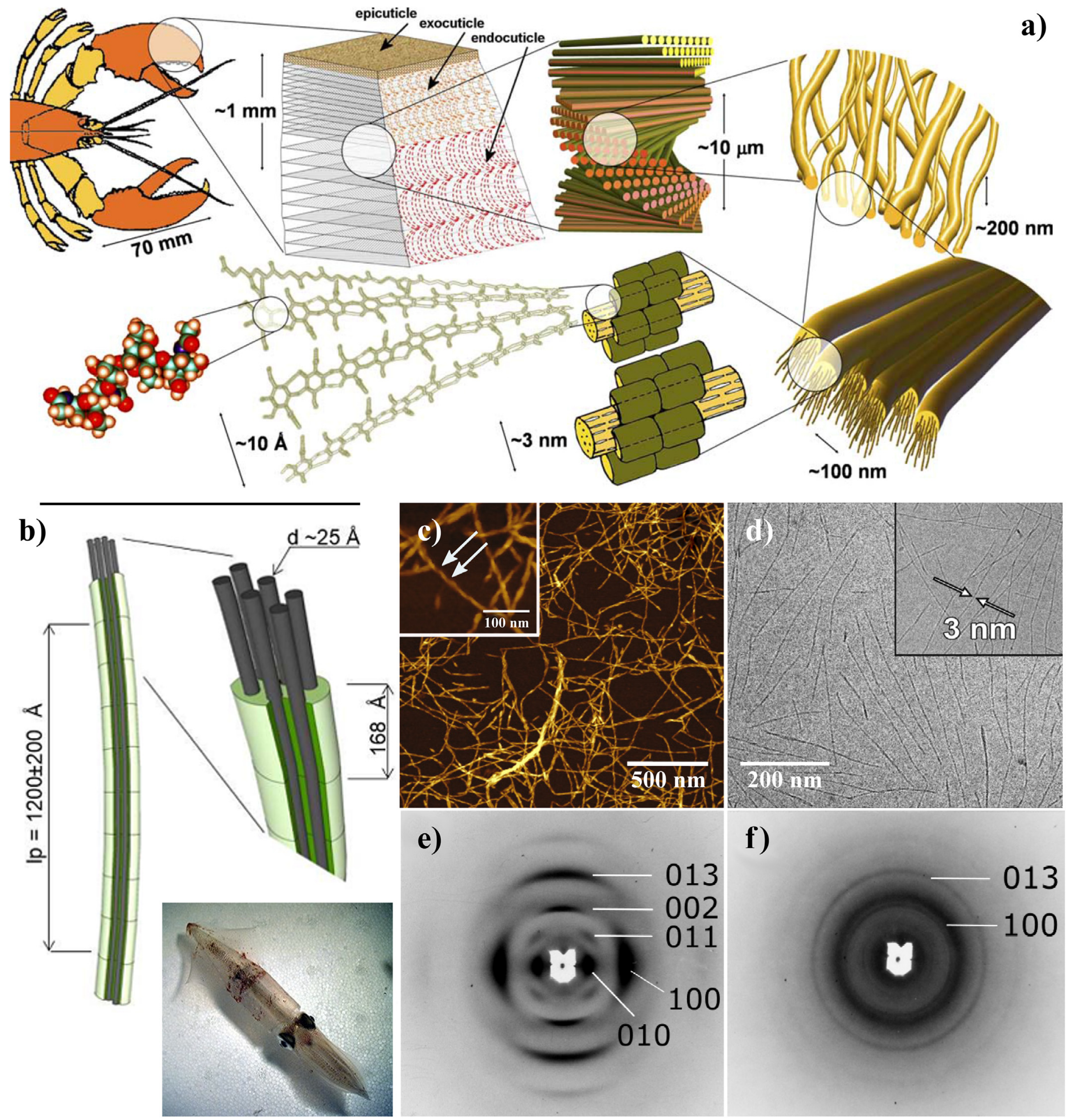

Fig. 15. Illustration of the hierarchical structure of crab claw, from macroscopic to molecular scale (a); model of squid pen chitin–protein complex (b); AFM image of dried chitin dispersion (c) and cryo-TEM images of frozen chitin dispersions (d) obtained by TEMPO oxidation. WAXS patterns of (e) deproteinized and (f) exfoliated chitin. Adapted from the studies by Raabe et al., and Bogdanova et al. [348] (a), [351] (b,e,f), and [352] (c,d) with permission from Elsevier, respectively.

study by Choi et al. [320], but nevertheless we dwell on the main achievements in this field.

Cellulose is the most abundant biopolymer in the world, and has a linear polysaccharide chain containing from hundreds to thousands of linked β-1,4-D-glucose monomers [321]. Cellulose (~40–50 wt%) is one of the main components of the tree cell wall. The hierarchical structure of tree results in the good mechanical properties of wood (Fig. 14a) [322]. Cellulose can be extracted from a broad range of plants and animals. Moreover, it can be obtained from various agricultural wastes such as wheat husks, straw, and soybeans [323,324], coconut husk fibers [325], mulberry [326] etc. Thus, people do not cause additional damage to the environment using waste for cellulose production. Cellulose contains two phases; they are amorphous and crystalline (Fig. 14b). Depending on the production resources and extraction processes, cellulose may have a different structure, including wood fiber, microfibrils, micro-or nanocrystals, etc., differing in morphology, length, aspect ratio, and degree of crystallinity [327–330] (Fig. 14c–e).

Commercially available rod-shaped particles of microcrystalline cellulose (MCC) exhibit an ER effect in silicone oil [331]. The yield stress and shear stress values were proportional to the weight fraction of MCC. The authors attribute this dependence to a proportionally increasing number of chains in the structure. However, the suspension does not form a gel at high mass fractions of cellulose particles. The micro-sized particles of MCC at a concentration of the order of 10 vol% allow to achieve the yield stress of several hundred Pa at low electric field strengths of 1–2 kV/mm [332,333]. Similar result was obtained for cellulose microfibers produced from rice husks. ERF achieved a yield stress exceeding 100 Pa under electric field of 2 kV/mm and filling degree of 10 vol% [334]. Such enhanced ER properties is obviously related with elongated particle morphology.

The ER activity of MCC suspensions in various oils has been studied as well. Devis et al. found that in ERFs based on biopolymer oils the yield stress was higher than in silicone oil both without and at electric field of 500 V/mm [335]. The ER characteristics of suspensions in almond oil as a dispersion medium differed slightly from silicone oil. However, suspensions in apricot, peanut, sesame, and safflower oils revealed higher yield stress due to oleic acid as the major constituent. It is important to note that the electric field strength was low because of lower resistance of biopolymer oils to electrical breakdown. This study was one of the first devoted to fully natural ERFs.

In addition to natural cellulose, its derivatives and composites were also used as fillers. Thus, the studies of ERFs filled by hydroxypropylcellulose [336], lithium cellulose carboxyl salt [142], cellulose phosphate [337], various cellulose composites both with polymers and inorganic particles [338,339] and surface-modified cellulose [340] were reported. It was shown that anhydrous ERFs filled with MCC phosphate showed an improved yield stress than pristine MCC suspensions [333,337]. Moreover, the concentration of phosphoric acid during esterification of cellulose has a critical role in ER performance of the particles. The best ER performance was observed for the ERF filled by particles treated with a 2.0 M phosphoric acid solution. Such result can also be explained by the optimum balance between conductivity and polarizability of particles. The polypyrrole-silica-methylcellulose nanocomposite particles dispersed in silicone oil also revealed ER activity. ER effect can be significantly influenced by the ratio of silica, polypyrrole, and methylcellulose in composite during the synthesis [338]. The higher silica/polypirrole ratio, the higher ER response. The yield stress of such fluids exceeds 1 kPa at electric field of 2 kV/mm and filler concentration of 7.2–9.9 vol%. Moreover, the yield stress initially increases, passes through a maximum, and then decreases with the methylcellulose amount. Thus, the interfacial particle polarization has a strong effect on ERF behavior. Partially modified to the Li-salt cellulose in corn oil revealed enhanced sedimentation stability and more intense ER effect. Suspensions of modified cellulose showed 5.5 times stronger yield stress compared to pristine cellulose. The optimum of the filler concentration was 20 wt% [142]. A cheap source of raw materials for ERFs can be spent coffee grounds, which consist mainly of hemicellulose, cellulose, and lignin [341]. Although such waste recycling without any further processing is rather attractive, the standardization of the composition of such filler will require additional costs, but achieving a high ER response of suspensions in silicone oil requires high filling up to 40–50 vol%.

It should be emphasized that cellulose nanocrystals or nanofibers have been considered as materials for ERFs only recently [342], although they have been investigated in many different aspects. The dynamic yield stress for 4 wt% suspensions in castor oil filled by nanocrystals and nanofibers was about 300 and 375 Pa at 4 kV/mm, respectively.

Starch is a natural mixture of two polysaccharides of amylose and amylopectin. The monomer of both these polysaccharides is α-glucose. Amylose is a linear biopolymer formed by α-1,4-D-glucopyranose monomers. Amylopectin is a branched polysaccharide also formed by α-1,4-D-glucopyranose units, with a branching rate of about 4–5% (every 20 units) [343]. Starch along with cellulose was considered as natural filler for ERFs. Thus, the ER behavior of suspensions filled by corn starch and its lithium carboxyl salt [344] in corn oil, potato starch phosphate [345], composite of feldspar/DMSO/carboxymethyl starch [346] in silicone oil was studied. ERFs filled by lithium carboxyl starch salt in silicone oil show a decrease in the yield stress with temperature reaching the minimum at 50 °C, then an increase of yield stress with a maximum at 80 °C was observed [344]. Such effect is related with the balance between polarization forces and the Brownian motion and strongly depends on the chemical nature of the filler.

Chitin is the second most abundant biopolymer in the world after cellulose. Chitin is a nitrogen-containing polysaccharide, polymer consisting of N-acetylglucosamine units linked by β-(1 → 4) - glycosidic bonds [347]. Chitin is a highly crystalline polymer; there are intra- and intermolecular bonds between hydroxyl groups, as well as between acetamide and hydroxyl groups. Chitin has three polymorphic modifications with different orientations of microfibrils: α, β, γ. The most common α-form is presented in the cell wall of fungi, the shell of crustaceans (Fig. 15a), and some mollusks, the cuticle of insects [348]. It is a tightly packed antiparallel polymer chains. In the case of β-form, the polymer chains are parallel and have greater solubility and swelling ability due to weaker intermolecular hydrogen bonds. This form is found in cuttlefish, in squid's gladius (Fig. 15b), the extracellular core of diatoms, and pogonophore tubes. A detailed analysis revealed that γ-chitin is a type of α-form [349].

Chitin can be produced at a relatively low price from organic resources especially shells of shellfish (shrimps, lobsters, crabs, and krill) and waste of sea food processing. The α-chitin forms nanofibrils with a diameter of 3 nm, containing 19 molecular chains with a length of about 0.3 μm. Natural crab chitin is not completely acetylated and contains ~12–13% glucosamine [350]. However, the both types, namely α- and β-forms, of chitin are used. The sizes, morphology, crystallinity degree, and particles aspect ratio, depends on production methods (Fig. 15c–f) [351,352]. Chitosan is an N-deacetylated chitin. The degree of deacetylation significantly affects various characteristics of chitosan, such as crystallinity, biocompatibility, bio- and thermal degradability, etc. Next, we consider the properties of some ERFs filled with chitin and chitosan.

One of the first studies of chitin ER activity showed that a suspension of chitin particles without a tendency to anisometry with size of about 10 μm in silicone oil does not reveal ER properties even at high electric field [353]. Glycerin was used as an activator of ER effect and the optimal concentration of chitin and activator was found ~11 and 5 wt%, respectively. It was observed that, the increase of the yield stress becomes less significant at high electric field strength (≥900 V/mm), due to the saturation of the particles' polarization. However, ER activity of pure chitin was discovered later.

As it turned out, the degree of deacetylation has a significant effect on the ER activity of chitin in ERF [47]. Pure chitin and chitosan (degree of deacetylation of more than 99%) exhibit the weakest ER effects. The best ER effect was observed in suspension of chitin with a degree of deacetylation of 87.3%, which was consistent with the data of dielectric spectroscopy. The dielectric permittivity changes depending on the degree of deacetylation. Thus, in the series of polymers the following dependence is

Table 6
Comparative analysis of ERFs filled by biodegradable polymer-based materials. Yield stress values are from experimental data unless an appropriate rheological model is specified.

Materials	Medium type and viscosity	Size	ρ_{filler}, g/cm^3	Conc.	$\Delta\varepsilon_{fluid}$, rel.un.	E_{max}, kV/mm	J_{max}, μA/cm^2	Yield stress at 3 kV/mm or E_{max}, Pa	Slope $\log(\tau_0) = f(\log(E))$	Sedimentation ratio, %	*Eff*, rel.un.
MCC [335]	Almond oil	< 45 μm	–	10 wt%	–	0.5	–	8.79[a] (Bingham)	–	–	1740
	Apricot oil							12.54[a] (Bingham)			2490
	Peanut oil							12.78[a] (Bingham)			2540
	Safflower oil							12.17[a] (Bingham)			2410
	Sesame seed oil							12.27[a] (Bingham)			2430
	SO							8.94[a] (Bingham)			1770
	Soybean oil							10.53[a] (Bingham)			2090
MCC [332]	SO, 50 cSt	< 20 μm	1.59	10 vol%	15.58	1.3	–	272[a] (Bingham)	1.5	–	20900
MCC phosphate [333]	SO, 50 cSt	25.3 μm	1.61	10 vol%	0.35	3	–	400 (CCJ, Bingham)	2	–	2660
MCC phosphate [337]	SO, 50 cSt	< 38 μm	1.3	10 vol%	~2	3	–	~400	1.7 ± 0.5	–	13330
Cellulose nanocrystal [342]	Castor oil, 550 ± 30 mPa s; 574 ± 27 cSt	Diameter of 10–20 nm and length of 0.3–0.9 μm	–	2 wt%	~0.75	4	~1.25	~100[a]	1.43 (< 0.94 kV) 0.72 (> 0.94 kV)	–	15000
				4 wt%	~1.1		~0.88	~300[a]	1.33		23000
				6 wt%	~1.4		~0.55	~350[a]			16700
Cellulose nanofiber [342]		Diameter of 10–20 nm and length of 2–3 μm		2 wt%	~0.75		~1.25	~70[a]	–//–		9980
				4 wt%	~1.1		~0.88	~350[a]			31200
				6 wt%	~1.4		~0.55	~400[a]			22200
Potato starch phosphate [345]	SO, 50 cSt	Egg-shaped particles with length of 60 μm	–	15 wt%	~0.7	2.5	–	~180[a]	1.60	–	4800
MCC phosphate [345]		50–100 μm			~2	5		~600	1.67		13300
HOPC [336]	SO, 50 cSt	50 μm	–	20 vol%	–	2	< 0.005	~160[a]	> 2	–	4000
MCC [142]	Corn oil, 45 mPa s	23.48 μm	1.35	20 wt%	–	4	–	~80	1.76	56	1330
Li-MCC [142]		46.30 μm					< 30	~500	1.69		8330
TiO_2–HOPC [339]	SO, 20 cSt	Spherical particles of 10–30 nm	–	30 wt%	–	4	–	< 200	–	–	2220
Polypyrrole-SiO_2-Methylcellulose [338]	SO, 100 mPa s	80–200 nm	–	9.9 vol%	~1.5	2	–	~1600[a]	1.24	–	11500
1-(3-(trimethoxysilyl) propyl) urea/MCC [340]	SO, 50 cSt	Rod-like particles of 20–100 μm	–	15 wt%	1.38	4	–	532 (CCJ)	2	–	1970
Microfibrillated cellulose [334]	SO, 100 cSt	Rod-like particles with diameter of 0.5–1 μm and length of < 10 μm	–	10 vol%	–	2	–	~100[a]	1.5	–	5000
Corn starch [344]	Corn oil, 45 mPa s	16.26 μm	–	30 wt%	–	3.5	–	~240	–	56	2670
Li-corn starch [344]		19.29 μm						~600			6670
Chitosan [345]	SO, 50 cSt	50–100 μm	–	15 wt%	~2	2.5	–	~250[a]	1.48	–	6660
Chitin doped by glycerin [353]	SO	10 μm	–	11.8 wt%	–	1	–	~550[a]	–	–	46600
Carboxylmethyl chitosan [353]								~180[a]			15250
Chitosan [357]	SO, 50 cSt	25 μm	–	30 vol%	–	4	< 0.04	~250	2	–	2780
Chitosan [363]	SO, 100 mPa s	Porous particles of ~50–60 μm	1.56	1 wt% (0.62 vol%)	9.55	7	–	169	1.3	98	5400
Chitosan [364]	Olive oil, 80 mPa s	Porous particles of ~50–60 μm	1.56	0.1 wt%	1.1	7	< 0.5	~60	1.4 (< 3 kV) 1.2 (> 3 kV)	18	200000
				0.2 wt%	2.5	7	–	~130	1.4 (< 3 kV) 1.2 (> 3 kV)	40	217000
				0.5 wt%	4.8	2	< 0.8	~180[a]	1.2	61	180000
				1 wt%	10.5	1.2	< 0.8	~130[a]	1	89	21600
											94
Chitosan, DD = 87.3% [47]	SO, 30 cSt	5–50 μm	–	30 vol%	–	3	< 0.017	~60	1.803	–	131
Chitosan, DD = 95% [358]	Soybean oil, 50 cSt	50–200 μm	–	15 wt%	–	3.5	–	~350	2 (< 1 kV) 1.5 (> 1 kV)	–	400
Chitosan, DD = 95% [359]	Corn oil, 50 cSt	< 100 μm	–	25 wt%	~3	3	–	~450	1.5	–	286
Chitosan, DD = 95% [109]	Soybean oil, 50 cSt	50–300 μm	–	25 wt%	–	2.5	–	~250–280[a]	1.86		416
	Corn oil, 50 cSt					3		470 (CCJ)	1.72		478
	SO, 50 cSt					2.5		~350–550[a]	1.78		8570

Chitosan succinate (I) [360]		SO, 30 cSt	5–20 μm	–	30 vol%	–	3.5	~0.6	~900	2.12	–	3810
Chitosan succinate (II) [360]								–	~400	1.50		8890
Chitosan adipate [360]							3	~1	~800	2.46		8570
Chitosan malonate [360]							3.5	~0.55	~900	2.03		8570
Chitosan sebacate [360]								~0.5	~900	1.90		2780
Chitosan succinate [369]	Al^{3+}	SO, 30 cSt	5–30 μm	–	30 vol%	–	3	~75	~250	1.579	–	310
	Cu^{2+}							~65	~28	0.061		221
	Ni^{2+}							~30	~20	0.297		54
	Fe^{2+}							~5	~5	0.004		88
	Zn^{2+}							~87	~8	0.008		1000
Chitosan phosphate [361]		SO, 100 cSt	25 μm	–	40 vol%	–	3	< 0.0001	~1700	1.67	–	60
Chitosan nitrate [362]		SO, 98 mPa s	–	–	25 wt%	–	4.2	–	~1000	–	–	1
Chitosan/perlite [354]		SO, 100 mPa s	10 μm	0.80	10 vol%	–	3	–	1250	–	–	3730
Chitosan phosphate-$Y(NO_3)_3$ [367]		SO, 98 mPa s	–	–	25%	–	4.2	–	~1400 (at 100 s^{-1})	–	–	96800
α-Chitin nanorods [355]		SO, 100 mPa s	Rods with diameter of 5.7 ± 1.6 nm and length of 380 ± 130 nm	1.4	0.5 wt%	0.82	7	0.55	39	1.4 ± 0.1	23	74100
					1 wt%	1.01		0.87	69	1.2 ± 0.1		6
Chitosan/Tb sulphosalicylic acid [368]		SO, 98 mPa s	< 200 μm	0.65	30 wt%	–	4	–	500	–	–	43
Chitosan DD = 92.3%/alginic acid [132]		SO, 50 cSt	5–50 μm	–	30 vol%	–	3	~0.05	~140	–	–	41
Oleic acid stabilized agarose [370]		SO, 350 cSt	1–9 μm	–	60 wt%	–	2	–	~1000[a]	–	–	1738

[a] Yield stress values are given at maximum electric field.

observed: 87.3% > 83.8% > 73.2% ≫ 93.4% > 99.3% ≫ 0% (chitin). Thus, the polarizability of particles and ER effect of their suspensions substantially depends on the balance of the aceto- and amino groups in chitin structure. The yield stress reaches about 60 Pa at 3 kV/mm and 30 vol% for a sample with a deacetylation degree of 87.3%. The leakage current density does not exceed 0.017 μA/cm^2. However, the particle sizes were relatively large and varied in the range of 5–50 μm. It was observed that the shear stress increases with temperature under electric field. The results suggest that the polarizability increases with temperature more extent and dominates over the contribution of Brownian motion due to the large particle size. Similar dependences were observed in suspensions filled by composite of chitosan with bentonite [192], perlite [354], and a mixture of chitosan with alginic acid [132]. Recently, it was shown that highly anisometric α-chitin nanocrystals in silicone oil exhibit a pronounced ER effect with stable response under switching electric field [355]. The static yield stress reaches 220 Pa at 7 kV/mm for the 1.0 wt% suspension. Nevertheless, the sedimentation stability of suspensions remains relatively low indicating the poor compatibility between filler and dispersion media, but theoretically can be improved by α-chitin nanoparticles modification. Obviously, this is a subject for further research.

Effect of chitosan concentration [356] and nature of dispersion medium, namely silicone [357], soybean [358], and corn [359] oil were studied as well and comparative analysis of the dispersion medium effect was also performed [109]. A change in the chemical nature of the chitosan side groups leads to a corresponding change in the ER activity of its suspensions, which depends on the structure of the group and the degree of chitosan substitution. Thus, trisubstituted chitosan succinate shows a higher ER effect compared to monosubstituted. The yield stress reaches about 900 Pa compared to 400 Pa at electric field of 3 kV/mm and particles concentration of 30 vol% [360]. Also, suspensions of chitosan phosphate [361] and nitrate [362] in silicone oil revealed the yield stress exceeding 1 kPa at electric field strength of 3 kV/mm and filling degree of 25 wt% and 30 vol%, respectively.

Recent advantages show the high efficiency of porous chitosan particles as fillers. Thus, freeze-dried micro particles revealed high ER response at extremely low concentrations (lower than 1 wt%) [363]. The yield stress reaches about 540 Pa at electric field of 7 kV/mm. High porosity allows the dispersion medium to penetrate into the particles. Thus, the structure of such particles acts as a pre-organized template of the percolation network leading to a high efficiency of ERFs. The porous particles can deform both under electric field and shear flow. However, these deformations are reversible, which is confirmed by the good reproducibility of the rheological behavior of fluids. The prospects of using highly porous particles were shown. Additionally, olive oil was used as the dispersion medium to produce nature-friendly ERFs filled by porous chitosan particles [364]. The data were fitted by the Bingham, Seo–Seo, and CCJ models. CCJ model reveals the best fit, but not ideal. A feature of such fluids is the narrowing of the electric field strengths before breakdown. Nonetheless, the fluids showed high yield stress values and stable operation in on/off mode.

Other promising ways to modify chitosan for ERFs are the production of polymer and inorganic composites, and complexes with metal ions. Thus, composites with previously considered multi-walled CNTs [243], graphene nanosheets [365], bentonite [192], perlite [354], and cellulose [366] are known. The nanoclay as the basis of the composite allows to improve the sedimentation stability of fluid, and to obtain a synergistic ER effect. The low amount of chitin on the surface of bentonite (1 wt%) results in enhanced yield stress under electric field. The yield stress increases by more than two times: compare 150 Pa and 73 Pa at 25 wt% particles content and at electric field of 1 kV/mm. A further increase in the

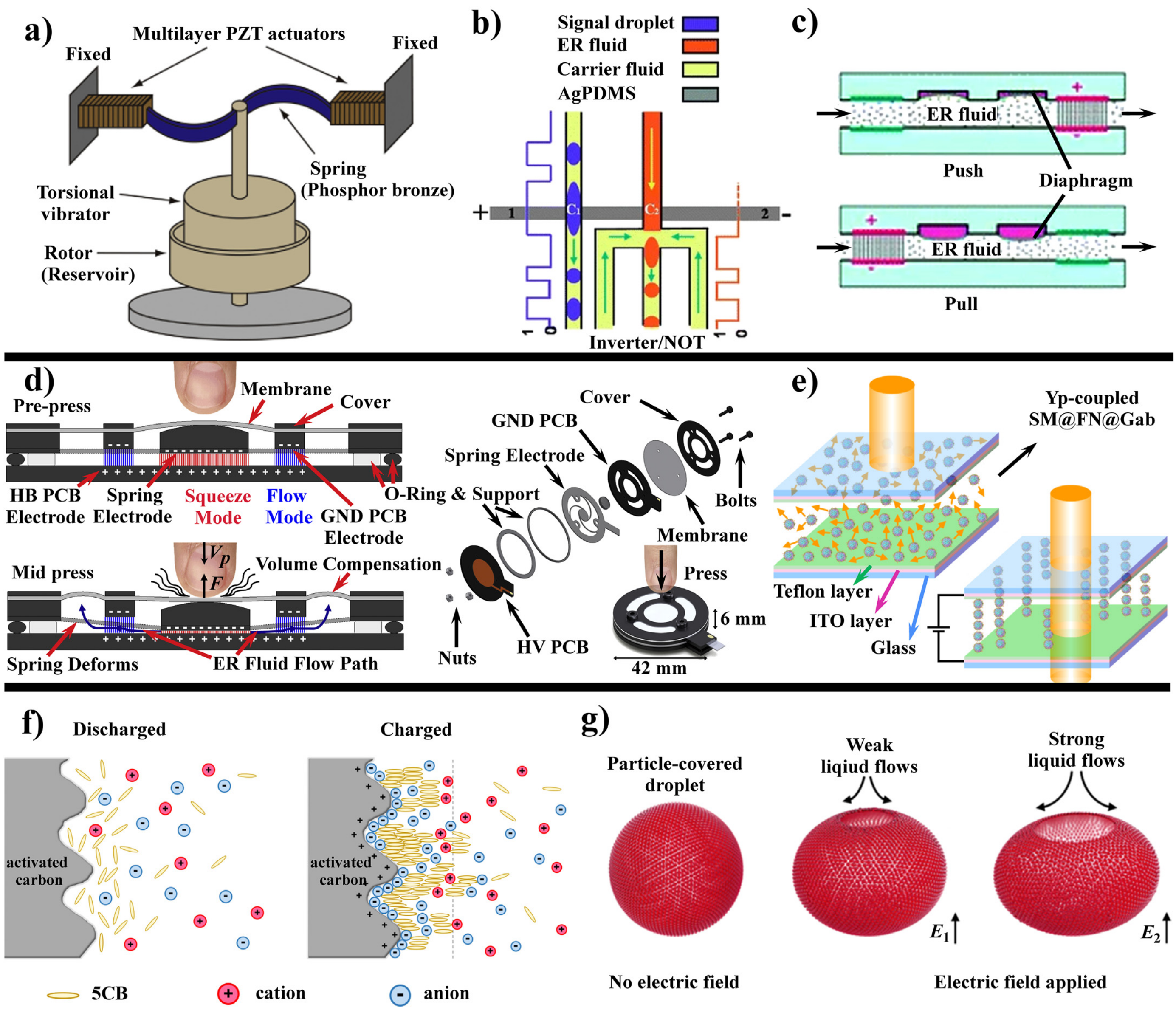

Fig. 16. Several modern applications of ERFs and developments based on ER effect. A principle of non-contact rotary motor using a piezoelectric torsional vibrator (a) [384], basic working principle of microfluidic switches (b,c) [412]. When the carrier flow is flowing between the signal electrodes, ERF flows continuously. When the signal droplet passes by the signal electrodes, ERF solidifies and is cut into droplets. Microfluidic logic switch is built by adding a capacitor (in the present case, another microfluidic channel filled with steady ERF) to the logic inverter (b). Operation of the mixer chip relies on the perturbation of the main channel flow by cross-stream side-channel flows. The side-channel flows are perturbated by pressure changes through thin membranes affected by an ERF micro valve (c). Cross-section view of the working principle of the button element operating in pre-press mid press modes and exploded view drawing of device (d) [416]. ER sensor consisting of transparent indium tin oxide glasses with an insulating layer (Teflon) and spacer operated at various frequencies under an alternating electric field in various transmittance state (e) [420]. The ER effect on the discharge of a supercapacitor (f) [422], and schematics of particle shell opening under electric field E_2>E_1 (electrosensitive microdiaphragm) (g) [423]. Adapted from the studies by Qiu et al., Wang et al., Mazursky et al., Chou et al., Xia et al., and Rozynek et al. [384] (a), [412] (b,c), [416] (d), [420] (e), [422] (f), and [423] (g) with permission from Elsevier, Springer Nature, IOP Publishing and American Chemical Society, respectively.

chitosan content in the composite leads to a decrease in the ER effect; however, the ER properties remain higher than on pure bentonite [192]. Composites of chitosan with perlite (50%) in silicone oil can reveal the yield stress of 1.25 kPa at 3 kV/mm and a filling degree of 10 vol% [354]. Li et al. studied the ER properties of chitosan phosphate suspensions and its complex with yttrium nitrate in silicone oil [367]. The particles concentration was 25%. The authors note, the drop of the yield stress value at zero-field for ERF filled by chitosan phosphate rare earth complex. Meanwhile, the ER activity of such complex was enhanced. The shear stress reaches about 2.5 kPa at electric field of 4.2 kV/mm and shear rate of 100 s^{-1}. The relative effectiveness of such filler was in 13.6 times higher than that of pristine chitosan. In addition, the yttrium complex material had better thermal stability. Later, the ER activity of chitosan complexes with sulfosalicylic acid and rare earth metals, namely terbium and lanthanum was studied [368]. The most effective was complex formed by Tb^{3+}. An extra property of studied fillers was their luminescence. However, such rare earth contain fillers will be quite expensive, which negates the economic attractiveness of chitin and chitosan. Disubstituted chitosan succinate dispersed in silicone oil showed good ER properties under electric field. However, its complex with metals, such as Al^{3+}, Cu^{2+}, Ni^{2+}, Fe^{2+}, or Zn^{2+} the fluid reveals decreased ER activity, increased current, and conductivity values [369]. The authors supposed that the applied electric current flows through the surface of particle with metal complex and polarization does not occur.

A comparison of the ER behavior of suspensions filled by chitosan, cellulose phosphate, and potato starch phosphate was studied by Sung and Choi [345]. Thus, chitosan particles in silicone oil at a concentration of 15 wt% showed the highest ER effect compare to cellulose phosphate and starch phosphate particles. The main reason of this dependence is in fact that chitosan contains more polar groups than other biocompatible polysaccharides.

A recent study shows the ER activity of agarose in silicone oil [370]. Oleic acid was used to stabilize the suspension. The authors managed to achieve the yield stress of 1 kPa at a field strength of 2 kV/mm. However, the filler concentration remains extremely high and is 60 wt%. The comprehensive analysis of ERFs with biodegradable fillers and their composites is presented in Table 6.

Finally, biocompatible and biodegradable polymer particles are promising fillers in terms of "green" chemistry and allow to create nature-friendly materials. The ER properties of fluids can be tuned by surface modifications of biopolymer particles and by using of various composite materials with inorganic compounds. The most studied are ERFs filled by cellulose and chitosan particles.

5. Areas of application of the ER effect

Some of recent practical applications of the ER effect were considered by Dong et al. [371]. Nevertheless, in the framework of this review, we note the long-known fields of ER effect application and briefly dwell on the modern ones, highlighting the basic generalizing principles.

Aerospace, robotics, clutches, shock-absorbers, valves, and quick-response dampers were one of the first areas of application for ERFs. The ability to damp external vibrations due to the rapid transition of ERF from a viscous to elastic state is quite obvious and was implemented in various ways [372]. Next, let's consider some recent studies in the field of damping. An important aspect of damping is the durability of the ERF. Bilyk and Korobko examined the effect of temperature on performance of ER shock absorber. It was shown that the adsorber heats up while operating due to the dissipative heating mainly during the first 40–50 min. Moreover, electric field led to an additional heating. Thus, it is important to take into account temperature factor for the development of controlled vibroprotective systems due to effect of viscosity decrease under increased temperature [373]. Unfortunately, only a few studies are devoted to this issue for quite long operational times. Thus, Bauerochs et al. showed the possibility of 2000 h ER valve operating under alternating voltage. However, the authors used circulation unit for mixing the fluid in order to avoid sedimentation [374]. Heinken et al. proposed ER drive to improve the operation of servo-valve in hydraulic linear actuator. The actuator consists of two stages. The first stage (pilot) provides constant volume flow of ERF. Under electric field, the ER effect is revealed and the pressure increased rapidly and passed through internal channels into the pressure chambers and the valve spool in the second stage (main) is deflected. The developed ER servo valve operates with a flow rate of 18 L/min at 180 bar and a high dynamic range over 233 Hz. The authors show that the performance of hydraulic drives can be increased by ER servo valve and combined with the high dynamics of ER systems [375,376]. Tan et al. created ER valves with parallel segmented electrodes and showed the ability of such damper to switch between stick- and slip-states to reduce selective vibrations. The model describing the ER-valve behavior was proposed as well [377,378]. Djokoto et al. suggested effective way to change the vibration characteristics of the micro-cantilever beam by a layer of ERF at the tip [379,380]. Recently, the possibility of damping sound vibrations has been shown by example of sandwich plate and sandwich cylindrical shell structures [381,382].

One of the earliest applications of ERF was a sound reinforcing device [383], proposed by Winslow, discoverer of ER effect. The diaphragm of the loudspeaker is connected to the clutch of a coaxial cylindrical type. There is ERF between the cylinders, which act as electrodes. Potential is supplied from the transformer secondary winding. Changes in sound frequency affect current and potential. The cylinders are coupled at sufficiently high voltage, and when the voltage is weakened, the external cylinder rotates in the opposite direction under the action of the diaphragm. A similar principle has been used in recent years to create a non-contact rotary motor using a piezoelectric torsional vibrator [384]. The electric pulses were synchronized with the vibration velocity of the torsional vibrator connected with inner cylinder and were applied to the ERF between coaxial cylinders to drive the motor. The external electric field applied to the ERF and large torque from piezo-element transmitted to inner cylinder (Fig. 16a). The electric field switched off and inner rotor slip, when the torsional vibration velocity low or opposite to the rotation direction. As a result, the output torque was always positive and the motor rotated in one direction. The operational characteristics of the ERF are sufficient to convert electrical energy (piezo) into mechanical one with an extremely high efficiency.

ERFs can be used in electrically controllable acoustic prisms, lenses, and prism/lens combinations [385]. The columnar structures formation under electric field results in increase of sound speed in the fluid. The acoustic index of refraction can be tuned analogously to varying the index of refraction of an optical lens or prism. This effect allows controlling the angles of refraction of acoustic beams and focal lengths of lenses.

It is expected that the 21st century will be the time of technological breakthrough, and today scientists are thinking about creating so-called "soft" and small-scale robots in which the mobility can be implemented using ER- or magnetorheological fluids [386–388]. One of the tasks in the space industry is the remote control of robotics, in particular the quantitative transmission of momentum. Thus, various options for actuators and robotic arms were proposed. The sensitivity of such arms can be transferred to the operator due to the elements built into the feedback system containing ERF as an operating agent. In the simplest version, this was a set of cylinders. Elasticity of cylinders was controlled by the viscosity of the ERF in electric field [389,390]. Later, the haptic device using an ER fluid was applied to the robot-assisted surgery. Two actuating ER-based mechanisms were introduced in haptic master to achieve both rotational and translational motions. The spherical joint generating 3 rotational motions and the linear actuator generating translation force reflection. Thus, the ERFs can be used as the operating fluid of the robot arms, hinges, etc. Fine-tuning the elasticity of the material allows to accurately position the limb of the robot and at the same time has high sensitivity. It was shown that operator precisely manipulates the slave robot based on the haptic sensation feedback signal. Thus, the remote of the proposed or similar haptic master is matter of time [391]. A similar idea was used for artificial muscles and exoskeletons to help the movement of people with musculoskeletal system diseases [392,393]. The set of electro-sensitive elements is contracted by electric field, which helps a person bend his leg at the knee, and thus facilitate movement.

The transition of rehabilitation robots from the clinic to the home could increase the accessibility of therapy to stroke survivors. Many stroke survivors have partial paralysis and significant upper extremity impairment likely. Upper extremity rehabilitation robots are one of candidates in order to extend therapy at home. Davidson and Krebs proposed a way to reduce the cost of robots of this type through the incorporation of ER clutches [394].

The general principle of the ER effect formed the basis for modifying or creating new technological processes for separating heavy emulsions, transporting oil, producing chocolate, and optimizing the process of starting diesel engines in cold weather. For example, when two mesh electrodes are inserted into the flow, the transmittance of the pipelines does not decrease; however, different processes are observed under electric field. In the case of oil, columnar structures are formed from electrosensitive impurities, which on the one hand results in purification or separation (suitable for water-in-crude oil emulsions) [271,395], and on the other hand reduces viscosity leading to appearing of slip planes of pure oil along the columnar structures formed by impurities [396]. However, further development of the ER effect application for oil transport has occurred in recent years. Huang et al. found that the viscosity of waxy crude oil reduces under applied electric field either perpendicular or parallel to the flow direction [397]. Such effect indicates that the viscosity reduction upon electric field caused not only by formation of wax particle chains but also due to interactions between wax particles or impurities. Cold flowability of waxy crude oil is improved by weakening the wax structure at higher electric field strength and lower temperature [398]. The mechanism of structure violation on the brink of discovery. It is hypothesized that charged colloidal particles such as asphaltenes and resins adsorb onto the surface of wax crystals leading to their repulse and reducing the van der Waals interaction [399]. In chocolate production, the aggregation of cocoa solids occurs, decreasing fat content and thus more delicious chocolate can be obtained [400]. One of important aspects for diesel engines is fuel injection during start-up. The combustion chamber should be saturated by fuel vapor to ignite them; however, in the cold season, the viscosity of the fuel decreases leading to an increase in the size of injected fuel droplets and, as a consequence, an increase in the time of their evaporation. It turned out that the principle of reducing the viscosity also works in this case. The application of an electric field leads to the separation of impurities in separate columnar structures and the viscosity of the fuel is reduced due to slipping along. Thus, injecting droplets are smaller and evaporate faster [401,402].

It is also reported on the application of the ER effect in polishing processes, in particular the movement of an abrasive mass under the influence of an electric field [403]. The mixture of the ERF and the abrasive particles shows excellent polishing performance. Moreover, ERFs can be used to the active control of the friction coefficient by electric field [342,404].

In recent decades, microfluidics and a laboratory on a chip have developed rapidly [405]. Mixing liquids in microchannels allows to provide quick and directional chemical reactions with a minimum consumption of reagents and is widely used in inkjet printing, in biology and medicine, *in vitro* diagnostics [406–408], and protein sequencing [409]. ERFs found their application in microfluidics as well [410,411]. In particular, the implementation of two principles is possible (Fig. 16b and c). One is the use for immiscible with ERF flows. Then, accurate drop dosing can be carried out via controlling the flow of ERF by an electric field. The second method is the organization of two parallel-flowing streams separated by a soft membrane. In this case, the acting of an electric field leads to ERF solidification and extrusion of the membrane, which in turn leads to the shutoff of the adjacent flow [412]. A similar principle formed the basis of elements with a changing relief, for example Braille books for the blind people [413,414]. Thus, the device consists of a capacity for ERF and a plate with holes, covered by elastic membrane. When pressure is created in the capacity with ERF, the relief in the hole depends on viscosity of ERF. Any relief can be formed by a suitable number of holes or by pins as well. Then the value of the ERF yield stress should be high enough to hold the pin in the upper position.

The modern development of the computer games industry and virtual reality requires a new approach to devices for transferring stimuli from computer visualization to humans. The transmission of tactile sensations was realized on the example of ERF. In the basic version, as a button element with a controlled response force, and in the future, probably in the form of whole suits (Fig. 16d) [415,416]. It should be noted, that tactile displays can be considered as prototypes of such development [417,418]. Furthermore, the fitness bike and flexible self-phone can be noticed among household applications of ER effect. In particular, an ERF with high yield stress can be used as a smart bike transmission [86]. Thus, the force you need to apply to pedal rotation will depend on the magnitude of the supplied electric potential. The idea of devices miniaturization remains relevant today; so, in 2006 a prototype of a flexible keyboard device based on the ERF was proposed [419]. In a compact state, keyboard is folded into a "tube" and is deployed when electric potential is applied (for example, from a battery).

One of the recent applications of ER effect in medicine become the plague diagnostics [420]. Plague is a disease infected by an etiological agent, Gram-negative bacterium *Yersinia pestis*. Thus, a sensor based on composite particles of polystyrene microspheres covered by ferroferric oxide nanoparticles with a core/shell structure was developed. Such particles were additionally coated by protein G and an antibody of *Y. pestis.* The sensor made from two transparent glasses filled by ERF and displayed in a low transmittance state without electric field. Commercially transparent indium tin oxide was used as electrodes. In an alternating electric field, particles are aligned along the electric field (ER effect) and the transmittance of the sensor increases (Fig. 16e). The field frequency corresponding to the maximum of transmittance is different for the particles with various compositions and can be used as an indicator to detect the antigen of *Y. pestis*. Thus, the plague can be quickly diagnosing by particles with protein G, which binds to *Y. pestis*. The maximum transmittance of particles with protein and with bacteria will be observed at various frequencies. Furthermore, the selectivity of such diagnostics was shown using *Escherichia coli* and *Staphylococcus aureus* as example. The great attractiveness of such displays loading by ER media is in rapid label-free biosensing that could diagnose plague simply by the naked eye. Obviously, the proposed approach opens the opportunity for other diseases diagnostics, if specific reactions of viruses or bacteria with proteins are known [421].

One of the latest practical applications of the ER effect include improving the performance of supercapacitors [422]. One of the problems in this area is the process of self-discharge of inoperative devices. When electro-sensitive particles are added to the capacitor's working medium, the particles are oriented along the lines of force during self-discharge (ER effect) and "clog" the surface of one of the electrode plates, thereby complicating the migration of ions and charge leakage (Fig. 16f).

Additionally, the study by Rozynek et al. should be pointed [423]. The authors showed the possibility of reversible opening and closing of particle shells on droplets realized by electric fields. The silicon oil droplets with various particles (silica, dyed polyethylene, paramagnetic particles, Li- fluorohectorite and laponite clays) were dispersed in castor oil. Thus, the authors obtained a complex disperse system consisting of two liquid phases and one solid. That is, the final material is emulsion of silicon oil-based suspension in castor oil. The initial droplet (silicon oil suspension) densely covered by particles is spherical and under electric field is warped due to the electric stress. Then the electrohydrodynamic flows are induced and shift particles away from the droplet's electric pole. Thus, the aperture in the particle layer appears (Fig. 16g). After removing the electric field, the drop again becomes uniform. These droplets can be used as a millimeter-sized diaphragm with an adjustable aperture, and open opportunities for numerous attractive applications such as manipulation of droplet liquid without harming its particle shell, online diagnostic for studying the content of droplets, microfluidic or microelectromechanical systems, and to control the colors of materials through structural manipulation (structural coloration) as well.

Finally, the ER effect and ERFs find wide practical application both in serious technological processes and in everyday life indicating the relevance of the further development of this scientific field. In this way, we are waiting for a large number of new and unexpected fields of application. Some additional valuable information in the field of electrorheology provided by Liang et al. [424].

6. Conclusion

ERFs are a promising modern material for various practical applications. It is possible to obtain materials in a wide range of specified operational characteristics depending on the electric field strength, nature, size and concentration of the filler, as well as viscosity and nature of dispersion media. The yield stress can be varied from units of Pa to hundreds of kPa. It is shown that, modern chemistry allows to directionally obtaining materials with predetermined properties by controlling the chemical nature of the filler and its morphology. The main promising areas in modern electrorheology were demonstrated. Most of them are related with the selection of a suitable disperse phase. However, the modern trends can also be associated with the new dispersion media or their modifications. Particles able for facile modification are an excellent template for creating hybrid fillers with improved characteristics. Particle shape effect on the ER behavior of fluids was considered during the review. It is clear that particle anisometry is an additional factor of enhancing the ER activity of suspensions. The review highlights a separate section devoted to the effect of temperature on ER behavior of fluids; this issue was covered during analysis of various filler types. The influence of temperature may be considered as balance of two contributions: one is the electrostatic polarization force, which may either strengthen or weaken the ER effect; the other is the intensification of Brownian motion at high temperature, which tends to weaken the ER effect. Thus, the effect of temperature is really dependent on which factor would become dominant and related with chemical nature and size of the filler particles.

However, despite a large number of studies in the field of electrorheology, fundamental questions regarding the influence of the nature of the dispersion medium and filler, the polarizability and concentration of the dispersed phase, as well as the influence of the shape of the particles and their structural organization on the ER effect still remain. To date, there is still no unified theory explaining the mechanisms of ER effect on whole filler types. According to the generally accepted approach, the yield stress values are a quadratic function of the electric field, which follows from the proportionality of the electrostatic energy, defined as the polarization of particles multiply by field strength. The deviation of the yield stress from the quadratic law is associated with various polarization mechanisms and conductivity contribution. The influence of conductivity with increasing electric field strength becomes significant and eventually leads to electrical breakdown. The deviation from a quadratic dependence up to a linear one for particles with a core/shell structure is related with the appearance of an additional component in the polarization process, namely, dipole interaction of the surface layer molecules.

Finally, the low-concentrated ERFs with high performance will reduce the cost of the material and will increase the commercial availability of the developed fluids. This is one of the urgent tasks today. The use of fillers that do not have a negative and destructive effect on the environment is extremely important in the context of the paradigm of nature-like materials and of restoring the balance between the biosphere and the technosphere.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this article.

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